



**PT SAPTAINDRA SEJATI**

**ENGINE**

**CAT ENGINE C32 DISSASSEMBLING  
CHECKSHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

**WO NO :**

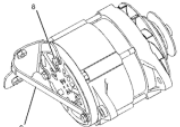
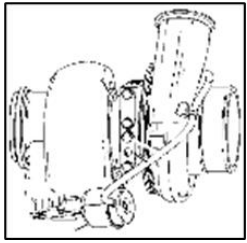
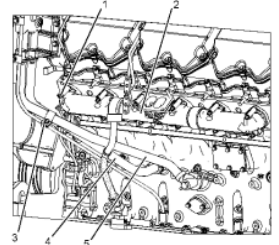
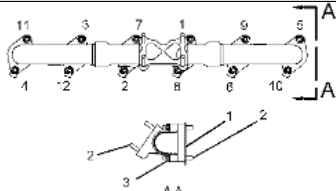
**ENGINE SECTION**

| CHECK SHEET                   |  | WO NUMBER  | DOCUMENT NO |
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| CAT ENGINE C32 DISSASSEMBLING |  | COMP.MODEL | PUBLIC DATE |
| ENGINE CAT C32                |  | COMP.S/N   | REVISION NO |
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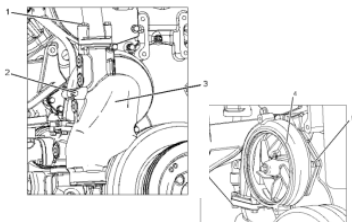
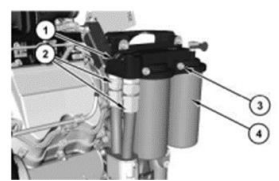
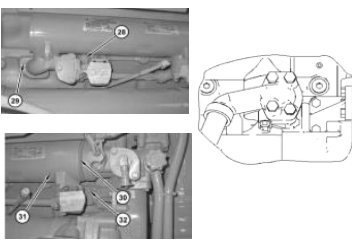
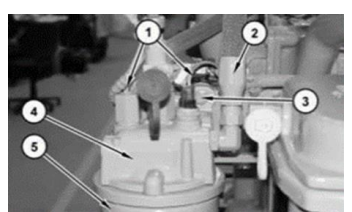
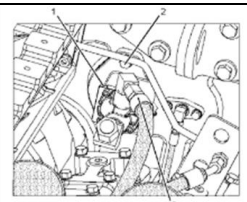
  

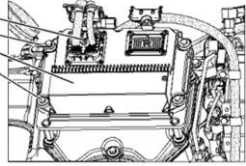
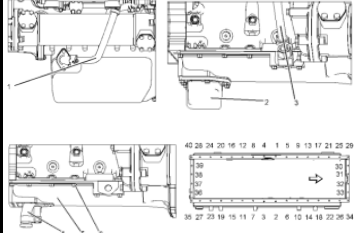
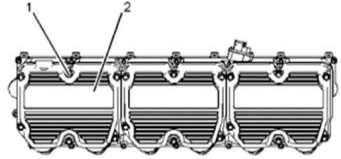
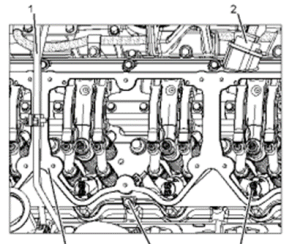
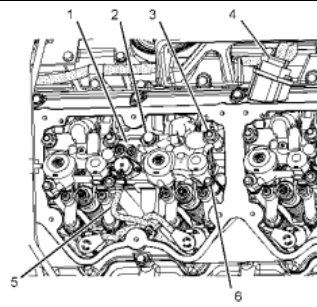
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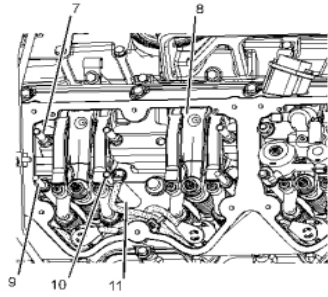
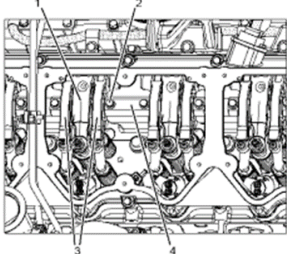
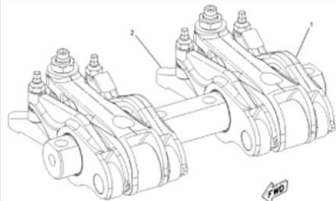
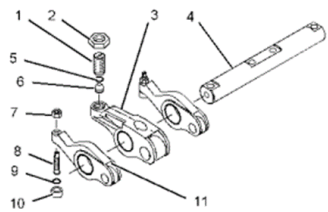
  

| NO | PART TO INSTALL OR ASSEMBLY      | CONDITION TOBE INSTALL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SPEC | SIGN |      | SKETCH DRAWING OR PHOTHOES                                                                                                                                                         |
|----|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      | MECH | GL/Q |                                                                                                                                                                                    |
| 1  | ENGINE ASSY                      | * Setting engine pada engine stand<br>* Posisikan Engine dengan baik dan benar atur agar stabil dan tidak mudah jatuh saat pelepasan dan perakitan komponen engine.<br>* Drain engine oil dari oil pan (Jumlah banyaknya Oli berbeda sesuai model Engine).<br><b>Putar Crankshaft 2xPutaran untuk memastikan kondisi nya (jamed or not).</b><br><div style="display: flex; align-items: center;"> <div style="border: 1px solid black; padding: 2px; margin-right: 5px;">CHECK</div> <div style="display: flex; flex-direction: column; align-items: center;"> <div style="border: 1px solid black; padding: 2px;">YES</div> <div style="border: 1px solid black; padding: 2px;">NO</div> </div> </div>                                                   |      |      |      |                                                                                                 |
| 2  | STARTING MOTOR<br><br>ALTERNATOR | * Perhatikan terpasang apa tidak.<br><br><b>Starting Motor AssyS/N :</b><br><br><b>ALTERNATOR ASSY S/N:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |      |      | <br><br>    |
| 3  | TURBO CHARGER                    | * Lepas Turbocharger, Lepas semua Bolt mounting dan adapters.<br><br>* Lepas Turbocharger Line dan conectionnya. angkat turbocharer.<br><br><b>TURBOCHARGER ASSY S/N:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |      |      |                                                                                               |
| 4  | EXHAUST MANIFOLD                 | * Lepas Clamp (3)<br>* Lepas Tube (4 & 5)<br>Kendorkan dan lepaskan Locknut (1) beserta washer dan specernya.<br><br><div style="display: flex; align-items: center;"> <div style="border: 1px solid black; padding: 2px; margin-right: 5px;">FRONT</div> <div style="border: 1px solid black; padding: 2px;">CENTRE</div> <div style="border: 1px solid black; padding: 2px;">REAR</div> </div><br><div style="display: flex; align-items: center;"> <div style="border: 1px solid black; padding: 2px; margin-right: 5px;">FRONT</div> <div style="border: 1px solid black; padding: 2px;">CENTRE</div> <div style="border: 1px solid black; padding: 2px;">REAR</div> </div><br>* Lepas exhaust manifold kendorkan baut sesuai Nut yang harus dilepas. |      |      |      | <br><br> |

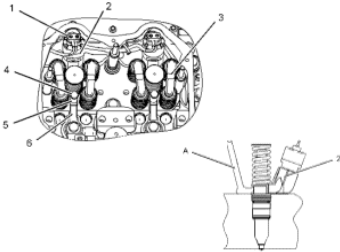
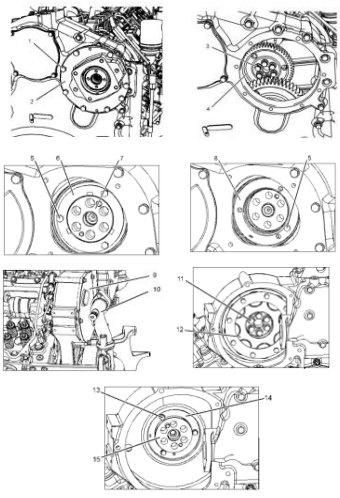
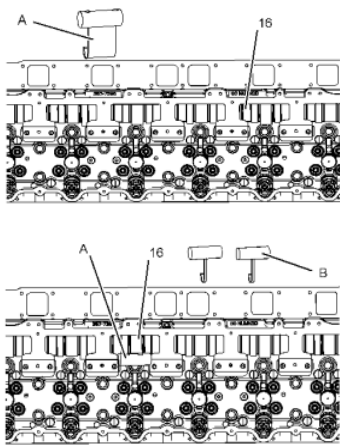
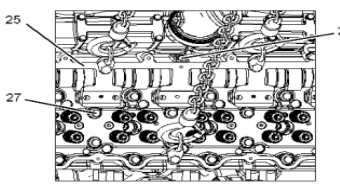
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| <b>CHECK SHEET</b><br><b>CAT ENGINE C32 DISSASSEMBLING</b><br><b>ENGINE CAT C32</b> | WO NUMBER  |  | DOCUMENT NO |        |
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| PROCESS |                         | START                                                                                                                                                                                                                                                                                                     | ASSEMBLY BY |  |                                                                                       |  |
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|         |                         | FINISH                                                                                                                                                                                                                                                                                                    |             |  |                                                                                       |  |
| 5       | <b>WATER PUMP</b>       | <ul style="list-style-type: none"> <li>* Lepas bolt (1) dan bolt (2)</li> <li>* Lepas Cover (3)</li> <li>* Lepas bolt (6) dan nuts (5)</li> <li>* Lepas Water Pump (4)</li> </ul>                                                                                                                         |             |  |    |  |
| 6       | <b>OIL FILTER BASE</b>  | <ul style="list-style-type: none"> <li>* Pisahkan hose Assy (2)</li> <li>* Gunakan kunci Filter untuk melepas Oil Filter (4) dari Base nya (1)</li> <li>* Lepas bolt (3).</li> </ul>                                                                                                                      |             |  |    |  |
| 7       | <b>OIL COOLER</b>       | <ul style="list-style-type: none"> <li>• Lepaskan Bolt (28 &amp; 29)</li> <li>• Lepas 2 bolt (32) Lepas Engine Oil Cooler (31) dan kedua O-Ring Seal nya.</li> </ul>                                                                                                                                      |             |  |   |  |
| 8       | <b>FUEL FILTER BASE</b> | <ul style="list-style-type: none"> <li>• Putar Supplay Pump Posisi Close/Off</li> <li>• Lepas Harness Assemblies (1)</li> <li>• Lepas Hose Assemblies (2)</li> <li>• Menggunakan Kunci Filter lepas Fuel Filter (5) dari dan Filter Base (4)</li> <li>• Lepas Bolt (3) dan Lepas Fuel Base (4)</li> </ul> |             |  |  |  |
| 9       | <b>TRANSFER PUMP</b>    | <ul style="list-style-type: none"> <li>• Tutup kran dari Fuel Tank agar Fuel tidak mengalir.</li> <li>• Lepas Hose (3) Lepas Bolt (2) Lepas Fuel Transfer Pump (1) beserta O-Ring nya.</li> </ul>                                                                                                         |             |  |  |  |

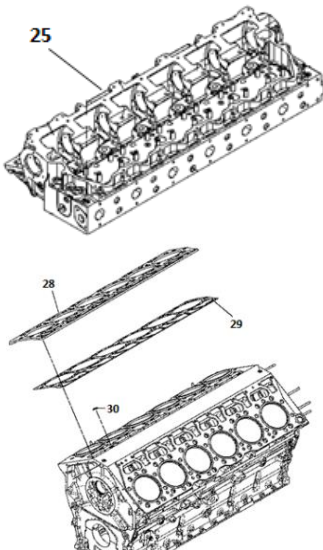
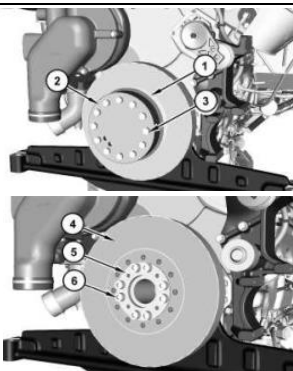
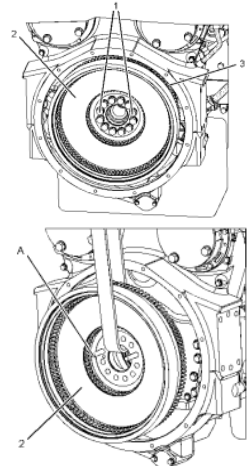
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| PROCESS                       | START<br>FINISH               | ASSEMBLY BY                                                                                                                                                                                                                                                                                       |             |                                                                                       |
| 10                            | ELECTRONIC<br>CONTROL MODUL   | <ul style="list-style-type: none"> <li>Kendorkan allen head screw (1) kemudian lepas</li> <li>Harness Assy dari ECM (3)</li> <li>Lepas bolt (4) Ambil ECM (3) dan ambil Ground Strap (2)</li> <li>Lepas washer dan isolasi mounting dari ECM.</li> </ul>                                          |             |    |
| 11                            | OIL PAN                       | <ul style="list-style-type: none"> <li>Lepas drain tube assy (1)</li> <li>Pisahkan Hose Assy (3)</li> <li>Lepas Engine Oil Pan (2) beserta Gasket nya.</li> <li>Tahan Oil Pan dengan Sling belt dan crank, berat oil pan 105 Kg.</li> <li>Lepas bolt (6) lepas Oil Pan dan Gasket nya.</li> </ul> |             |    |
| 12                            | VALVE MECHANISM<br>COVER      | <ul style="list-style-type: none"> <li>Lepas Bolt (1) dan Lepas Valve Mechanism Cover (2) beserta sealnya.</li> </ul>                                                                                                                                                                             |             |   |
| 13                            | VALVE MECHANISM<br>COVER BASE | <ul style="list-style-type: none"> <li>Lepas Tube Assy (1)</li> <li>Pisahkan Harness assy (5) dan Harness Ass (2)</li> <li>Lepas Bolt (4)</li> <li>Lepas Vale Mechanism Cover Base (3) dan Seal dari Cylinder Head.</li> </ul>                                                                    |             |  |
| 14                            | VARIABLE VALVE<br>ACTUATOR    | <ul style="list-style-type: none"> <li>Pisahkan Harness assy (4) dan Harness Ass (5)</li> <li>Lepas bolt (2) dan Lepas empat nut (3) dan dua nut (6).</li> <li>Lepas Variable Valve Actuator.</li> </ul>                                                                                          |             |  |

| CHECK SHEET                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | WO NUMBER   | DOCUMENT NO                                                                                                                                                                |
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| CAT ENGINE C32 DISSASSEMBLING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | COMP.MODEL  | PUBLIC DATE                                                                                                                                                                |
| ENGINE CAT C32                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | COMP.S/N    | REVISION NO                                                                                                                                                                |
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | UNIT MODEL  | PAGE                                                                                                                                                                       |
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|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             | 1 OF 3                                                                                                                                                                     |
| PROCESS                       | START                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ASSEMBLY BY |                                                                                                                                                                            |
|                               | FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                                                                                                                                                            |
|                               | <ul style="list-style-type: none"> <li>Lepas support (11) dan Stud (10)</li> <li>Lepas dua stud (9) dan tiga stud (7)</li> <li>Lepas Rocker Arm</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                                                                                         |
| 15                            | <b>ROCKER ARM &amp; SHAFT</b> <ul style="list-style-type: none"> <li>Lepas bolt (2) dan yang lainnya.</li> <li>Angkat Rocker Arm dan Shaft (4) Valve Rocker</li> <li>Arm (3) dan Injector Arm (1) sebagai satu kesatuan.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |   |
| 16                            | <b>ROCKER ARM &amp; SHAFT</b> <ol style="list-style-type: none"> <li>Lepas Valve Rocker Arms (11) dan Injector Rocker Arm (3) dari Rocker Shaft Assy (4)</li> <li></li> <li>Lepas Jam Nut (7) dari adjustment screw (8)</li> <li>Lepas adjustment screw (8) dari Valve Rocker Arm (11)</li> <li>Lepas Button (10) dari adjustment screw (8) Lepas O-Ring Seal (9) dari adjustment screw.</li> <li>Lepas Jam Nut (2) dari Unit Injector adjustment Screw (1)</li> <li>Lepas Injector adjustment screw (1) dari Injector rocker Arm (3)</li> <li>Lepas Button (6) dari Injector adjustment screw (1)</li> <li>Lepas O-Ring Seal (5) dari Unit Injector Screw.</li> </ol> |             |                                                                                       |

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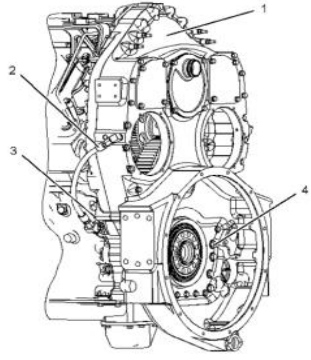
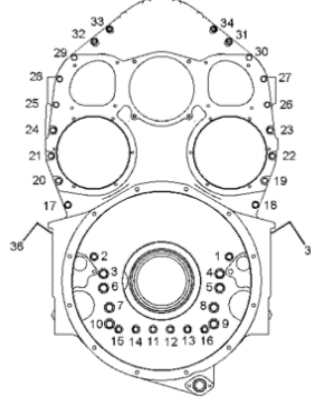
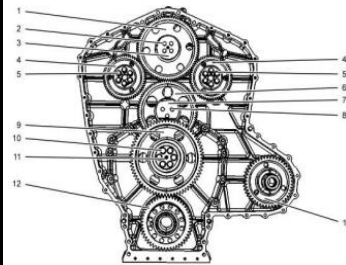
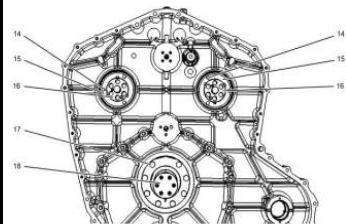
| PROCESS        |                             | START                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ASSEMBLY BY    |                                                                                                  |  |  |      |             |                  |     |   |         |           |   |   |         |               |   |  |                                                                                                  |
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|                |                             | FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                |                                                                                                  |  |  |      |             |                  |     |   |         |           |   |   |         |               |   |  |                                                                                                  |
| 17             | ELECTRONIC<br>UNIT INJECTOR | <div>1 Lepas Harness Injector Assy (1)</div> <div>2 Lepas Bridge Assy (3)</div> <div>3 Lepas Bolt (4), Spacer (5) dan Clamp (6) dan kendorkan Bolt (4)</div> <div>Tandai Injector untuk memudahkan dalam pemasangan, Injector tidak boleh tertukar dalam pemasangannya.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                | <div></div>   |  |  |      |             |                  |     |   |         |           |   |   |         |               |   |  |                                                                                                  |
| 18             | CAMSHAFT                    | <div>1 Lepas Bolt (1) dan Fan Drive (2)</div> <div>2 Beri Tanda pada timing gear untuk kemudahan pemasangan kembali.</div> <div>3 Lepas Bolt (3) Lepas Gear (4) dari Flywheel Housing.</div> <div>4 Lepas Bolt (5), thrust plate (6), and adapter Assy (7)</div> <div>5 Gunakan Bolt (5) untuk melepas sealing plate (8) dan Lepas O-Ring seal.</div> <div>6 Lepas Elbow (10) dan Cover (9)</div> <div>7 Lepas Bolt (11) dan Damper (12)</div> <div>8 Lepas Bolt (13) dan Lepas Sealing Plat (14), Adapter Assy(14) dan O-ring Seal.</div> <div>9 Install Tool (A) pada tiga lokasi di Cylinder Head.</div> <div>10 Hati - hati saat menggeser Campshaft (16) keluar dari Cylinder Head, Gunakan Tool (B) harus dua Orang unt menggeer Champshaft keluar dari Cylinder Head.</div> |                | <div></div>  |  |  |      |             |                  |     |   |         |           |   |   |         |               |   |  |                                                                                                  |
| 19             | CAMSHAFT                    | <div><h3>C32 Removal Procedure</h3><table><tr><th colspan="4">Required Tools</th></tr><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr><tr><td>A</td><td>9U-7257</td><td>Cradle As</td><td>3</td></tr><tr><td>B</td><td>9U-7240</td><td>Camshaft Hook</td><td>2</td></tr></table><div>Note :</div><div>Berat Champshaft : 35 Kg.</div></div>                                                                                                                                                                                                                                                                                                                                                                                                         | Required Tools |                                                                                                  |  |  | Tool | Part Number | Part Description | Qty | A | 9U-7257 | Cradle As | 3 | B | 9U-7240 | Camshaft Hook | 2 |  | <div></div> |
| Required Tools |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                |                                                                                                  |  |  |      |             |                  |     |   |         |           |   |   |         |               |   |  |                                                                                                  |
| Tool           | Part Number                 | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Qty            |                                                                                                  |  |  |      |             |                  |     |   |         |           |   |   |         |               |   |  |                                                                                                  |
| A              | 9U-7257                     | Cradle As                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3              |                                                                                                  |  |  |      |             |                  |     |   |         |           |   |   |         |               |   |  |                                                                                                  |
| B              | 9U-7240                     | Camshaft Hook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2              |                                                                                                  |  |  |      |             |                  |     |   |         |           |   |   |         |               |   |  |                                                                                                  |
|                |                             | <div>• Pasang Tool (A) dan Lifting Bracket ke Cylinder Head 25.</div> <div>• Lepas Bolt (26) dan Boolt (27)</div> <div>• Lepas Cylinder Head (25) Lepas Gasket, O-Ring Seal dan Spacer plate.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                | <div></div> |  |  |      |             |                  |     |   |         |           |   |   |         |               |   |  |                                                                                                  |

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| PROCESS        |                  | START                                                                                                                                                                                                                                                                                                                                                                                                                                        | ASSEMBLY BY    |                                                                                       |                  |     |      |             |                                    |     |   |                                                                                       |              |   |   |         |                    |   |  |                                                                                     |
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|                |                  | FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |                                                                                       |                  |     |      |             |                                    |     |   |                                                                                       |              |   |   |         |                    |   |  |                                                                                     |
| 20             | CYLINDER HEAD    | <p><b>Note :</b><br/>Berat Cylinder Head 365 Kg.</p> <p><b>Removal Procedure</b></p> <p>Table 1</p> <table><tr><th colspan="4">Required Tools</th></tr><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr><tr><td>A</td><td>138-7573</td><td>Link Bracket</td><td>3</td></tr><tr><td>B</td><td>1U-9200</td><td>Lever Puller Hoist</td><td>1</td></tr></table>                                                    | Required Tools |                                                                                       |                  |     | Tool | Part Number | Part Description                   | Qty | A | 138-7573                                                                              | Link Bracket | 3 | B | 1U-9200 | Lever Puller Hoist | 1 |  |  |
| Required Tools |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                |                                                                                       |                  |     |      |             |                                    |     |   |                                                                                       |              |   |   |         |                    |   |  |                                                                                     |
| Tool           | Part Number      | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                             | Qty            |                                                                                       |                  |     |      |             |                                    |     |   |                                                                                       |              |   |   |         |                    |   |  |                                                                                     |
| A              | 138-7573         | Link Bracket                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3              |                                                                                       |                  |     |      |             |                                    |     |   |                                                                                       |              |   |   |         |                    |   |  |                                                                                     |
| B              | 1U-9200          | Lever Puller Hoist                                                                                                                                                                                                                                                                                                                                                                                                                           | 1              |                                                                                       |                  |     |      |             |                                    |     |   |                                                                                       |              |   |   |         |                    |   |  |                                                                                     |
| 21             | VIBRATION DAMPER | <ul style="list-style-type: none"><li>Lepas bootlt (3), adapter (2) dan pulley (1)</li><li>Pasang Lifting bracket pada Dumper (4) Berat Vibration Damper 35 Kg.</li><li>Lepas bolt (6) kemudian angkat Damper dan adapter (5).</li></ul>                                                                                                                                                                                                     |                |  |                  |     |      |             |                                    |     |   |                                                                                       |              |   |   |         |                    |   |  |                                                                                     |
| 22             | FLYWHEEL         | <ul style="list-style-type: none"><li>Lepas dua bolt (1), Pasang Tool (A) pada Crankshaft</li><li>Lepas semua bolt yang tersisa (1) dan Flywheel spline jika ada kemudian geser dan angkat Flywheel menggunakan Sling belt.</li></ul> <p>Required Tools</p> <table><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr><tr><td>A</td><td>-</td><td>Guide Stud 5/8 - 18 UNF by 11 inch</td><td>2</td></tr></table> | Tool           | Part Number                                                                           | Part Description | Qty | A    | -           | Guide Stud 5/8 - 18 UNF by 11 inch | 2   |   |  |              |   |   |         |                    |   |  |                                                                                     |
| Tool           | Part Number      | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                             | Qty            |                                                                                       |                  |     |      |             |                                    |     |   |                                                                                       |              |   |   |         |                    |   |  |                                                                                     |
| A              | -                | Guide Stud 5/8 - 18 UNF by 11 inch                                                                                                                                                                                                                                                                                                                                                                                                           | 2              |                                                                                       |                  |     |      |             |                                    |     |   |                                                                                       |              |   |   |         |                    |   |  |                                                                                     |

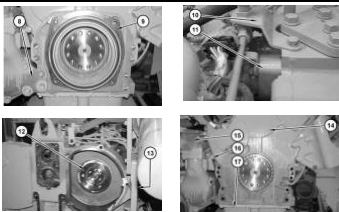
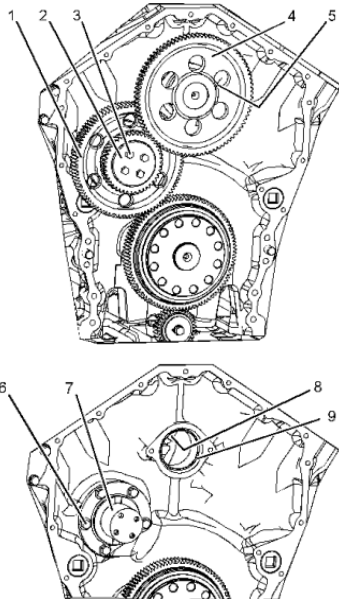
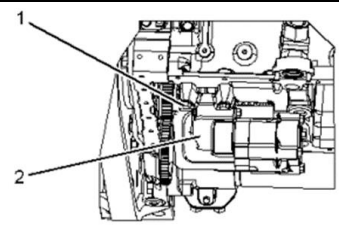
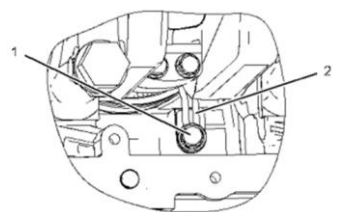


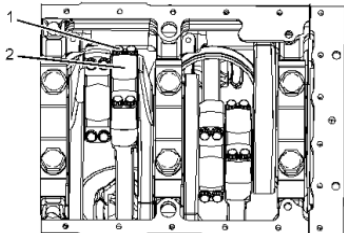
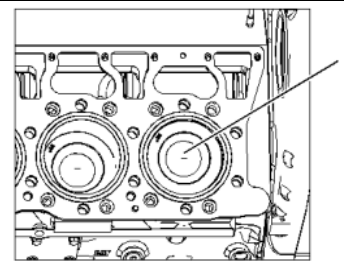
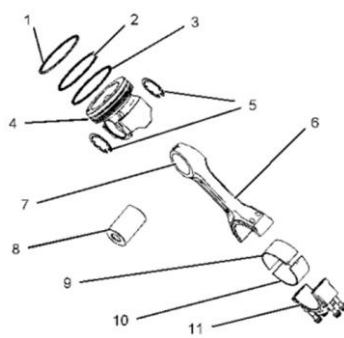
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| CHECK SHEET<br>CAT ENGINE C32 DISSASSEMBLING<br>ENGINE CAT C32 | WO NUMBER  |  | DOCUMENT NO |        |
|                                                                | COMP.MODEL |  | PUBLIC DATE |        |
|                                                                | COMP.S/N   |  | REVISION NO | REV 01 |
|                                                                | UNIT MODEL |  | PAGE        | 1 OF 3 |

| PROCESS        |                  | START                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                | ASSEMBLY BY |  |  |      |             |                  |     |   |          |              |   |   |   |                                                                                                                                                                                |   |  |  |                                                                                                                                                                             |
|----------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------|--|--|------|-------------|------------------|-----|---|----------|--------------|---|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                |                  | FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |             |  |  |      |             |                  |     |   |          |              |   |   |   |                                                                                                                                                                                |   |  |  |                                                                                                                                                                             |
| 23             | Flywheel Housing | <ul style="list-style-type: none"><li>Pisahkan Hose Assy (2) dari Flywheel housing (1)</li><li>Lepas Bolt (4) dan Pasang Tool (B)</li><li>Gunakan Tool (A) sebagai pengganti untuk mengangkat Flywheel Housing, Berat Flywheel 145 Kg.</li><li>Lepas Bolt Pengganti (4) dan Bolt (3).</li><li>Lepas Flywheel Housing (1) dan Gasket dari Cylinder Block.</li></ul> <p><b>Removal Procedure</b></p> <p>Table 1</p> <table><tr><th colspan="4">Required Tools</th></tr><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr><tr><td>A</td><td>138-7574</td><td>Link Bracket</td><td>2</td></tr><tr><td>B</td><td>-</td><td>Guide Stud (1/2 - 13 NC by 5 inch)</td><td>2</td></tr></table>                                                                                                                                                                                                                                                                                       | Required Tools |             |  |  | Tool | Part Number | Part Description | Qty | A | 138-7574 | Link Bracket | 2 | B | - | Guide Stud (1/2 - 13 NC by 5 inch)                                                                                                                                             | 2 |  |  | <br> |
| Required Tools |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |             |  |  |      |             |                  |     |   |          |              |   |   |   |                                                                                                                                                                                |   |  |  |                                                                                                                                                                             |
| Tool           | Part Number      | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Qty            |             |  |  |      |             |                  |     |   |          |              |   |   |   |                                                                                                                                                                                |   |  |  |                                                                                                                                                                             |
| A              | 138-7574         | Link Bracket                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2              |             |  |  |      |             |                  |     |   |          |              |   |   |   |                                                                                                                                                                                |   |  |  |                                                                                                                                                                             |
| B              | -                | Guide Stud (1/2 - 13 NC by 5 inch)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2              |             |  |  |      |             |                  |     |   |          |              |   |   |   |                                                                                                                                                                                |   |  |  |                                                                                                                                                                             |
| 24             | REAR GEAR GROUP  | <ul style="list-style-type: none"><li>Lepas Bolt (10), Retainer Ass (11) O-Ring Seal &amp; Gear Assy (9).</li><li>Gunkan Tool (A) dan pasang pada bearing gear (9)</li><li>Lepas Bolt (8), Gear Assy (6) dan Shaft Assy (7)</li><li>Gunakan Tool (A) sebagai pengganti untuk mengangkat bearing dari Gear Assy (6).</li><li>Lepas bolt (2) Gear Assy (1) dan Shaft Assy (3).</li><li>Gunakan Tool (A) sebagai pengganti untuk mengangkat Bearing dari Gear Assy (1).</li><li>Lepas Gear Assy (12)</li><li>Lepas Bolt (5) dan Gear Assy (4)</li><li>Lepas Gear Assy (13) dan Thrust washers.</li><li>Lepas Bolt (17) dan Shaft Assy (18).</li><li>Lepas Bolt (14) Thrust plate (15) dan Adapter Assy (16)</li><li>Gunakan Bolt (14) untuk membuka Sealing Plate dan Lepas O-Ring.</li></ul> <table><tr><th colspan="4">Required Tools</th></tr><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr><tr><td>A</td><td>1P-0520</td><td>Driver Group</td><td>1</td></tr></table> | Required Tools |             |  |  | Tool | Part Number | Part Description | Qty | A | 1P-0520  | Driver Group | 1 |   |   | <br> |   |  |  |                                                                                                                                                                             |
| Required Tools |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |             |  |  |      |             |                  |     |   |          |              |   |   |   |                                                                                                                                                                                |   |  |  |                                                                                                                                                                             |
| Tool           | Part Number      | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Qty            |             |  |  |      |             |                  |     |   |          |              |   |   |   |                                                                                                                                                                                |   |  |  |                                                                                                                                                                             |
| A              | 1P-0520          | Driver Group                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1              |             |  |  |      |             |                  |     |   |          |              |   |   |   |                                                                                                                                                                                |   |  |  |                                                                                                                                                                             |

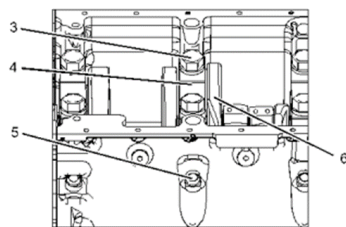
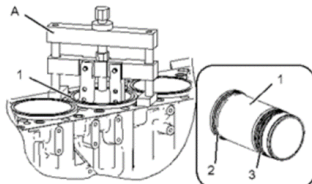


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| <b>CHECK SHEET</b><br><b>CAT ENGINE C32 DISSASSEMBLING</b><br><b>ENGINE CAT C32</b> | WO NUMBER  |  | DOCUMENT NO |        |
|                                                                                     | COMP.MODEL |  | PUBLIC DATE |        |
|                                                                                     | COMP.S/N   |  | REVISION NO | REV 01 |
|                                                                                     | UNIT MODEL |  | PAGE        | 1 OF 3 |

| PROCESS |                        | START                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ASSEMBLY BY |  |                                                                                       |
|---------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--|---------------------------------------------------------------------------------------|
|         |                        | FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |  |                                                                                       |
| 25      | FRONT HOUSING          | <ul style="list-style-type: none"> <li>Remove bracket (8) dan Engine Trunnion (9)</li> <li>Pisahkan bracket (10) dan Elbow (11)</li> <li>Lepas adapter (12) dan bolt (13)</li> <li>Pasang Lifting Bracket pada Front Housing (14)</li> <li>Berat Front Housing : 70 Kg</li> <li>Lepas bonnet (15), Bolt (16) dan Bolt (17)</li> <li>Lepas Front Housing (14) beserta gasket nya.</li> </ul>                                                                                        | 4           |  |    |
| 26      | FRONT GEAR GROUP       | <ul style="list-style-type: none"> <li>Lepas Bolt (5) dan Gear Assy (4)</li> <li>Lepas Bolt (3) dan Plate (2)</li> <li>Lepas Gear Assy (10)</li> <li>Lepas Bolt (6) dan Shaft (7)</li> </ul>                                                                                                                                                                                                                                                                                       |             |  |   |
| 27      | ENGINE OIL PUMP REMOVE | <ul style="list-style-type: none"> <li>Lepas Bolt (1)</li> <li>Lepas Engine Oil Pump (2)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                |             |  |  |
| 28      | PISTON COOLING JET     | <ul style="list-style-type: none"> <li>Rebahkan Cylinder Block distand, gunakan Sling belt dan over head crank, pastikan posisi Cyl Block kuat agar tidak bergeser saat pengendoran bolt dan pelepasan piston dr liner.</li> <li>Gunakan dua bolt Vibration Damper pasang sejajar pada depan Crankshft dan Putar Crankshaft menggunakan pipa untuk mencari posisi yang mudah dalam membuka Piston Cooling Jet.</li> <li>Lepas Bolt (1) dan Lepas Piston Cooling Jet (2)</li> </ul> |             |  |  |
|         |                        | <b>Check Piston Cooling Jets :</b><br>Check Kebengkokan,<br>buntu dan groove/alur.                                                                                                                                                                                                                                                                                                                                                                                                 |             |  |                                                                                       |

| CHECK SHEET                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | WO NUMBER               |     | DOCUMENT NO |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
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| CAT ENGINE C32 DISSASSEMBLING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | COMP.MODEL              |     | PUBLIC DATE |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
| ENGINE CAT C32                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | COMP.S/N                |     | REVISION NO |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | UNIT MODEL              |     | PAGE        |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |     | REV 01      |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |     | 1 OF 3      |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
| PROCESS                       | START<br>FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ASSEMBLY BY             |     |             |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
|                               | <table border="1"> <tr> <th>Cylinder NO</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table> <p>√ Good X No Good</p> <table border="1"> <tr> <th>Cylinder NO</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Cylinder NO             | 1   | 2           | 3                                                                                    | 4    | 5           | 6                |     |   |         |        |   |   |                        | Cylinder NO             | 7 | 8                      | 9                       | 10 | 11 | 12 |                                                                                       |  |  |  |  |  |  |  |  |  |  |
| Cylinder NO                   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2                       | 3   | 4           | 5                                                                                    | 6    |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |     |             |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
| Cylinder NO                   | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 8                       | 9   | 10          | 11                                                                                   | 12   |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |     |             |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
| 29                            | <b>PISTON &amp; CONROD</b> <ul style="list-style-type: none"> <li>Gunakan dua bolt Vibration Damper pasang sejajar pada depan Crankshft dan Putar Crankshaft menggunakan pipa untuk mencari posisi bolt Connecting Rod agar mudah dibuka.</li> <li>Kendorkan 4 Bolt Cap Connecting Rod dengan cara menyilang kemudian lepas connecting rod cap.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |     |             |   |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
|                               | <b>PISTON &amp; CONROD</b> <ul style="list-style-type: none"> <li>Dorong Connecting Road ke atas sampai Ring Piston Lepas kemudian tarik Piston (3) dan Con Rod nya.</li> <li>Lakukan cara serupa untuk membuka Cap Conrod dan melepas Piston yang lain sampai Piston terlepas semuanya.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                         |     |             |  |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
| 30                            | <b>PISTON RING &amp; CONROD</b> <ul style="list-style-type: none"> <li>Lepas bagian bawah Connecting Rod Bearing (10) dari Conrod Cap (11).</li> <li>Lepas bagian atas Connecting Rod Bearing (9) dari Conrod (6).</li> <li>Gunakan Tool (A) untuk membuka Snap Ring Pin Piston (5) dari Piston Body Assy (4).</li> <li>Lepas Connecting Rod Pin (8) dan Conrod (6) dari Piston Body Assy.</li> <li>Gunakan Tool (B) untuk membuka Piston Ring 1, 2, 3. dari Piston Body Assy.</li> <li>Jangan membuka bearing (7) pada saat ini.</li> </ul> <p><b>NOTE :</b><br/><i>Penggantian Bushing Conrod dilakukan dgn cara dipress sehingga pelepasannya harus bersamaan dengan pemasangan bushing baru</i></p> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>6V-6192</td> <td>Pliers</td> <td>1</td> </tr> <tr> <td rowspan="2">B</td> <td>7M-3978<sup>(1)</sup></td> <td>Piston Ring Expander Gp</td> <td>1</td> </tr> <tr> <td>7B-7974<sup>(2)</sup></td> <td>Piston Ring Expander Gp</td> <td>1</td> </tr> </tbody> </table> <p><small>(1) For use with 137 mm (5.4 inch) bores.<br/>(2) For use with 145 mm (5.7 inch) bores.</small></p> | Required Tools          |     |             |                                                                                      | Tool | Part Number | Part Description | Qty | A | 6V-6192 | Pliers | 1 | B | 7M-3978 <sup>(1)</sup> | Piston Ring Expander Gp | 1 | 7B-7974 <sup>(2)</sup> | Piston Ring Expander Gp | 1  |    |    |  |  |  |  |  |  |  |  |  |  |  |
| Required Tools                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                         |     |             |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
| Tool                          | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Part Description        | Qty |             |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
| A                             | 6V-6192                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Pliers                  | 1   |             |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
| B                             | 7M-3978 <sup>(1)</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Piston Ring Expander Gp | 1   |             |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |
|                               | 7B-7974 <sup>(2)</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Piston Ring Expander Gp | 1   |             |                                                                                      |      |             |                  |     |   |         |        |   |   |                        |                         |   |                        |                         |    |    |    |                                                                                       |  |  |  |  |  |  |  |  |  |  |

|                                                                                     |            |  |             |        |
|-------------------------------------------------------------------------------------|------------|--|-------------|--------|
| <b>CHECK SHEET</b><br><b>CAT ENGINE C32 DISSASSEMBLING</b><br><b>ENGINE CAT C32</b> | WO NUMBER  |  | DOCUMENT NO |        |
|                                                                                     | COMP.MODEL |  | PUBLIC DATE |        |
|                                                                                     | COMP.S/N   |  | REVISION NO | REV 01 |
|                                                                                     | UNIT MODEL |  | PAGE        | 1 OF 3 |

| PROCESS          |             | START                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ASSEMBLY BY    |                                                                                     |  |  |      |             |                  |     |   |         |                          |   |                  |         |                                |   |          |                                |   |  |                                                                                     |
|------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|--|--|------|-------------|------------------|-----|---|---------|--------------------------|---|------------------|---------|--------------------------------|---|----------|--------------------------------|---|--|-------------------------------------------------------------------------------------|
|                  |             | FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |                                                                                     |  |  |      |             |                  |     |   |         |                          |   |                  |         |                                |   |          |                                |   |  |                                                                                     |
| 31               | CRANKSHAFT  | <ul style="list-style-type: none"><li>Lepas Bolt (3) dan Bolt (5).</li><li>Lepas Main Bearing Cap (4)</li><li>Gunakan Sling Belt dan Over head crank untuk mengangkat Crankshaft, Berat Crankshaft : 188 Kg.</li><li>Lepas Thrust Plate dari sisi samping Main Bearing No.4</li><li>Lepas Lower main bearing dari main bearing cap (4)</li><li>Remove Upper main bearing dari cylinder block.</li><li>Tempatkan Crankshaft pada stand posisi aman.</li></ul>                                                                                                                                                                                                            |                |  |  |  |      |             |                  |     |   |         |                          |   |                  |         |                                |   |          |                                |   |  |                                                                                     |
| 32               | LINER       | <ul style="list-style-type: none"><li>Gunakan Tool (A) untuk membuka Liner Putar Bolt Shaft Tool (A) kencangkan sampai menarik Liner keatas. Lepas O-Ring Seal (3) dan Filler Band dari Liner (1).</li></ul> <table border="1"><thead><tr><th colspan="4">Required Tools</th></tr><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr></thead><tbody><tr><td>A</td><td>5P-8665</td><td>Cylinder Liner Puller Gp</td><td>1</td></tr><tr><td rowspan="2">B<sup>(1)</sup></td><td>6V-9448</td><td>Cylinder Liner Puller Gp (C27)</td><td>1</td></tr><tr><td>373-1438</td><td>Cylinder Liner Puller Gp (C32)</td><td>1</td></tr></tbody></table> | Required Tools |                                                                                     |  |  | Tool | Part Number | Part Description | Qty | A | 5P-8665 | Cylinder Liner Puller Gp | 1 | B <sup>(1)</sup> | 6V-9448 | Cylinder Liner Puller Gp (C27) | 1 | 373-1438 | Cylinder Liner Puller Gp (C32) | 1 |  |  |
| Required Tools   |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |                                                                                     |  |  |      |             |                  |     |   |         |                          |   |                  |         |                                |   |          |                                |   |  |                                                                                     |
| Tool             | Part Number | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Qty            |                                                                                     |  |  |      |             |                  |     |   |         |                          |   |                  |         |                                |   |          |                                |   |  |                                                                                     |
| A                | 5P-8665     | Cylinder Liner Puller Gp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1              |                                                                                     |  |  |      |             |                  |     |   |         |                          |   |                  |         |                                |   |          |                                |   |  |                                                                                     |
| B <sup>(1)</sup> | 6V-9448     | Cylinder Liner Puller Gp (C27)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1              |                                                                                     |  |  |      |             |                  |     |   |         |                          |   |                  |         |                                |   |          |                                |   |  |                                                                                     |
|                  | 373-1438    | Cylinder Liner Puller Gp (C32)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1              |                                                                                     |  |  |      |             |                  |     |   |         |                          |   |                  |         |                                |   |          |                                |   |  |                                                                                     |

**INSPECTION RESULT :**[illegible]

**INSPECTION & APPROVAL BY :**

INSPECTION BY

**APPROVE BY**

---

**MECHANIC**

### QUALITY ASSURANCE (QA)

**GROUP LEADER (GL)**



**PT SAPTAINDRA SEJATI**

**ENGINE**

**DISASSY CYL HEAD**

**CHECKSHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

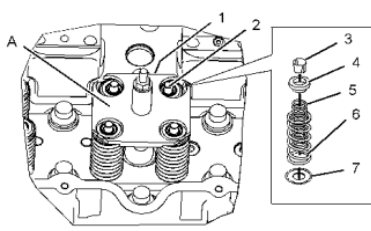
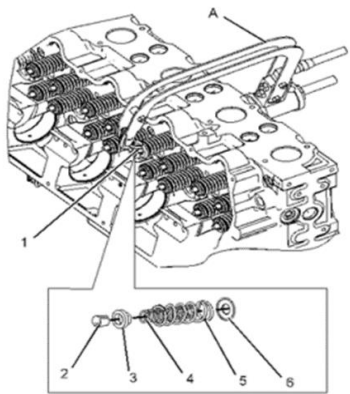
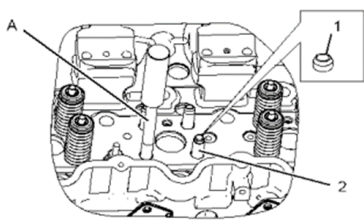
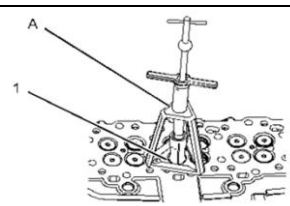
**CAT777E**

**WO NO :**

**ENGINE SECTION**

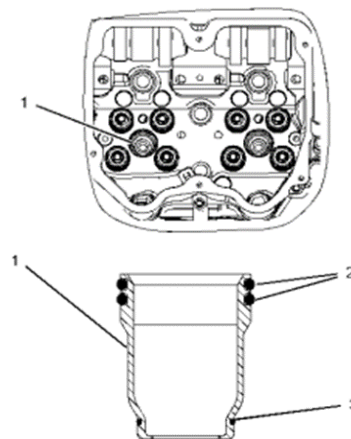
|                                                                            |            |  |             |        |
|----------------------------------------------------------------------------|------------|--|-------------|--------|
| <b>CHECK SHEET<br/>CAT ENGINE C32<br/>DISASSEMBLING<br/>ENGINE CAT C32</b> | WO NUMBER  |  | DOCUMENT NO |        |
|                                                                            | COMP.MODEL |  | PUBLIC DATE |        |
|                                                                            | COMP.S/N   |  | REVISION NO | REV 01 |
|                                                                            | UNIT MODEL |  | PAGE        | 1 OF 3 |

|         |        |  |             |  |  |
|---------|--------|--|-------------|--|--|
| PROCESS | START  |  | ASSEMBLY BY |  |  |
|         | FINISH |  |             |  |  |

| NO               | PART TO INSTALL OR ASSEMBLY | CONDITION TOBE INSTALL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | SPEC | SIGN        |                  | SKETCH DRAWING OR PHOTHOES                                                                                                                                                                                                                                                                                                                |                  |             |                                 |     |   |         |                         |                                                                                       |
|------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------|---------------------------------|-----|---|---------|-------------------------|---------------------------------------------------------------------------------------|
|                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      | MECH        | GL/Q             |                                                                                                                                                                                                                                                                                                                                           |                  |             |                                 |     |   |         |                         |                                                                                       |
| 1                |                             | <ul style="list-style-type: none"><li>Beri Pelumas Valve Stem dengan Oli Engine bersih 15W40.</li><li>Pasang Tool (A) pada Sleeve Injector. Amankan Tooling (A) dengan Unit Injector clamp (1).</li><li>Kencangkan Nut sampai Valve Spring keepers (3) kendor di Valve (2).</li><li>Lepas Vave Spring Keeper (3) dari Valve.</li><li>Berhati -hati saat melepas Tooling (A).</li><li>Lepas Valve Rotator (4).</li><li>Lepas Outer Valve Spring (6) dan Inner Valve spring (5) dari Valve.</li><li>Lepas Washer (7) dari Valve Guide.</li><li>Gunakan Tool (B) untuk cek Valve Spring ( Spring Force).</li></ul>                                                                                                          |      |             |                  |  <table border="1"><caption>Required Tools</caption><thead><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr></thead><tbody><tr><td>A</td><td>9U-7241</td><td>Valve Spring Compressor</td><td>1</td></tr></tbody></table> | Tool             | Part Number | Part Description                | Qty | A | 9U-7241 | Valve Spring Compressor | 1                                                                                     |
| Tool             | Part Number                 | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Qty  |             |                  |                                                                                                                                                                                                                                                                                                                                           |                  |             |                                 |     |   |         |                         |                                                                                       |
| A                | 9U-7241                     | Valve Spring Compressor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1    |             |                  |                                                                                                                                                                                                                                                                                                                                           |                  |             |                                 |     |   |         |                         |                                                                                       |
| 2                |                             | <table border="1"><caption>Required Tools</caption><thead><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr></thead><tbody><tr><td>A</td><td>5S-1330</td><td>Valve Spring Compressor</td><td>1</td></tr></tbody></table> <ul style="list-style-type: none"><li>Pelepasan Valve juga bisa menggunakan Tool (A) Valve Spring Compressor.</li><li>Seperti biasanya Tool (A) dipasang satu-satu pada setiap Valve yg akan dilepas, menggunakan Angin bertekanan / Compressor, setelah Tool dioperasikan dan menekan Spring Valve (5) Lepas spring Keeper dan kendorkan perlahan-lahan Kompresinya sampai Rotatornya lepas.</li><li>Lakukan berulang kali sampai semua Valve terlepas.</li></ul> | Tool | Part Number | Part Description | Qty                                                                                                                                                                                                                                                                                                                                       | A                | 5S-1330     | Valve Spring Compressor         | 1   |   |         |                         |   |
| Tool             | Part Number                 | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Qty  |             |                  |                                                                                                                                                                                                                                                                                                                                           |                  |             |                                 |     |   |         |                         |                                                                                       |
| A                | 5S-1330                     | Valve Spring Compressor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1    |             |                  |                                                                                                                                                                                                                                                                                                                                           |                  |             |                                 |     |   |         |                         |                                                                                       |
| 3                |                             | <ul style="list-style-type: none"><li>Lepas Semua Valve Steam Seal.</li><li>Lepas Valve Guide Inlet &amp; Exhaust menggunakan Tool (A) Valve Guide Driver bisa dengan cara memukul pakai Palu / Hammer atau dengan di Jack menggunakan Hydraulic Jack.</li></ul> <table border="1"><caption>Required Tools</caption><thead><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr></thead><tbody><tr><td>A<sup>(1)</sup></td><td>9U-6895</td><td>Valve Guide Driver</td><td>1</td></tr></tbody></table> <p><sup>(1)</sup> Part of 9U-7530 Engine Tool Group</p>                                                                                                                                  | Tool | Part Number | Part Description | Qty                                                                                                                                                                                                                                                                                                                                       | A <sup>(1)</sup> | 9U-6895     | Valve Guide Driver              | 1   |   |         |                         |  |
| Tool             | Part Number                 | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Qty  |             |                  |                                                                                                                                                                                                                                                                                                                                           |                  |             |                                 |     |   |         |                         |                                                                                       |
| A <sup>(1)</sup> | 9U-6895                     | Valve Guide Driver                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1    |             |                  |                                                                                                                                                                                                                                                                                                                                           |                  |             |                                 |     |   |         |                         |                                                                                       |
|                  |                             | <ul style="list-style-type: none"><li>Gunakan Tool (A) untuk melepas Valve Seat Inset (1) dari Cylider Head.</li><li>Bersihkan lubang dudukan Valve Seat Insert</li><li>Ulangi cara yang sama untuk melepas Valve Seat Insert.</li></ul> <table border="1"><caption>Required Tools</caption><thead><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr></thead><tbody><tr><td>A</td><td>166-7441</td><td>Valve Seat Extractor Tool Group</td><td>1</td></tr></tbody></table>                                                                                                                                                                                                                  | Tool | Part Number | Part Description | Qty                                                                                                                                                                                                                                                                                                                                       | A                | 166-7441    | Valve Seat Extractor Tool Group | 1   |   |         |                         |  |
| Tool             | Part Number                 | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Qty  |             |                  |                                                                                                                                                                                                                                                                                                                                           |                  |             |                                 |     |   |         |                         |                                                                                       |
| A                | 166-7441                    | Valve Seat Extractor Tool Group                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1    |             |                  |                                                                                                                                                                                                                                                                                                                                           |                  |             |                                 |     |   |         |                         |                                                                                       |

- Rendam Cylinder Head & Bersihkan sebelum melakukan perakitan ulang Component Cylinder head.
- Pasang Tool (A) pada Sleeve Injector (1)
- Kencangkan Nut yang ada di Tool (A) sampai Sleeve Injector ketarik keluar pada Cylinder Head.
- Lepaskan O-Ring Seals (2), O-Ring Seal (3) dari Sleeve Injector.

| Required Tools |             |                     |     |
|----------------|-------------|---------------------|-----|
| Tool           | Part Number | Part Description    | Qty |
| A              | 9U-6891     | Injector Tool Group | 1   |



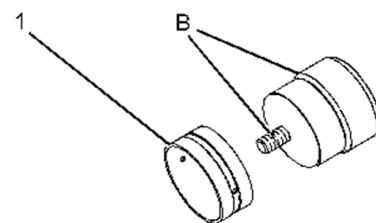
- Lepas No.7 Camshaft bearing / bushing (Belakang) Pelepasan Bushing dimulai dari belakang engine menuju ke bagian depan engine.
- Pasang bushing Campshaft (1) pada bagian kecil dari Cylinder Head Tool (B)
- Posisikan Tool (C) diatas Tool (A), Pasang Tool (A) melalui Cyl head pada Tool (B).

**Note :**

**Tool (C) dipasang pada Outside dari Cyl Head Tool (C) digunakan untuk melepas semua bushing Camshaft.**

- Gunakan Tool (A) untuk melepas Campshaft Bushing (1) dari Cyl Head.
- Lepas Tool (B) dari Tool (A) dan Lepas Campshaft Bushing/bearing.

| Required Tools |             |                                        |     |
|----------------|-------------|----------------------------------------|-----|
| Tool           | Part Number | Part Description                       | Qty |
| A              | 8S-2241     | Camshaft Bearing Tool Group            | 1   |
| B              | 9U-7222     | Camshaft Bearing Pilot                 | 1   |
|                | 8M-8778     | Taperlock Stud 1/2 - 13 by 1 9/16 inch | 1   |
| C              | 9U-7210     | Puller Plate                           | 1   |



#### INSPECTION RESULT :

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|--|
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|  |
|  |
|  |

INSPECTION & APPROVAL BY :

INSPECTION BY

APPROVE BY

MECHANIC

QUALITY ASSURANCE (QA)

GROUP LEADER (GL)





**PT SAPTAINDRA SEJATI**

**ENGINE**

**CYLINDER HEAD AFTER MACHINING AND  
MEASUREMENT CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

**WO NO :**

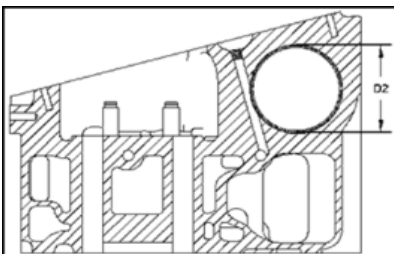
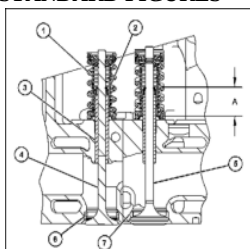
**ENGINE SECTION**

## CYLINDER HEAD AFTER MACHINING AND MEASUREMENT CHECK SHEET

|               |  |                    |  |                |  |
|---------------|--|--------------------|--|----------------|--|
| JOB NUMBER    |  | PART NAME          |  | DOCUMENT NO.   |  |
| ENGINE MODEL  |  | PART NUMBER        |  | PUBL. DATE     |  |
| ENGINE S/N    |  | START DATE / TIME  |  | REVISION NO. 4 |  |
| MACHINE MODEL |  | FINISH DATE / TIME |  | PAGE           |  |

| MEASURED BY |  | QUANTITY |  |
|-------------|--|----------|--|
|-------------|--|----------|--|

## STANDARD FIGURES



| No | Valve Spring | standard Size (mm) |                | Test Force(N) |
|----|--------------|--------------------|----------------|---------------|
|    |              | Free lenght        | Install lenght |               |
| 1  | outer        | 76.7               | 67.12          | 320 ± 25      |
| 2  | inner        | 71.7               | 60.14          | 150 ± 12      |

| No | Measuring items                 |                           | Std Tolerance (mm) |  |
|----|---------------------------------|---------------------------|--------------------|--|
| 4  | Ex Valve                        | Angle of Face             | $45.75 \pm 0.25^0$ |  |
|    |                                 | Steam Diameter            | $9,441 \pm 0.01$   |  |
|    |                                 | Valve Head Diameter       | $41.81 \pm 0.13$   |  |
| 5  | In Valve                        | Angle of Face             | $60.75 \pm 0.25^0$ |  |
|    |                                 | Steam Diameter            | $9,441 \pm 0.01$   |  |
|    |                                 | Valve Head Diameter       | $47.00 \pm 0.13$   |  |
| 6  | Valve Seat Insert               | Angle of Face             | $45.25 \pm 0.25^0$ |  |
|    |                                 | Outside Diameter          | $42.84 \pm 0.0013$ |  |
|    |                                 | Depth of Bore In Cyl Head | $13.90 \pm 0.05$   |  |
|    |                                 | Bore in the Cyl Head      | $42.774 \pm 0.025$ |  |
| D2 | Bore of camshaft Sleeve bushing |                           | $90.985 \pm 0.020$ |  |

### A.Result of Measurement

## VALVE SPRING

[illegible][illegible]





**PT SAPTAINDRA SEJATI**

**ENGINE**

**CYLINDER BLOCK MEASURING  
INSPECTION CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

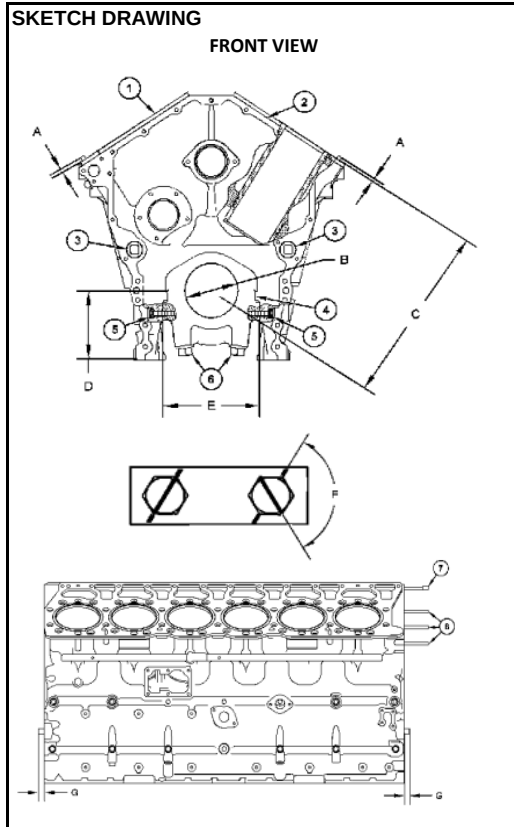
**CAT777E**

**WO NO :**

**ENGINE SECTION**

## CYLINDER BLOCK MEASURING INSPECTION CHECKSHEET

|                |  |                    |  |              |  |
|----------------|--|--------------------|--|--------------|--|
| JOB NUMBER     |  | PART NAME          |  | DOCUMENT NO. |  |
| ENGINE MODEL   |  | PART NUMBER        |  | PUBL. DATE   |  |
| ENGINE SER.NO. |  | START.DATE / TIME  |  | REVISION NO. |  |
| MACHINE MODEL  |  | FINISH DATE / TIME |  | PAGE         |  |



|                |                       |
|----------------|-----------------------|
| MACHINIST NAME | Cyl. Block Inspection |
|                | Machining Process     |

| Standard Size                  |                                   | Unit : mm |
|--------------------------------|-----------------------------------|-----------|
| DESCRIPTION                    | STANDARD SIZE                     |           |
| 1 THICKNESS OF SPACER PLATE    | 8.484 ± 0.025                     |           |
| 2 THICKNESS OF PLATE GASKET    | 0.20 ± 0,02                       |           |
| 3 TAPERLOCK PLUG               | 80 ± 15 N.m                       |           |
| A PROTUTION OF DOWEL           | 18.5 ± 0,5                        |           |
| 4 CRANKSHAFT CAP               | Harus terpasang sesuai urutannya. |           |
| B BORE CYL BLOCK MAIN BEARING  | 129.891 ± 0.0135                  |           |
| C CENTER BORE & TOP CYL BLOCK  | 419.10 ± 0.15                     |           |
| D CENTER BORE & BOTM CYL BLOCK | 165.10 ± 0.10                     |           |
| E LEBAR MAIN CAP               | 236.00 ± 0.05                     |           |
| 8 TAPERLOCK STUD               | 35 ± 5                            |           |
| G PANJANG DOWEL DARI CYL BLOCK | 19.0 ± 0.5                        |           |

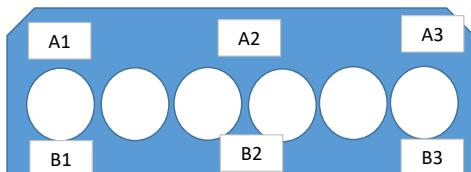
|                                       |                      |   |                |
|---------------------------------------|----------------------|---|----------------|
| TIGHTENING TORQUE OF MAIN BEARING CAP | 1 <sup>st</sup> step | = | 50 ± 5 lb.ft   |
|                                       | 2 <sup>nd</sup> step | = | 190 ± 10 lb.ft |
|                                       | 3 <sup>rd</sup> step | = | 120° ± 5°      |

| VISUAL CHECK           |       | OK     | Not OK    |
|------------------------|-------|--------|-----------|
| CRACK HAS BEEN CHECKED |       |        |           |
| MANHOURS UTILIZATION   |       |        |           |
|                        | START | FINISH | MAN HOURS |
| Measuring Process      |       |        |           |
| TOTAL                  |       |        |           |

INSPECTION SALVAGE PROCESS BY MACHINIST

RESULT : ☒ GOOD CONDITION    ☐ X : BAD CONDITION

THICKNESS OF SPACER PLATE : 8.484 ± 0.025 mm

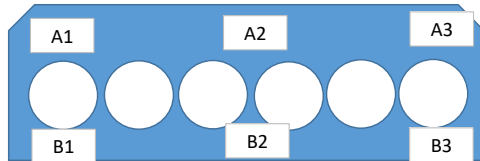


| 1 | NO | 1 | 2 | 3 | REMARK |
|---|----|---|---|---|--------|
|   | A  |   |   |   |        |
|   | B  |   |   |   |        |
| 2 | NO | 1 | 2 | 3 | REMARK |
|   | A  |   |   |   |        |
|   | B  |   |   |   |        |

## CYLINDER BLOCK MEASURING INSPECTION CHECKSHEET

|                |  |                    |  |              |  |
|----------------|--|--------------------|--|--------------|--|
| JOB NUMBER     |  | PART NAME          |  | DOCUMENT NO. |  |
| ENGINE MODEL   |  | PART NUMBER        |  | PUBL. DATE   |  |
| ENGINE SER.NO. |  | START.DATE / TIME  |  | REVISION NO. |  |
| MACHINE MODEL  |  | FINISH DATE / TIME |  | PAGE         |  |

THICKNESS OF GASKET :  $0.20 \pm 0.02$  mm



| 1 | NO | 1 | 2 | 3 | REMARK |
|---|----|---|---|---|--------|
|   | A  |   |   |   |        |
|   | B  |   |   |   |        |

| 2 | NO | 1 | 2 | 3 | REMARK |
|---|----|---|---|---|--------|
|   | A  |   |   |   |        |
|   | B  |   |   |   |        |

PROTUTION OF DOWEL :  $18.5 \pm 0.5$  mm

| CYL | ACTUAL | REMARK |
|-----|--------|--------|
| 1   |        |        |
| 2   |        |        |
| 3   |        |        |
| 4   |        |        |
| 5   |        |        |
| 6   |        |        |

| CYL | ACTUAL | REMARK |
|-----|--------|--------|
| 7   |        |        |
| 8   |        |        |
| 9   |        |        |
| 10  |        |        |
| 11  |        |        |
| 12  |        |        |

### Specifications

Table 1

| C13B Main Bore Alignment Specification |                                |
|----------------------------------------|--------------------------------|
| Item                                   | Specification                  |
| Main bearing bore alignment            | Less than 0.08 mm (0.003 inch) |

### MAIN BEARING BORE

|   |                   |                      |
|---|-------------------|----------------------|
| K | MAIN BEARING BORE | $129.891 \pm 0.0135$ |
|---|-------------------|----------------------|

|   | Main Bore No. | X - X' | Y - Y' | Z - Z' |
|---|---------------|--------|--------|--------|
| K | 1 A           |        |        |        |
|   | 1 B           |        |        |        |
|   | 2 A           |        |        |        |
|   | 2 B           |        |        |        |
|   | 3 A           |        |        |        |
|   | 3 B           |        |        |        |
|   | 4 A           |        |        |        |
|   | 4 B           |        |        |        |
|   | 5 A           |        |        |        |
|   | 5 B           |        |        |        |
|   | 6 A           |        |        |        |
|   | 6 B           |        |        |        |
|   | 7 A           |        |        |        |
|   | 7 B           |        |        |        |

INSPECTION RESULT :

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |

INSPECTION BY :

INSPECTION & APPROVAL BY

MECHANIC(MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



**PT SAPTAINDRA SEJATI**

**ENGINE**

**CRANKSHAFT MEASUREMENT**

**CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

**WO NO :**

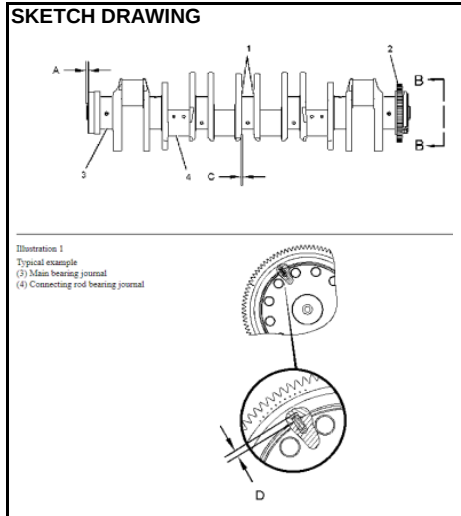
**ENGINE SECTION**



## CRANKSHAFT MEASUREMENT CHECK SHEET

|               |  |                    |  |              |  |
|---------------|--|--------------------|--|--------------|--|
| JOB NUMBER    |  | PART NAME          |  | DOCUMENT NO. |  |
| ENGINE MODEL  |  | PART NUMBER        |  | PUBL. DATE   |  |
| ENGINE S/N    |  | START DATE / TIME  |  | REVISION NO. |  |
| MACHINE MODEL |  | FINISH DATE / TIME |  | PAGE         |  |

|             |  |          |  |
|-------------|--|----------|--|
| MEASURED BY |  | QUANTITY |  |
|-------------|--|----------|--|



**Physical Data**

| No. | Part Number | Part Name  | Qty | New | Reuse |
|-----|-------------|------------|-----|-----|-------|
| 1   |             | Crankshaft | 1   |     |       |
| 2   |             | Gear       | 1   |     |       |
| 3   |             |            | 1   |     |       |
| 4   |             | Dowel Pin  | 1   |     |       |
| 5   |             | Key        | 1   |     |       |
| 6   |             | Spacer     | 1   |     |       |

|                |  |
|----------------|--|
| MACHINIST NAME |  |
|----------------|--|

| No | Measuring items                           | Std Tolerance (mm) | Repair limit |
|----|-------------------------------------------|--------------------|--------------|
| 3  | Main bearing journal                      | 120,650 ± 0,020    |              |
|    | Main bearing journal (US 0,025 inch)      | 120,015 ± 0,020    |              |
|    | Main bearing journal (US 0,050 inch)      | 119,380 ± 0,020    |              |
|    | Clearance between bearing and journal     | 0,070 ~ 0,178      |              |
| 4  | ConRod bearing journal                    | 97.028 ± 0,020     |              |
|    | ConRod bearing journal (US 0,025 inch)    |                    |              |
|    | ConRod bearing journal (US 0,050 inch)    |                    |              |
|    | Clearance between new bearing and journal | 0,070 ~ 0,178      |              |
| A  | Protution dowel                           | 8,4 mm             |              |
| C  | End Play New Thrust Plate                 | 0,10 ~ 0,60 mm     | Max 0.89 mm  |
| D  | Protution dowel gear crankshaft           | 4,1 ± 0,5          |              |

| FILLET RADIUS CHECK |                 | OK | NOT |
|---------------------|-----------------|----|-----|
| Main Journal        | = 2.54 ± 0.2 mm |    |     |
| Pin Journal         | = 3.00 ± 0.2 mm |    |     |

| VISUAL CHECK           |  | OK | NOT |
|------------------------|--|----|-----|
| CRACK HAS BEEN CHECKED |  |    |     |

| MANHOURS UTILIZATION |       |        |           |
|----------------------|-------|--------|-----------|
|                      | START | FINISH | MAN HOURS |
| Machining Process    |       |        |           |
| Measuring Process    |       |        |           |
| TOTAL                |       |        |           |

| INSPECTION SALVAGE PROCESS |                                       |    |    |     |                         |   |       |       |   |       |       |   |       |  |                    |  |
|----------------------------|---------------------------------------|----|----|-----|-------------------------|---|-------|-------|---|-------|-------|---|-------|--|--------------------|--|
| CRITICAL INSPECTION        |                                       |    |    |     | CHECK POINT (DIMENSION) |   |       |       |   |       |       |   |       |  | ROUNDNE<br>SS MAX. |  |
|                            |                                       |    |    |     | 1                       |   |       | 2     |   |       |       | 3 |       |  |                    |  |
|                            |                                       |    |    |     | X - X                   |   | Y - Y | X - X |   | Y - Y | X - X |   | Y - Y |  |                    |  |
| 2                          | CON ROD<br>BEARING<br>JOURNAL         | 1  | 2  | MAC |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       |    |    | QA  |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       | 3  | 4  | MAC |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       |    |    | QA  |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       | 5  | 6  | MAC |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       |    |    | QA  |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       | 7  | 8  | MAC |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       |    |    | QA  |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       | 9  | 10 | MAC |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       |    |    | QA  |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       | 11 | 12 | MAC |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            |                                       |    |    | QA  |                         |   |       |       |   |       |       |   |       |  |                    |  |
|                            | Clearance between bearing and journal |    |    |     |                         | 1 | 2     | 3     | 4 | 5     | 6     | 7 |       |  |                    |  |



|               |  |                    |  |              |  |
|---------------|--|--------------------|--|--------------|--|
| JOB NUMBER    |  | PART NAME          |  | DOCUMENT NO. |  |
| ENGINE MODEL  |  | PART NUMBER        |  | PUBL. DATE   |  |
| ENGINE S/N    |  | START DATE / TIME  |  | REVISION NO. |  |
| MACHINE MODEL |  | FINISH DATE / TIME |  | PAGE         |  |

| INSPECTION SALVAGE PROCESS |                      |                                       |     |                         |       |       |       |       |       |                |       |   |
|----------------------------|----------------------|---------------------------------------|-----|-------------------------|-------|-------|-------|-------|-------|----------------|-------|---|
| CRITICAL INSPECTION        |                      |                                       |     | CHECK POINT (DIMENSION) |       |       |       |       |       | ROUNDNESS MAX. |       |   |
|                            |                      |                                       |     | 1                       |       | 2     |       | 3     |       |                |       |   |
|                            |                      |                                       |     | X - X                   | Y - Y | X - X | Y - Y | X - X | Y - Y |                | X - Y |   |
| 3                          | MAIN BEARING JOURNAL | 1                                     | MAC |                         |       |       |       |       |       |                |       |   |
|                            |                      |                                       | QA  |                         |       |       |       |       |       |                |       |   |
|                            |                      | 2                                     | MAC |                         |       |       |       |       |       |                |       |   |
|                            |                      |                                       | QA  |                         |       |       |       |       |       |                |       |   |
|                            |                      | 3                                     | MAC |                         |       |       |       |       |       |                |       |   |
|                            |                      |                                       | QA  |                         |       |       |       |       |       |                |       |   |
|                            |                      | 4                                     | MAC |                         |       |       |       |       |       |                |       |   |
|                            |                      |                                       | QA  |                         |       |       |       |       |       |                |       |   |
|                            |                      | 5                                     | MAC |                         |       |       |       |       |       |                |       |   |
|                            |                      |                                       | QA  |                         |       |       |       |       |       |                |       |   |
|                            |                      | 6                                     | MAC |                         |       |       |       |       |       |                |       |   |
|                            |                      |                                       | QA  |                         |       |       |       |       |       |                |       |   |
|                            |                      | 7                                     | MAC |                         |       |       |       |       |       |                |       |   |
|                            |                      |                                       | QA  |                         |       |       |       |       |       |                |       |   |
|                            |                      | Clearance between bearing and journal |     |                         |       | 1     | 2     | 3     | 4     | 5              | 6     | 7 |
|                            |                      |                                       |     |                         |       |       |       |       |       |                |       |   |

|   |                                 |   |    |   |    |
|---|---------------------------------|---|----|---|----|
| A | Gap between gear and crankshaft | X | X' | Y | Y' |
|   |                                 |   |    |   |    |

|                 |   |   |
|-----------------|---|---|
| PROTUTION DOWEL | A | D |
|                 |   |   |

[illegible]

|                                                                         |                                                                         |                                                                         |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| INSPECTION BY :                                                         | INSPECTION & APPROVAL BY :                                              |                                                                         |
| <hr style="border: 0; border-top: 1px solid black; margin-top: 80px;"/> | <hr style="border: 0; border-top: 1px solid black; margin-top: 80px;"/> | <hr style="border: 0; border-top: 1px solid black; margin-top: 80px;"/> |
| MECHANIC (MEC)                                                          | QUALITY ASSURANCE (QA)                                                  | GL/SPV PRODUCTION                                                       |



**PT SAPTAINDRA SEJATI**

**ENGINE**

**CAMSHAFT PROCESS AND  
MEASUREMENT CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

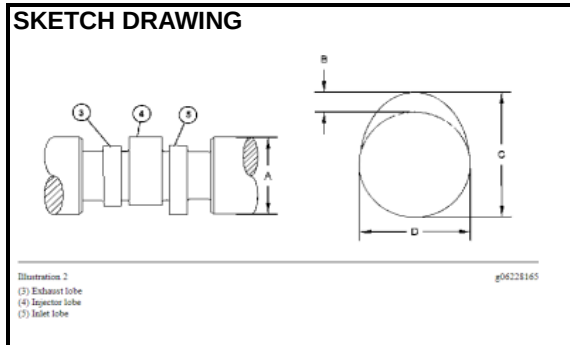
**CAT777E**

**WO NO :**

**ENGINE SECTION**

## CAMSHAFT PROCESS AND MEASUREMENT CHECK SHEET

|               |      |                  |  |              |  |
|---------------|------|------------------|--|--------------|--|
| JOB NUMBER    |      | PART NAME        |  | DOCUMENT NO. |  |
| ENGINE MODEL  | C 32 | PART NUMBER      |  | PUBL. DATE   |  |
| ENGINE S/N    |      | START DATE/TIME  |  | REVISION NO. |  |
| MACHINE MODEL |      | FINISH DATE/TIME |  | PAGE         |  |



|                |  |
|----------------|--|
| MACHINIST NAME |  |
|----------------|--|

| No | Measuring items                      | Std Tolerance (mm) | Repair limit |
|----|--------------------------------------|--------------------|--------------|
| A  | Diameter camshaft Journal            | 84.850 ± 0,020     |              |
| 3  | Max exhaust lobe lift at zero (C-D)  | 8.5152             | 0.13         |
| 4  | Max Injector lobe lift at zero (C-D) | 10.33685           | 0.13         |
| 5  | Max intake lobe lift at zero (C-D)   | 9.6185             | 0.13         |

### CAMSHAFT 1

| NO | CAMSHAFT JOURNAL |   | MAX LIFT CAM LUBE EXHAUST |   | MAX LIFT CAM LUBE INJECTOR<br>B=(C-D) |   | MAX LIFT CAM LUBE INTAKE<br>B=(C-D) |   |
|----|------------------|---|---------------------------|---|---------------------------------------|---|-------------------------------------|---|
|    | X                | Y | X                         | Y | X                                     | Y | X                                   | Y |
|    |                  |   |                           |   |                                       |   |                                     |   |
| 1  |                  |   |                           |   |                                       |   |                                     |   |
| 2  |                  |   |                           |   |                                       |   |                                     |   |
| 3  |                  |   |                           |   |                                       |   |                                     |   |
| 4  |                  |   |                           |   |                                       |   |                                     |   |
| 5  |                  |   |                           |   |                                       |   |                                     |   |
| 6  |                  |   |                           |   |                                       |   |                                     |   |

### CAMSHAFT 2

| NO | CAMSHAFT JOURNAL |   | MAX LIFT CAM LUBE EXHAUST |   | MAX LIFT CAM LUBE INJECTOR<br>B=(C-D) |   | MAX LIFT CAM LUBE INTAKE<br>B=(C-D) |   |
|----|------------------|---|---------------------------|---|---------------------------------------|---|-------------------------------------|---|
|    | X                | Y | X                         | Y | X                                     | Y | X                                   | Y |
|    |                  |   |                           |   |                                       |   |                                     |   |
| 1  |                  |   |                           |   |                                       |   |                                     |   |
| 2  |                  |   |                           |   |                                       |   |                                     |   |
| 3  |                  |   |                           |   |                                       |   |                                     |   |
| 4  |                  |   |                           |   |                                       |   |                                     |   |
| 5  |                  |   |                           |   |                                       |   |                                     |   |
| 6  |                  |   |                           |   |                                       |   |                                     |   |

| DESCRIPTION          | OK | Not OK |
|----------------------|----|--------|
| CRACK HAS BEEN CHECK |    |        |

Lubricate the camshaft lobes with a 50/50 mixture of 8T-2998 Lubricant and clean engine oil.

### MECHANIC INSPECTION COMMENT :

|                                                               |                                                                          |
|---------------------------------------------------------------|--------------------------------------------------------------------------|
| INSPECTION BY : _____<br><br><br><br><br><br><br><br><br><br> | INSPECTION & APPROVED BY : _____<br><br><br><br><br><br><br><br><br><br> |
| _____<br>MACHINIST (MACH)                                     | _____<br>QUALITY ASSURANCE (QA)                                          |
| _____<br>GL/SPV PRODUCTION                                    |                                                                          |



**PT SAPTAINDRA SEJATI**

**ENGINE**

**CONNECTING ROD PROCESS AND  
MEASUREMENT CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

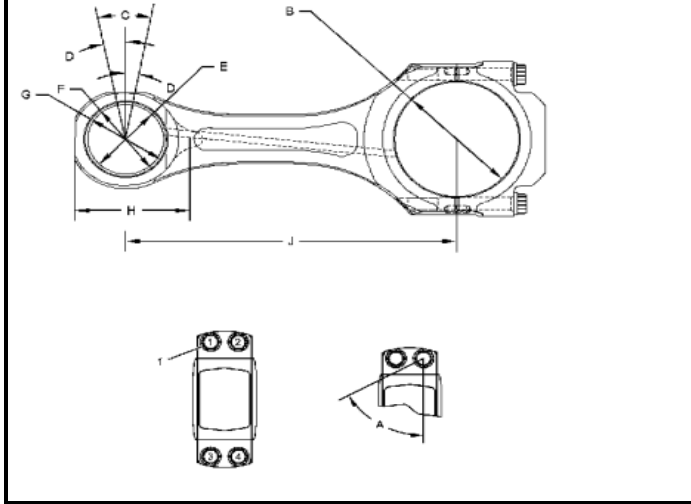
**WO NO :**

**ENGINE SECTION**

## CONNECTING ROD PROCESS AND MEASUREMENT CHECK SHEET

|               |      |                  |         |              |  |
|---------------|------|------------------|---------|--------------|--|
| JOB NUMBER    |      | PART NAME        | CON ROD | DOCUMENT NO. |  |
| ENGINE MODEL  | C 32 | PART NUMBER      |         | PUBL. DATE   |  |
| ENGINE S/N    |      | START DATE/TIME  |         | REVISION NO. |  |
| MACHINE MODEL |      | FINISH DATE/TIME |         | PAGE         |  |

### SKETCH DRAWING



### MACHINIST NAME

| No | Measuring items                                                                              | Std Tolerance (mm) | Repair limit |
|----|----------------------------------------------------------------------------------------------|--------------------|--------------|
| B  | Diameter dalam Conrod Crankshaft Journal                                                     | $103.5 \pm 0.013$  |              |
| D  | Sudut Max Horizontal Centerline                                                              | $\pm 5 \%$         |              |
| E  | Diameter dalam Conrod Bearing untuk Pin                                                      | $64,592 \pm 0,013$ |              |
| F  | Diameter Lubang Pin Piston Bearing                                                           | $60,035 \pm 0,008$ |              |
| G  | Diameter Luar Bearing Pin Piston sebelum dipasang ke dalam conrod                            | $64,717 \pm 0,013$ |              |
| H  | Panjang maksimum Bearing Conrod saat terpasang bearing Piston Pin                            | 95 mm              |              |
| J  | Diameter antara Center Lubang Bearing Pin Piston dan Center Lubang Crankshaft Journal Conrod | 274.91             |              |

### ID SMALL END WITHOUT BEARING

| No | CON-ROD NO. |    | TIGHTENING BOLTS                  | CHECK POINT | Ø SMALL END WITHOUT BEARING |    |     |    |        |    |        |    |
|----|-------------|----|-----------------------------------|-------------|-----------------------------|----|-----|----|--------|----|--------|----|
|    |             |    |                                   |             | X - X'                      |    |     |    | Y - Y' |    | Z - Z' |    |
|    |             |    |                                   |             | MAC                         | QA | MAC | QA | MAC    | QA | MAC    | QA |
| E  | 1           | 7  | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                             |    |     |    |        |    |        |    |
|    | 2           | 8  | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                             |    |     |    |        |    |        |    |
|    | 3           | 9  | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                             |    |     |    |        |    |        |    |
|    | 4           | 10 | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                             |    |     |    |        |    |        |    |
|    | 5           | 11 | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                             |    |     |    |        |    |        |    |
|    | 6           | 12 | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                             |    |     |    |        |    |        |    |

### ID SMALL END WITH BEARING

| No | CON-ROD NO. |    | TIGHTENING BOLTS                  | CHECK POINT | Ø SMALL END WITH BEARING |    |     |    |        |    |        |    |
|----|-------------|----|-----------------------------------|-------------|--------------------------|----|-----|----|--------|----|--------|----|
|    |             |    |                                   |             | X - X'                   |    |     |    | Y - Y' |    | Z - Z' |    |
|    |             |    |                                   |             | MAC                      | QA | MAC | QA | MAC    | QA | MAC    | QA |
| F  | 1           | 7  | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                          |    |     |    |        |    |        |    |
|    | 2           | 8  | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                          |    |     |    |        |    |        |    |
|    | 3           | 9  | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                          |    |     |    |        |    |        |    |
|    | 4           | 10 | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                          |    |     |    |        |    |        |    |
|    | 5           | 11 | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                          |    |     |    |        |    |        |    |
|    | 6           | 12 | $70 \pm 4 \text{ N.m} + 60^\circ$ | A           |                          |    |     |    |        |    |        |    |

### ID BIG END WITHOUT BEARING

| No | CON-ROD NO. |   | TIGHTENING BOLTS  | CHECK POINT | Ø BIG END WITHOUT BEARING |    |     |    |        |    |        |    |
|----|-------------|---|-------------------|-------------|---------------------------|----|-----|----|--------|----|--------|----|
|    |             |   |                   |             | X - X'                    |    |     |    | Y - Y' |    | Z - Z' |    |
|    |             |   |                   |             | MAC                       | QA | MAC | QA | MAC    | QA | MAC    | QA |
| R  | 1           | 7 | 7 Kgm + 60 degree | A           |                           |    |     |    |        |    |        |    |
|    | 2           | 8 | 7 Kgm + 60 degree | A           |                           |    |     |    |        |    |        |    |
|    | 3           | 9 | 7 Kgm + 60 degree | A           |                           |    |     |    |        |    |        |    |



## CONNECTING ROD PROCESS AND MEASUREMENT CHECK SHEET

|               |   |      |                   |                  |         |  |  |              |              |  |  |  |  |  |  |
|---------------|---|------|-------------------|------------------|---------|--|--|--------------|--------------|--|--|--|--|--|--|
| JOB NUMBER    |   |      | PART NAME         |                  | CON ROD |  |  | DOCUMENT NO. |              |  |  |  |  |  |  |
| ENGINE MODEL  |   | C 32 |                   | PART NUMBER      |         |  |  |              | PUBL. DATE   |  |  |  |  |  |  |
| ENGINE S/N    |   |      |                   | START DATE/TIME  |         |  |  |              | REVISION NO. |  |  |  |  |  |  |
| MACHINE MODEL |   |      |                   | FINISH DATE/TIME |         |  |  |              | PAGE         |  |  |  |  |  |  |
|               | 4 | 10   | 7 Kgm + 60 degree | A                |         |  |  |              |              |  |  |  |  |  |  |
|               | 5 | 11   | 7 Kgm + 60 degree | A                |         |  |  |              |              |  |  |  |  |  |  |
|               | 6 | 12   | 7 Kgm + 60 degree | A                |         |  |  |              |              |  |  |  |  |  |  |

MECHANIC INSPECTION COMMENT :

|  |
|--|
|  |
|  |
|  |
|  |
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|  |
|  |
|  |

INSPECTION BY :

INSPECTION & APPROVED BY :

\_\_\_\_\_  
MACHINIST (MACH)

\_\_\_\_\_  
QUALITY ASSURANCE (QA)

\_\_\_\_\_  
GL/SPV PRODUCTION



**PT SAPTAINDRA SEJATI**

**ENGINE**

**ROCKER ARM MEASURING AND  
POLISHING CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

**WO NO :**

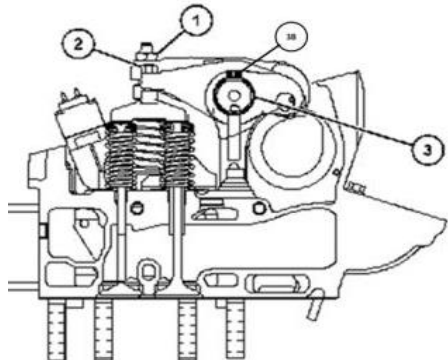
**ENGINE SECTION**



## ROCKER ARM MEASURING AND POLISHING CHECK SHEET

|               |  |                  |  |             |  |
|---------------|--|------------------|--|-------------|--|
| JOB NUMBER    |  | PART NAME        |  | DOCUMENT NO |  |
| ENGINE MODEL  |  | PART NUMBER      |  | PUBL. DATE  |  |
| ENGINE S/N    |  | START DATE/TIME  |  | REVISION NO |  |
| MACHINE MODEL |  | FINISH DATE/TIME |  | PAGE        |  |

### SKETCH DRAWING



MACHINIST NAME

| No | Measuring items              | Std Tolerance (mm)  | Repair limit |
|----|------------------------------|---------------------|--------------|
| 1  | Rocker Arm Injector Locknut  | Torque 100 ± 10 N.m | -            |
| 2  | Valve Rocker Arm Locknut     | Torque 30 ± 7 N.m   | -            |
| 3  | Diameter New shaft           | 40                  | ± 0,01       |
|    | Inner Diameter rocker arm    | 40.065              | ± 0,015      |
|    | Inlet Valve lash (clearance) | 0.38 ± 0.08 N.m     | -            |
|    | Exh. Valvelash (clearance)   | 0.76 ± 0.08 N.m     | -            |

| INSPECTION MEASURING OUTER DIAMETER SHAFT |    |    |                   |          |    |          |    |          |    |     |                     |
|-------------------------------------------|----|----|-------------------|----------|----|----------|----|----------|----|-----|---------------------|
| ROCKER<br>ARM NO.                         |    |    | CHECK<br>POSITION | 1        |    | 2        |    | 3        |    | SPV | Actual<br>Clearance |
|                                           |    |    |                   | MACHANIC | QA | MECHANIC | QA | MECHANIC | QA |     |                     |
| 3                                         | 1  | 2  | X – X'            |          |    |          |    |          |    |     |                     |
|                                           |    |    | Y – Y'            |          |    |          |    |          |    |     |                     |
|                                           | 3  | 4  | X – X'            |          |    |          |    |          |    |     |                     |
|                                           |    |    | Y – Y'            |          |    |          |    |          |    |     |                     |
|                                           | 5  | 6  | X – X'            |          |    |          |    |          |    |     |                     |
|                                           |    |    | Y – Y'            |          |    |          |    |          |    |     |                     |
|                                           | 7  | 8  | X – X'            |          |    |          |    |          |    |     |                     |
|                                           |    |    | Y – Y'            |          |    |          |    |          |    |     |                     |
|                                           | 9  | 10 | X – X'            |          |    |          |    |          |    |     |                     |
|                                           |    |    | Y – Y'            |          |    |          |    |          |    |     |                     |
|                                           | 11 | 12 | X – X'            |          |    |          |    |          |    |     |                     |
|                                           |    |    | Y – Y'            |          |    |          |    |          |    |     |                     |

# ROCKER ARM MEASURING AND POLISHING CHECK SHEET

|               |  |                  |  |             |  |
|---------------|--|------------------|--|-------------|--|
| JOB NUMBER    |  | PART NAME        |  | DOCUMENT NO |  |
| ENGINE MODEL  |  | PART NUMBER      |  | PUBL. DATE  |  |
| ENGINE S/N    |  | START DATE/TIME  |  | REVISION NO |  |
| MACHINE MODEL |  | FINISH DATE/TIME |  | PAGE        |  |

| INSPECTION MEASURING INNER DIAMETER ROCKER ARM |    |    |                |          |    |          |    |          |    |     |                  |
|------------------------------------------------|----|----|----------------|----------|----|----------|----|----------|----|-----|------------------|
| ROCKER ARM NO.                                 |    |    | CHECK POSITION | 1        |    | 2        |    | 3        |    | SPV | Actual Clearance |
|                                                |    |    |                | MACHANIC | QA | MECHANIC | QA | MECHANIC | QA |     |                  |
| 3B                                             | 1  | 2  | X - X'         |          |    |          |    |          |    |     |                  |
|                                                |    |    | Y - Y'         |          |    |          |    |          |    |     |                  |
|                                                | 3  | 4  | X - X'         |          |    |          |    |          |    |     |                  |
|                                                |    |    | Y - Y'         |          |    |          |    |          |    |     |                  |
|                                                | 5  | 6  | X - X'         |          |    |          |    |          |    |     |                  |
|                                                |    |    | Y - Y'         |          |    |          |    |          |    |     |                  |
|                                                | 7  | 8  | X - X'         |          |    |          |    |          |    |     |                  |
|                                                |    |    | Y - Y'         |          |    |          |    |          |    |     |                  |
|                                                | 9  | 10 | X - X'         |          |    |          |    |          |    |     |                  |
|                                                |    |    | Y - Y'         |          |    |          |    |          |    |     |                  |
|                                                | 11 | 12 | X - X'         |          |    |          |    |          |    |     |                  |
|                                                |    |    | Y - Y'         |          |    |          |    |          |    |     |                  |

Page 2

## MACHINIST INSPECTION RESULT :

|  |
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|  |
|  |

INSPECTION BY :

INSPECTION & APPROVED BY :

MACHINIST (MACH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



**PT SAPTAINDRA SEJATI**

**ENGINE**

**CYLINDER LINER MEASURING AND  
POLISHING CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

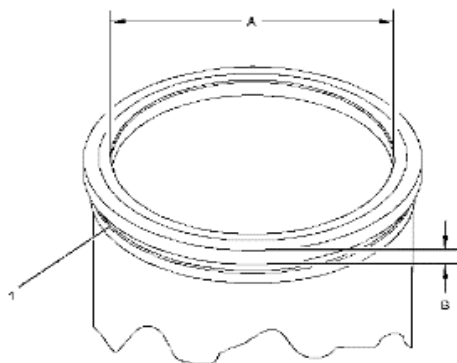
**WO NO :**

**ENGINE SECTION**

## CYLINDER LINER MEASURING AND POLISHING CHECK SHEET

|               |      |                  |           |              |   |
|---------------|------|------------------|-----------|--------------|---|
| JOB NUMBER    |      | PART NAME        | CYL LINER | DOCUMENT NO  |   |
| ENGINE MODEL  | C 32 | PART NUMBER      |           | PUBL. DATE   |   |
| ENGINE S/N    |      | START DATE/TIME  |           | REVISION NO. |   |
| MACHINE MODEL |      | FINISH DATE/TIME |           | PAGE         | 1 |

### SKETCH DRAWING



MACHINIST NAME

| No       | Measuring items  | Std Tolerance (mm) | Repair limit |
|----------|------------------|--------------------|--------------|
| <b>A</b> | CYL LINER BORE   | 137.185            | ± 0,025      |
| <b>B</b> | FLANGE THICKNESS | 8,890 ± 0,020      | 8.870        |

### Ikuti Prosedur Pemasangan Cylinder Liner:

|   |                                                                                                                        |
|---|------------------------------------------------------------------------------------------------------------------------|
| 1 | Oleskan sabun cair pada permukaan lubang silinder blok liner dan seal karet yang berada di ujung bawah liner silinder. |
| 2 | Celupkan Filler Band (1) pada Oli Engine yang bersih.                                                                  |
| 3 | Segera pasang Filler Band (1) di alur yang berada di bawah liner Flange                                                |
| 4 | Pasang 3 O-Ring Seal pada Liner kemudian Pasang Liner kedalam Cylinder Block .                                         |

### INSPECTION MEASURING FLANGE THICKNESS

| CYL LINER NO. |    |    | CHECK POSITION | 1        |    | 2        |    | 3        |    | REMARK |
|---------------|----|----|----------------|----------|----|----------|----|----------|----|--------|
|               |    |    |                | MACHANIC | QA | MACHANIC | QA | MACHANIC | QA |        |
| <b>A</b>      | 1  | 2  | X - X'         |          |    |          |    |          |    |        |
|               |    |    | Y - Y'         |          |    |          |    |          |    |        |
|               | 3  | 4  | X - X'         |          |    |          |    |          |    |        |
|               |    |    | Y - Y'         |          |    |          |    |          |    |        |
|               | 5  | 6  | X - X'         |          |    |          |    |          |    |        |
|               |    |    | Y - Y'         |          |    |          |    |          |    |        |
|               | 7  | 8  | X - X'         |          |    |          |    |          |    |        |
|               |    |    | Y - Y'         |          |    |          |    |          |    |        |
|               | 9  | 10 | X - X'         |          |    |          |    |          |    |        |
|               |    |    | Y - Y'         |          |    |          |    |          |    |        |
|               | 11 | 12 | X - X'         |          |    |          |    |          |    |        |
|               |    |    | Y - Y'         |          |    |          |    |          |    |        |

### INSPECTION MEASURING CYL LINER BORE

| CYL LINER NO. |    |    | X |  | X' |  | Y |  | Y' |  | REMARK |
|---------------|----|----|---|--|----|--|---|--|----|--|--------|
|               |    |    |   |  |    |  |   |  |    |  |        |
|               | 1  | 2  |   |  |    |  |   |  |    |  |        |
|               | 3  | 4  |   |  |    |  |   |  |    |  |        |
|               | 5  | 6  |   |  |    |  |   |  |    |  |        |
|               | 7  | 8  |   |  |    |  |   |  |    |  |        |
|               | 9  | 10 |   |  |    |  |   |  |    |  |        |
|               | 11 | 12 |   |  |    |  |   |  |    |  |        |

### MACHINIST INSPECTION RESULT :

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INSPECTION BY :

INSPECTION & APPROVED BY :

\_\_\_\_\_  
MACHINIST (MACH)

\_\_\_\_\_  
QUALITY ASSURANCE (QA)

\_\_\_\_\_  
GL/SPV PRODUCTION



**PT SAPTAINDRA SEJATI**

**ENGINE**

**PIN PISTON MEASURING AND POLISHING  
CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

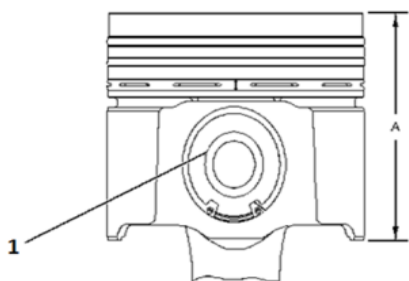
**WO NO :**

**ENGINE SECTION**

## PIN PISTON MEASURING AND POLISHING CHECK SHEET

|               |  |                  |            |              |  |
|---------------|--|------------------|------------|--------------|--|
| JOB NUMBER    |  | PART NAME        | PIN PISTON | DOCUMENT NO. |  |
| ENGINE MODEL  |  | PART NUMBER      |            | PUBL. DATE   |  |
| ENGINE S/N    |  | START DATE/TIME  |            | REVISION NO. |  |
| MACHINE MODEL |  | FINISH DATE/TIME |            | PAGE         |  |

SKETCH DRAWING



| NO | Measuring items                            | Std Tolerance (mm) | Repair limit |
|----|--------------------------------------------|--------------------|--------------|
| 1  | Bore in the new piston body piston pin     | 60,095             | ± 0.005      |
| 2  | Length of the new piston pin is 89.8 ± 0.2 | 89.8               | ± 0.2        |
| A  | Lumuri Oli Piston dgn Oli Engine Bersih    |                    |              |

| INSPECTION MEASURING PROCESS |          |    |        |    |     |    |     |    |                  |  |  |  |
|------------------------------|----------|----|--------|----|-----|----|-----|----|------------------|--|--|--|
| PIN PISTON NO.               | CHECK    |    | 1      |    | 2   |    | 3   |    | Clearance actual |  |  |  |
|                              | POSITION |    | MAC    | QA | MAC | QA | MAC | QA |                  |  |  |  |
| 1                            | 1        | 2  | X - X' |    |     |    |     |    |                  |  |  |  |
|                              |          |    | Y - Y' |    |     |    |     |    |                  |  |  |  |
|                              | 3        | 4  | X - X' |    |     |    |     |    |                  |  |  |  |
|                              |          |    | Y - Y' |    |     |    |     |    |                  |  |  |  |
|                              | 5        | 6  | X - X' |    |     |    |     |    |                  |  |  |  |
|                              |          |    | Y - Y' |    |     |    |     |    |                  |  |  |  |
|                              | 7        | 8  | X - X' |    |     |    |     |    |                  |  |  |  |
|                              |          |    | Y - Y' |    |     |    |     |    |                  |  |  |  |
|                              | 9        | 10 | X - X' |    |     |    |     |    |                  |  |  |  |
|                              |          |    | Y - Y' |    |     |    |     |    |                  |  |  |  |
|                              | 11       | 12 | X - X' |    |     |    |     |    |                  |  |  |  |
|                              |          |    | Y - Y' |    |     |    |     |    |                  |  |  |  |

| INSPECTION MEASURING LENGTH OF PIN PISTON |    |    |   |  |    |  |   |  |    |  |  |  |
|-------------------------------------------|----|----|---|--|----|--|---|--|----|--|--|--|
| PIN PISTON NO.                            |    |    | X |  | X' |  | Y |  | Y' |  |  |  |
| 2                                         | 1  | 2  |   |  |    |  |   |  |    |  |  |  |
|                                           | 3  | 4  |   |  |    |  |   |  |    |  |  |  |
|                                           | 5  | 6  |   |  |    |  |   |  |    |  |  |  |
|                                           | 7  | 8  |   |  |    |  |   |  |    |  |  |  |
|                                           | 9  | 10 |   |  |    |  |   |  |    |  |  |  |
|                                           | 11 | 12 |   |  |    |  |   |  |    |  |  |  |
|                                           |    |    |   |  |    |  |   |  |    |  |  |  |

**MACHINIST INSPECTION RESULT :**

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INSPECTION BY :

\_\_\_\_\_

MACHINIST (MACH)

INSPECTION & APPROVED BY :

\_\_\_\_\_

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



**PT SAPTAINDRA SEJATI**

**ENGINE**

**PISTON MEASURING AND POLISHING  
CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

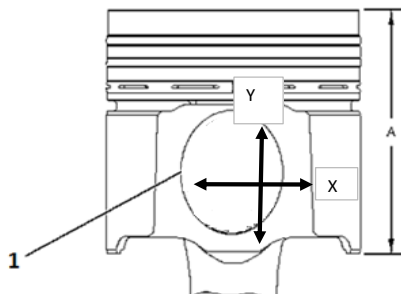
**WO NO :**

**ENGINE SECTION**

# PISTON MEASURING AND POLISHING CHECK SHEET

|               |  |                  |  |              |  |
|---------------|--|------------------|--|--------------|--|
| JOB NUMBER    |  | PART NAME        |  | DOCUMENT NO. |  |
| ENGINE MODEL  |  | PART NUMBER      |  | PUBL. DATE   |  |
| ENGINE S/N    |  | START DATE/TIME  |  | REVISION NO. |  |
| MACHINE MODEL |  | FINISH DATE/TIME |  | PAGE         |  |

## SKETCH DRAWING



| Measuring items            | Std Tolerance | Repair limit |
|----------------------------|---------------|--------------|
| Bore in the piston for pin | 60.0925       | ± 0,0075     |

| INSPECTION MEASURING PROCESS |    |    |          |     |    |     |    |     |    |   |  |  |                  |
|------------------------------|----|----|----------|-----|----|-----|----|-----|----|---|--|--|------------------|
| PISTON NO                    |    |    | CHECK    | 1   |    |     | 2  |     |    | 3 |  |  | Clearance actual |
|                              |    |    | POSITION | MAC | QA | MAC | QA | MAC | QA |   |  |  |                  |
| 1                            | 1  | 2  | X - X'   |     |    |     |    |     |    |   |  |  |                  |
|                              |    |    | Y - Y'   |     |    |     |    |     |    |   |  |  |                  |
|                              | 3  | 4  | X - X'   |     |    |     |    |     |    |   |  |  |                  |
|                              |    |    | Y - Y'   |     |    |     |    |     |    |   |  |  |                  |
|                              | 5  | 6  | X - X'   |     |    |     |    |     |    |   |  |  |                  |
|                              |    |    | Y - Y'   |     |    |     |    |     |    |   |  |  |                  |
|                              | 7  | 8  | X - X'   |     |    |     |    |     |    |   |  |  |                  |
|                              |    |    | Y - Y'   |     |    |     |    |     |    |   |  |  |                  |
|                              | 9  | 10 | X - X'   |     |    |     |    |     |    |   |  |  |                  |
|                              |    |    | Y - Y'   |     |    |     |    |     |    |   |  |  |                  |
|                              | 11 | 12 | X - X'   |     |    |     |    |     |    |   |  |  |                  |
|                              |    |    | Y - Y'   |     |    |     |    |     |    |   |  |  |                  |

## MACHINIST INSPECTION RESULT :

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INSPECTION BY :

INSPECTION & APPROVED BY :

\_\_\_\_\_  
MACHINIST (MACH)

\_\_\_\_\_  
QUALITY ASSURANCE (QA)

\_\_\_\_\_  
GL/SPV PRODUCTION





**PT SAPTAINDRA SEJATI**

**ENGINE**

**OIL COOLER LEAK TESTER**

**CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

**WO NO :**

**ENGINE SECTION**

# OIL COOLER LEAK TESTER

|                                                            |               |  |                |        |
|------------------------------------------------------------|---------------|--|----------------|--------|
| AIR LEAK TEST CHECK SHEET<br>OIL COOLER<br>ENGINE CAT C-32 | JOB NUMBER    |  | DOCUMENT NO    |        |
|                                                            | ENGINE MODEL  |  | PUBLISHED DATE |        |
|                                                            | ENGINE S/N    |  | REVISION NO    | 00     |
|                                                            | MACHINE MODEL |  | PAGE           | 1 of 1 |

|         |        |  |             |  |
|---------|--------|--|-------------|--|
| PROCESS | START  |  | ASSEMBLY BY |  |
|         | FINISH |  |             |  |

| NO | PARTS              | CHECK POINT     | CONDITION TO BE CHECKED                                                                                                                                                                                                                                                                                                                                                            | SPEC.                                                                                                       | SIGN |     |    | REMARKS<br>SKETCH, PHOTOS,<br>COMMENTS                                                                                                                                            |
|----|--------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------|-----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                    |                 |                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                             | MEC  | SPV | QA |                                                                                                                                                                                   |
| 1  | Oil Cooler Element | Air Leak Tester | Inspect Element By Visual Check<br><br>No Crack <input type="text"/>                                                                                                                                                                                                                                                                                                               | Check                                                                                                       |      |     |    |                                                                                                |
| 2  |                    |                 | Pasang adaptor dengan erat ke Port Oil Cooler.                                                                                                                                                                                                                                                                                                                                     |                                                                                                             |      |     |    |                                                                                                |
| 3  | Test Pressure      |                 | Cek Crack Element dengan Udara Bertekanan.<br>a).Air Pressure<br>Actual: <input type="text"/> Kpa<br><br>b).Water temperatur<br>Actual: <input type="text"/> °C<br><br>c).Time for Test<br>Actual: <input type="text"/> minutes<br><br>Saat oil cooler masih bertekanan, gunakan larutan sabun untuk membantu menemukan kebocoran yang lambat..<br>No Leakage <input type="text"/> | 586 Kpa- 792 Kpa<br>(85 psi - 115 psi)<br><br>Ambient Temperatur<br><br><br>3 MINUTES<br><br>Check use SOAP |      |     |    | <br><br> |

Comment :

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|                                             |                                                                    |
|---------------------------------------------|--------------------------------------------------------------------|
| ASSEMBLY BY<br><br><br><hr/> MECHANIC (MEC) | INSPECTION & APPROVED BY<br><br><br><hr/> QUALITY ASSURANCE ( QA ) |
|---------------------------------------------|--------------------------------------------------------------------|

GL/SPV PRODUCTION



**PT SAPTAINDRA SEJATI**

**ENGINE**

**WASHING OIL COOLER**

**CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

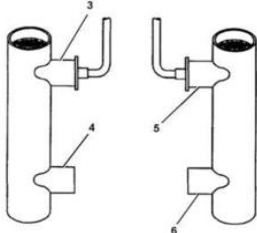
**WO NO :**

**ENGINE SECTION**

## WASHING OIL COOLER CHECK SHEET

|                                                                                   |            |  |             |  |
|-----------------------------------------------------------------------------------|------------|--|-------------|--|
| <b>WASHING CHECK SHEET</b><br><br><b>OIL COOLER</b><br><br><b>ENGINE CAT C-32</b> | WO NUMBER  |  | DOCUMENT NO |  |
|                                                                                   | COMP.MODEL |  | PUBLIC DATE |  |
|                                                                                   | COMP.S/N   |  | REVISION NO |  |
|                                                                                   | UNIT MODEL |  | PAGE        |  |

|         |        |  |             |  |  |
|---------|--------|--|-------------|--|--|
| PROCESS | START  |  | ASSEMBLY BY |  |  |
|         | FINISH |  |             |  |  |

| NO | PART TO<br>INSTALL OR<br>FIT | CONDITION TO BE ASSEMBLY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SPEC | SIGNATURE |       | SKETCH DRAWING OR PHOTOES                                                            |
|----|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------|-------|--------------------------------------------------------------------------------------|
|    |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      | MECH      | GL/QA |                                                                                      |
|    |                              | 1. Letakkan oil cooler dalam caustic tank yang mengaduk dengan port oli menghadap ke bawah. Kocok pendingin oli di tangki larutan kaustik panas setidaknya selama 4 jam.                                                                                                                                                                                                                                                                                                                                            |      |           |       |   |
|    |                              | Setelah pengadukan, letakkan oil cooler di atas dudukan pembilas. Dudukan pembilasan harus diisi dengan larutan kaustik yang sama dengan tangki agitasi. Dudukan harus memiliki kartrid filtrasi magnetis dan laju aliran 132 L / menit (35 US gpm), pada suhu minimum konstan 82 ° C (180 ° F). Membilas oil cooler untuk kedua kalinya dengan larutan kaustik yang bersih akan menghilangkan aluminium tambahan atau kotoran yang terlepas selama tahap agitasi. Bilas oil cooler di setiap arah selama 30 menit. |      |           |       |  |

|           |
|-----------|
| Comment : |
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|                                 |                                           |
|---------------------------------|-------------------------------------------|
| ASSEMBLY BY<br><br><br><br><br> | INSPECTION & APPROVED<br><br><br><br><br> |
| _____<br>MECHANIC (MEC)         | _____<br>QUALITY ASSURANCE (QA)           |
|                                 | _____<br>GL/SPV PRODUCTION                |



**PT SAPTAINDRA SEJATI**

**ENGINE**

**CYLINDER LINER PROJECTION  
MEASUREMENT CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

**CAT777E**

**WO NO :**

**ENGINE SECTION**

# CYLINDER LINER PROJECTION MEASUREMENT CHECK SHEET

|               |          |                    |  |              |  |
|---------------|----------|--------------------|--|--------------|--|
| JOB NUMBER    |          | PART NAME          |  | DOCUMENT NO. |  |
| ENGINE MODEL  | C32      | PART NUMBER        |  | PUBL. DATE   |  |
| ENGINE S/N    |          | START DATE / TIME  |  | REVISION NO. |  |
| MACHINE MODEL | CAT 777E | FINISH DATE / TIME |  | PAGE         |  |

|             |  |          |  |
|-------------|--|----------|--|
| MEASURED BY |  | QUANTITY |  |
|-------------|--|----------|--|

## STANDARD FIGURES

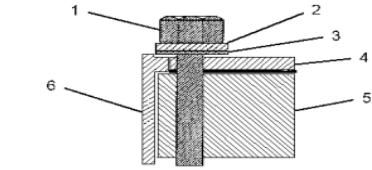


Illustration 1  
Liner Projection Components  
(1) Bolt  
(2) Washer  
(3) Washer  
(4) Spacer plate  
(5) Block  
(6) Cylinder liner

The 8T-0455 Liner Projection Tool Group

Table 1

| Needed Components |                                   |                             |                           |
|-------------------|-----------------------------------|-----------------------------|---------------------------|
| Item              | Part Number                       | Description                 | Quantity For One Cylinder |
| 1                 | 7H-3598                           | Track Bolt                  | 6                         |
| 2                 | 8F-1484 C27<br>or<br>7X-0564 C32  | Washer<br>or<br>Hard Washer | 6                         |
| 3                 | 7K-1977 C27<br>or<br>126-1454 C32 | Washer<br>or<br>Washer      | 6                         |

## A.Result of Measurement

|            |  |                          |                               |                                |
|------------|--|--------------------------|-------------------------------|--------------------------------|
| Date _____ |  | Miles, Km<br>Hours _____ | Engine<br>Serial Number _____ | Vehicle<br>Serial Number _____ |
|------------|--|--------------------------|-------------------------------|--------------------------------|

FRONT

Max Variation Each Cylinder

Max 0.051 mm (.002 in)

Max Variation of AVG Under One Cylinder

Head Max 0.051 mm (.002 in)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
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| <p><b>Liner Projection</b><br/>Max 0.15 mm (.006 in)<br/>Min 0.05 mm (.001 in)</p> <table style="width: 100%;"> <tr><td>1A</td><td>2A</td></tr> <tr><td>1B</td><td>2B</td></tr> <tr><td>1C</td><td>2C</td></tr> <tr><td>1D</td><td>2D</td></tr> <tr><td>SUM 1</td><td>SUM 2</td></tr> <tr><td>AVG 1</td><td>AVG 2</td></tr> <tr><td>3A</td><td>4A</td></tr> <tr><td>3B</td><td>4B</td></tr> <tr><td>3C</td><td>4C</td></tr> <tr><td>3D</td><td>4D</td></tr> <tr><td>SUM 3</td><td>SUM 4</td></tr> <tr><td>AVG 3</td><td>AVG 4</td></tr> <tr><td>5A</td><td>6A</td></tr> <tr><td>5B</td><td>6B</td></tr> <tr><td>5C</td><td>6C</td></tr> <tr><td>5D</td><td>6D</td></tr> <tr><td>SUM 5</td><td>SUM 6</td></tr> <tr><td>AVG 5</td><td>AVG 6</td></tr> <tr><td>7A</td><td>8A</td></tr> <tr><td>7B</td><td>8B</td></tr> <tr><td>7C</td><td>8C</td></tr> <tr><td>7D</td><td>8D</td></tr> <tr><td>SUM 7</td><td>SUM 8</td></tr> <tr><td>AVG 7</td><td>AVG 8</td></tr> <tr><td>9A</td><td>10A</td></tr> <tr><td>9B</td><td>10B</td></tr> <tr><td>9C</td><td>10C</td></tr> <tr><td>9D</td><td>10D</td></tr> <tr><td>SUM 9</td><td>SUM 10</td></tr> <tr><td>AVG 9</td><td>AVG 10</td></tr> <tr><td>11A</td><td>12A</td></tr> <tr><td>11B</td><td>12B</td></tr> <tr><td>11C</td><td>12C</td></tr> <tr><td>11D</td><td>12D</td></tr> <tr><td>SUM 11</td><td>SUM 12</td></tr> <tr><td>AVG 11</td><td>AVG 12</td></tr> </table> | 1A           | 2A | 1B | 2B | 1C | 2C | 1D | 2D | SUM 1 | SUM 2 | AVG 1 | AVG 2 | 3A | 4A | 3B | 4B | 3C | 4C | 3D | 4D | SUM 3 | SUM 4 | AVG 3 | AVG 4 | 5A | 6A | 5B | 6B | 5C | 6C | 5D | 6D | SUM 5 | SUM 6 | AVG 5 | AVG 6 | 7A | 8A | 7B | 8B | 7C | 8C | 7D | 8D | SUM 7 | SUM 8 | AVG 7 | AVG 8 | 9A | 10A | 9B | 10B | 9C | 10C | 9D | 10D | SUM 9 | SUM 10 | AVG 9 | AVG 10 | 11A | 12A | 11B | 12B | 11C | 12C | 11D | 12D | SUM 11 | SUM 12 | AVG 11 | AVG 12 | <table style="width: 100%;"> <tr><td>Max 1A-1D</td><td>Max 2A-2D</td></tr> <tr><td>Min 1A-1D</td><td>Min 2A-2D</td></tr> <tr><td>Variation</td><td>Variation</td></tr> <tr><td>Max 3A-3D</td><td>Max 4A-4D</td></tr> <tr><td>Min 3A-3D</td><td>Min 4A-4D</td></tr> <tr><td>Variation</td><td>Variation</td></tr> <tr><td>Max 5A-5D</td><td>Max 6A-6D</td></tr> <tr><td>Min 5A-5D</td><td>Min 6A-6D</td></tr> <tr><td>Variation</td><td>Variation</td></tr> <tr><td>Max 7A-7D</td><td>Max 8A-8D</td></tr> <tr><td>Min 7A-7D</td><td>Min 8A-8D</td></tr> <tr><td>Variation</td><td>Variation</td></tr> <tr><td>Max 9A-9D</td><td>Max 10A-10D</td></tr> <tr><td>Min 9A-9D</td><td>Min 10A-10D</td></tr> <tr><td>Variation</td><td>Variation</td></tr> <tr><td>Max 11A-11D</td><td>Max 12A-12D</td></tr> <tr><td>Min 11A-11D</td><td>Min 12A-12D</td></tr> <tr><td>Variation</td><td>Variation</td></tr> </table> <table style="width: 100%;"> <tr><td>AVG 1</td><td>AVG 2</td></tr> <tr><td>AVG 3</td><td>AVG 4</td></tr> <tr><td>Variation</td><td>Variation</td></tr> <tr><td>AVG 5</td><td>AVG 6</td></tr> <tr><td>AVG 7</td><td>AVG 8</td></tr> <tr><td>Variation</td><td>Variation</td></tr> <tr><td>AVG 9</td><td>AVG 10</td></tr> <tr><td>AVG 11</td><td>AVG 12</td></tr> <tr><td>Variation</td><td>Variation</td></tr> </table> <table style="width: 100%;"> <tr><td>Max AVG 1-11</td><td>Max AVG 2-12</td></tr> <tr><td>Min AVG 1-11</td><td>Min AVG 2-12</td></tr> <tr><td>Variation</td><td>Variation</td></tr> </table> | Max 1A-1D | Max 2A-2D | Min 1A-1D | Min 2A-2D | Variation | Variation | Max 3A-3D | Max 4A-4D | Min 3A-3D | Min 4A-4D | Variation | Variation | Max 5A-5D | Max 6A-6D | Min 5A-5D | Min 6A-6D | Variation | Variation | Max 7A-7D | Max 8A-8D | Min 7A-7D | Min 8A-8D | Variation | Variation | Max 9A-9D | Max 10A-10D | Min 9A-9D | Min 10A-10D | Variation | Variation | Max 11A-11D | Max 12A-12D | Min 11A-11D | Min 12A-12D | Variation | Variation | AVG 1 | AVG 2 | AVG 3 | AVG 4 | Variation | Variation | AVG 5 | AVG 6 | AVG 7 | AVG 8 | Variation | Variation | AVG 9 | AVG 10 | AVG 11 | AVG 12 | Variation | Variation | Max AVG 1-11 | Max AVG 2-12 | Min AVG 1-11 | Min AVG 2-12 | Variation | Variation |
| 1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2A           |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 1B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2B           |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
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| AVG 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AVG 2        |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
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| AVG 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AVG 4        |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
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| 5B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6B           |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 5C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6C           |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 5D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6D           |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| SUM 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SUM 6        |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| AVG 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AVG 6        |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 7A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 8A           |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 7B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 8B           |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 7C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 8C           |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 7D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 8D           |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| SUM 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SUM 8        |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| AVG 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AVG 8        |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 9A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10A          |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 9B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10B          |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 9C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10C          |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 9D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 10D          |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| SUM 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SUM 10       |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| AVG 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AVG 10       |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 11A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 12A          |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 11B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 12B          |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 11C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 12C          |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| 11D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 12D          |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| SUM 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | SUM 12       |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| AVG 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AVG 12       |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Max 1A-1D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Max 2A-2D    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Min 1A-1D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Min 2A-2D    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Variation    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Max 3A-3D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Max 4A-4D    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Min 3A-3D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Min 4A-4D    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Variation    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Max 5A-5D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Max 6A-6D    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Min 5A-5D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Min 6A-6D    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Variation    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Max 7A-7D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Max 8A-8D    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Min 7A-7D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Min 8A-8D    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Variation    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Max 9A-9D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Max 10A-10D  |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Min 9A-9D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Min 10A-10D  |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Variation    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Max 11A-11D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Max 12A-12D  |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Min 11A-11D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Min 12A-12D  |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Variation    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| AVG 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AVG 2        |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| AVG 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AVG 4        |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Variation    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| AVG 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AVG 6        |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| AVG 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AVG 8        |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Variation    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| AVG 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AVG 10       |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| AVG 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AVG 12       |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Variation    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Max AVG 1-11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Max AVG 2-12 |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Min AVG 1-11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Min AVG 2-12 |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |
| Variation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Variation    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |    |    |    |    |    |    |    |       |       |       |       |    |     |    |     |    |     |    |     |       |        |       |        |     |     |     |     |     |     |     |     |        |        |        |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |             |           |             |           |           |             |             |             |             |           |           |       |       |       |       |           |           |       |       |       |       |           |           |       |        |        |        |           |           |              |              |              |              |           |           |

## CYLINDER LINER PROJECTION MEASUREMENT CHECK SHEET

|               |          |                    |  |              |  |
|---------------|----------|--------------------|--|--------------|--|
| JOB NUMBER    |          | PART NAME          |  | DOCUMENT NO. |  |
| ENGINE MODEL  | C32      | PART NUMBER        |  | PUBL. DATE   |  |
| ENGINE S/N    |          | START DATE / TIME  |  | REVISION NO. |  |
| MACHINE MODEL | CAT 777E | FINISH DATE / TIME |  | PAGE         |  |

1. Pasang spacer plate gasket yang baru dan spacer plate harus bersih.
2. Pasang Liner pada Cyl Block tanpa O-Ring Seal dan Band.
3. Pasang Washer
4. Pasang semua bolt atau the six bolts disekeliling liner.  
Torque for bolts ... 95 N·m (70 lb ft)
5. Gunakan 8T-0455 Liner Projection Tool Group untuk mengukur proyeksi liner pada "A", "B", "C" dan "D".
6. Catat pengukuran untuk setiap cylinder.
7. Tambahkan empat bacaan untuk setiap cylinder. Bagi jumlahnya dengan empat untuk mencari rata-ratanya.

| Specifications                                    |                                           |
|---------------------------------------------------|-------------------------------------------|
| Liner Projection                                  | 0.025 to 0.152 mm (0.0010 to 0.0060 inch) |
| Maximum Variation In Each Liner                   | 0.051 mm (0.0020 inch)                    |
| Maximum Average Variation Between Adjacent Liners | 0.051 mm (0.0020 inch)                    |
| Maximum Variation Between All Liners              | 0.102 mm (0.0040 inch)                    |

**NOTE :**

*Jika proyeksi liner berubah di sekitar liner, putar liner ke posisi baru di dalam lubang.*

*Jika proyeksi liner tidak sesuai spesifikasi, pindahkan liner ke lubang lain.*

*Periksa bagian atas blok silinder.*

Spacer plate yang lebih tipis harus dipasang jika pengukuran proyeksi liner di bawah spesifikasi. Spacer plate juga harus dipasang jika ukuran proyeksi liner rendah dalam kisaran tertentu. Pelat ini lebih tipis 0,076 mm (0,0030 inci) dari pelat biasa. Pelat akan meningkatkan proyeksi liner. Ini akan meningkatkan penghancuran fire ring. Gunakan pelat pengatur jarak ini untuk mengimbangi proyeksi lapisan rendah yang kurang dari 0,076 mm (0,0030 inci). Gunakan pelat penjarak ini jika pemeriksaan dek atas tidak menemukan kerusakan yang dapat diukur langsung di bawah flange liner tetapi proyeksi liner rata-rata kurang dari 0,076 mm (0,0030 inci).

Jangan melebihi proyeksi liner maksimum 0,152 mm (0,0060 inci). Proyeksi liner yang berlebihan akan menyebabkan retaknya liner flange.

Jika proyeksi liner sudah benar, beri tanda sementara pada liner dan pelat spacer. Sisihkan liner

**NOTE :**

*Lihat Pembongkaran dan Perakitan, "Cylinder Liner - Install" untuk mengetahui prosedur pemasangan akhir yang benar dari liner.*

Saat Engine siap untuk perakitan akhir, item berikut harus dilumasi sebelum dipasang :

- \* O-ring seals
- \* Cylinder block
- \* Upper filler band

**NOTE :**

*Segera gunakan sabun cair dan / atau bersihkan oli mesin sebelum perakitan. Hindari mengoleskan sabun cair dan / atau oli mesin yang bersih ke seal terlalu dini.*

|                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <b>INSPECTION RESULT :</b>                                                                                                                                                                                                                                                                                                                                                                                  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |
| <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p><b>INSPECTION BY :</b></p> <br/><br/> <p>_____<br/>MECHANIC (MEC)</p> </div> <div style="width: 45%;"> <p><b>INSPECTION &amp; APPROVAL BY :</b></p> <br/><br/> <p>_____<br/>QUALITY ASSURANCE (QA)</p> </div> </div> <div style="text-align: right; margin-top: 20px;"> <p>_____<br/>GL/SPV PRODUCTION</p> </div> |  |  |



**PT SAPTAINDRA SEJATI**

**ENGINE**

**CAT ENGINE C32 ASSEMBLING**

**CHECK SHEET**

**ENGINE CAT C32**

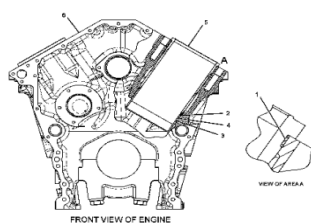
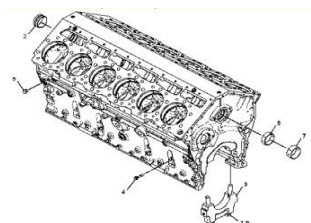
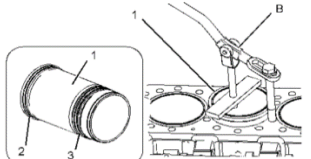
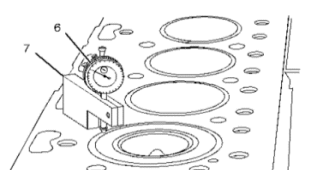
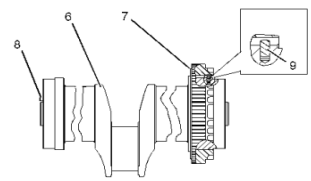
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**CAT777E**

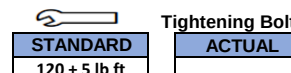
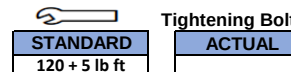
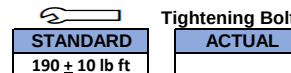
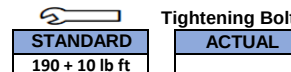
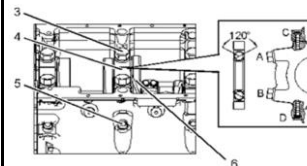
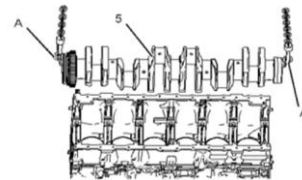
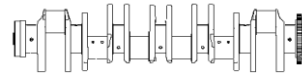
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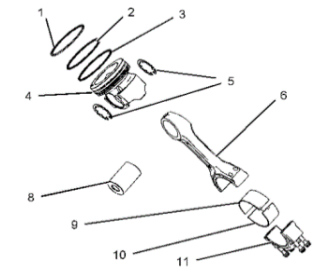
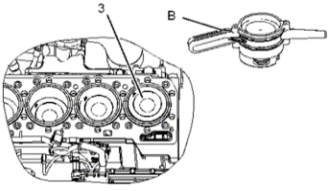
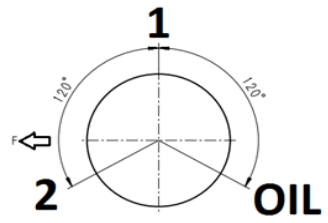
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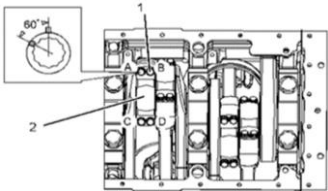
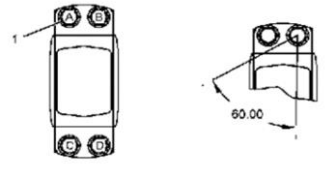
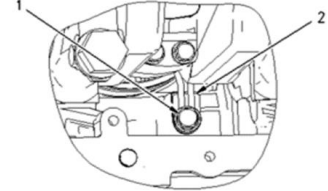


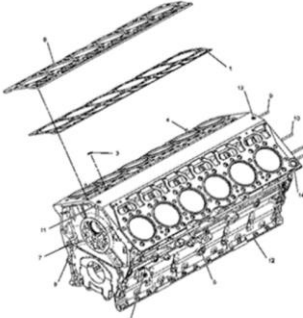
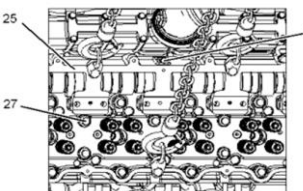
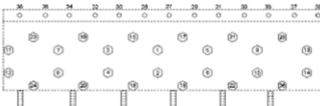
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|---------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------|------------------------|---------------------------------------------------|------------------------|--------------------------------------|------------------------|--|--|--|
| CAT ENGINE C32 ASSEMBLING                         |                                           | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | PUBLIC DATE    |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
| ENGINE CAT C32                                    |                                           | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | REVISION NO    |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
|                                                   |                                           | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | PAGE           |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
|                                                   |                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 OF 3         |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
| PROCESS                                           | START                                     | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
|                                                   | FINISH                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
| NO                                                | PART TO INSTALL OR ASSEMBLY               | CONDITION TOBE INSTALL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SPEC           | SIGN | SKETCH DRAWING OR PHOTOES                                                                                                                                                |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
|                                                   |                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                | MECH | GL/QA                                                                                                                                                                    |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
| 1                                                 | CYLINDER BLOCK                            | 1. Periksa apakah liner counterbore bersih sebelum memasang Liner.<br>2. Periksa apakah semua lubang oli bebas dari benda asing seperti serpihan atau kotoran.<br>3. Pastikan kondisi Cylinder Blocl sudah siap digunakan sepenuhnya.<br>4. Pastikan Sleeve No.2 sudah diganti.<br>5. Pastikan Bushing No.6 dan 7 sudah diganti.                                                                                                                                                                                                                                                                                                                                                                                        |                |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
|                                                   |                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |      |      |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
| 2                                                 | CYLINDER LINER                            | 1. Check cylinder liner projection. Referensi ke Testing and 'Adjusting,' "Cylinder Liner Projection - Inspect<br>2. Pasang Seal O-Ring Liner (3) pada Cyl liner (1)<br>3. Gunakan Tooling (C) pada permukaan lubang liner untuk 'memasukkan Liner pada lubang Cyl Block berhati hati saat terhadap O-ring saat Pemasangan.<br>4. Celupkan filler band (2) ke dalam oli mesin yang bersih sebentar. Pasang segera pita pengisi pada liner silinder.<br><br>1 Tempatkan Body Gauge(7) di atas pelat untuk blok silinder. Titik indikator harus berada di liner flange. Baca indikator dial untuk mengetahui jumlah proyeksi liner.Periksa proyeksi di empat lokasi (setiap 90 derajat) di sekitar setiap liner silinder. |                |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
|                                                   |                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |      |   |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
|                                                   |                                           | <table border="1"> <thead> <tr> <th colspan="2">Specifications</th> </tr> </thead> <tbody> <tr> <td>Liner Projection</td> <td>0.025 to 0.152 mm (0.0010 to 0.0060 inch)</td> </tr> <tr> <td>Maximum Variation In Each Liner</td> <td>0.051 mm (0.0020 inch)</td> </tr> <tr> <td>Maximum Average Variation Between Adjacent Liners</td> <td>0.051 mm (0.0020 inch)</td> </tr> <tr> <td>Maximum Variation Between All Liners</td> <td>0.102 mm (0.0040 inch)</td> </tr> </tbody> </table>                                                                                                                                                                                                                                 | Specifications |      | Liner Projection                                                                                                                                                         | 0.025 to 0.152 mm (0.0010 to 0.0060 inch) | Maximum Variation In Each Liner | 0.051 mm (0.0020 inch) | Maximum Average Variation Between Adjacent Liners | 0.051 mm (0.0020 inch) | Maximum Variation Between All Liners | 0.102 mm (0.0040 inch) |  |  |  |
| Specifications                                    |                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
| Liner Projection                                  | 0.025 to 0.152 mm (0.0010 to 0.0060 inch) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
| Maximum Variation In Each Liner                   | 0.051 mm (0.0020 inch)                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
| Maximum Average Variation Between Adjacent Liners | 0.051 mm (0.0020 inch)                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
| Maximum Variation Between All Liners              | 0.102 mm (0.0040 inch)                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
| 3                                                 | CRANKSHAFT                                | 1 Pasang dowel (8) pada crankshaft.<br>Protrusion of dowel (8) dari Permukaan crankshaft.<br><br>2 Pasang dowel (9) pada crankshaft.<br>Protrusion of dowel (9) dari permukaan crankshaft .<br><br>3 Pasang Crangshaft gear (7) pada crankshaft dengan (keyway)jalur pasak sejajar dengan batang kayu (9).<br>Timing Mark "V" dari gear harus menjauhi crankshaft (6)<br><br><b>NOTE :</b><br><b>Pasang Main Bearing Cap saat kering dan check clearance ,</b><br><b>Oleskan oli mesin yang bersih pada bagian depan main bering</b><br><b>saat perakitan akhir.</b>                                                                                                                                                    |                |      |                                                                                                                                                                          |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |
|                                                   |                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                |      |                                                                                      |                                           |                                 |                        |                                                   |                        |                                      |                        |  |  |  |

| CHECK SHEET               |                  | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DOCUMENT NO |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
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| CAT ENGINE C32 ASSEMBLING |                  | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PUBLIC DATE |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
| ENGINE CAT C32            |                  | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | REVISION NO |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
|                           |                  | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PAGE        |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
|                           |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1 OF 3      |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
| PROCESS                   | START<br>FINISH  | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
| 4                         | CRANKSHAFT       | <p>4 Pasang bagian atas main bearing cap dengan Cylinder Block.</p> <p><b>NOTE :</b><br/>Pastikan main bearing dipasang sehingga tab bearing cocok dengan Cyl blockr. Bagian atas main bearing memiliki alur oli dan lubang oli</p> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>439-3939</td> <td>Link Bracket</td> <td>2</td> </tr> <tr> <td>D</td> <td>8T-5096</td> <td>Dial Indicator</td> <td>1</td> </tr> <tr> <td>E</td> <td>8T-3052</td> <td>Degree Wheel</td> <td>1</td> </tr> </tbody> </table> <p>5 Pasang Tooling (A) dan perangkat pengangkat yang sesuai di setiap ujung crankshaft (6). Berat crankshaft (6) kira - kira 188 kg (415 lb).</p> <p>6 Pasang crankshaft (6) dengan Timing Mark "V" pada crankshaft gear sejajar dengan Timing Mark "V" pada cluster Gear.</p> <p>7 Pasang Thrust Plate di setiap sisi main bearing No. 4</p> <p><b>NOTE :</b><br/>Pasang Thrust Plate dengan tulisan " Block Side" ke arah Cylinder Block.</p> <p>8 Pasang bagian bawah Main Bearing dan Cap dgn Crankshaft.</p> <p>9 Posisikan main cap (4) pada crankshaft.</p> <p><b>NOTE:</b><br/>Pastikan angka pada main cap bearing sesuai dengan angka pada cylinder block. Pastikan juga bahwa "FRONT" pada main caps bearing dipasang ke arah depan cylinder block.</p> |             | Required Tools |  |  |  | Tool | Part Number | Part Description | Qty | A | 439-3939 | Link Bracket | 2 | D | 8T-5096 | Dial Indicator | 1 | E | 8T-3052 | Degree Wheel | 1 |
| Required Tools            |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
| Tool                      | Part Number      | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Qty         |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
| A                         | 439-3939         | Link Bracket                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2           |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
| D                         | 8T-5096          | Dial Indicator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1           |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
| E                         | 8T-3052          | Degree Wheel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1           |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |
| 5                         | MAIN BEARING CAP | <p>10 Lumasi ulir baut (3) dengan oli mesin yang bersih. Pasang baut dan washernya.</p> <p>11 Lumasi ulir baut (5) dan Seal washer dengan oli mesin yang bersih, Pasang bolt (5) dan Washer.</p> <p><b>Note :</b><br/>Inspect washer sebelum dimasukkan ke bolt.</p> <p>12 Kencangkan Bolt (3) dan Bolt (5) untuk main bearing cap</p> <p>ikuti langkah langkah sbb :</p> <p>a Kencangkan Bolt (A) Torque : 258.1 ± 14 Nm Note : Bolt (A) ada pada bearing tab samping main bearing cap. Kencangkan Bolt (B) Torque : 258.1 ± 14 Nm</p> <p>258.1 ± 14 Nm<br/>(190 ± 10 lb ft)</p> <p>b</p> <p><b>Note :</b><br/>Bolt (B) berada di sisi berlawanan dari tab main bearing cap .</p> <p>Kencangkan Bolt (B) 120 ± 5 derajat (2flat)</p> <p>120 ± 5 derajat</p> <p>c</p> <p><b>Note :</b><br/>Jika Tool (E) tidak dapat digunakan, pakai marker untuk menandai bolt dan main cap bearing.</p> <p>Kencangkan Bolt (A) 120 ± 5 derajat (2flat)</p> <p>120 ± 5 derajat</p> <p>d</p> <p><b>Note : Jika Tool (E) tidak dapat digunakan, pakai marker</b></p>                                                                                                                                                                                                                                                                                                                                                                                 |             |                |  |  |  |      |             |                  |     |   |          |              |   |   |         |                |   |   |         |              |   |

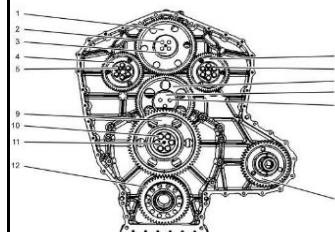
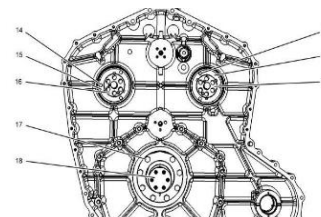
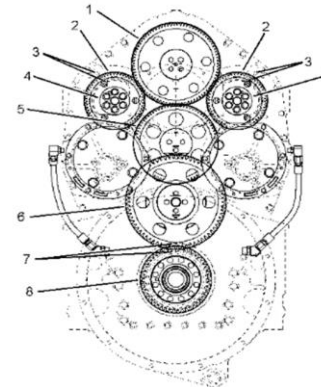


| CHECK SHEET               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | WO NUMBER                                                                                                       | DOCUMENT NO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                  |  |
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| CAT ENGINE C32 ASSEMBLING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | COMP.MODEL                                                                                                      | PUBLIC DATE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                  |  |
| ENGINE CAT C32            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | COMP.S/N                                                                                                        | REVISION NO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                  |  |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | UNIT MODEL                                                                                                      | PAGE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |        |                  |  |          |        |                  |  |
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| PROCESS                   | START<br>FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ASSEMBLY BY                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |        |                  |  |          |        |                  |  |
| 6                         | <p>e Kencangkan Bolt (D) Torque <math>80 \pm 10</math> Nm (<math>59 \pm 7</math> lb ft).</p> <p><b>Note :</b><br/><b>Bolt (D) tidak dapat digunakan, pakai marker untuk menandai bolt dan main cap bearing.</b></p> <p>f Kencangkan Bolt (C) Torque <math>80 \pm 10</math> Nm (<math>59 \pm 7</math> lb ft)</p> <p><b>Note :</b><br/><b>Bolt (C) ada pada bearing tab samping main bearing cap.</b></p> <p>g Putar Crankshaft untuk memastikan bisa berputar bebas</p> <p>Gunakan Tooling (D) untuk memeriksa End Play Crankshaft. Pastikan bahwa Tooling (D) berhadapan dengan permukaan mesin. Pergerakan akhir End Play dibatasi oleh Thrust Plat dari main bearing No. 4 (Center).<br/>crankshaft end play (New Thrust Plat) 0.10 to 0.60 mm.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p><math>80 \pm 10</math> Nm (<math>59 \pm 7</math> lb ft)</p> <p>0.10 to 0.60 mm<br/>0.004 to 0.024 (Inch)</p> | <p> <b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td><math>59 \pm 7</math> lb ft</td> <td></td> </tr> </table> <p> <b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td><math>59 \pm 7</math> lb ft</td> <td></td> </tr> </table> | STANDARD | ACTUAL | $59 \pm 7$ lb ft |  | STANDARD | ACTUAL | $59 \pm 7$ lb ft |  |
| STANDARD                  | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |        |                  |  |          |        |                  |  |
| $59 \pm 7$ lb ft          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |        |                  |  |          |        |                  |  |
| STANDARD                  | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |        |                  |  |          |        |                  |  |
| $59 \pm 7$ lb ft          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |        |                  |  |          |        |                  |  |
| 7                         | <p>1 Pasang Spring untuk ring kontrol oli di alur ring oli di unit bodi piston (3)</p> <p>2 Pasang Ring kontrol oli (3) di atas spring. Posisikan Ring kontrol oli agar jaraknya 180 derajat dari sambungan di spring. Pasang Ring Piston kontrol oli pada piston menggunakan alat Ring Piston Compression.</p> <p>Gunakan alat expander untuk memasang intermediate ring piston (2) dengan sisi yang bertanda "UP-2". dan ring piston atas (1) dengan sisi yang memiliki identifikasi "UP-1" ke arah atas piston.</p> <p>Posisikan perakitan piston (4) dengan Conrod (6). Oleskan oli mesin bersih ke pin Conrod (8) dan pasang pin conrod. Pasang penahan pin/retainer (5) dengan tang lancip.</p> <p>Oleskan oli mesin bersih ke lubang liner silinder, ke ring piston, dan ke permukaan luar piston.</p> <p><b>NOTE :</b><br/><b>Pastikan Connecting Roda dan Piston dipasang di Cylinder yang benar.</b></p> <p>Putar Crankshaft sampai jurnal Conrod berada di tengah bawah.</p> <p>6</p> <p>7 Posisikan ring piston (1), (2), dan (3) sehingga jaraknya 120 derajat satu sama lain.</p> <p>8 Gunakan Tool (B) untuk memasang piston (3) Conrod masuk ke Liner dan Codrod Jurnal.</p> <p>9 Posisikan Connection rod lain pada Crankshaft. Pasang Conrod bearing cap (2) pada Conrod.</p> |                                                                                                                 |                                                                                                                                                                                                                                                |          |        |                  |  |          |        |                  |  |

| CHECK SHEET               |                    | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DOCUMENT NO                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |
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| CAT ENGINE C32 ASSEMBLING |                    | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PUBLIC DATE                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |
| ENGINE CAT C32            |                    | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | REVISION NO                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |
|                           |                    | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PAGE                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |
|                           |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1 OF 3                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |
| PROCESS                   | START<br>FINISH    | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |
|                           |                    | <p><b>NOTE :</b><br/><i>Pastikan nomor pada sisi Main bearing Cap sama dengan nomer yang ada pada conrod. Tab dari bearing conrod cap berada di sisi yang berlawanan dengan Piston Cooling jet.</i></p> <p>10 Lumasi ulir baut (1) dengan oli mesin yang bersih. Pasang bautnya.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |
| 8                         | MAIN CAP           | <p>11 Kencangkan Bolt (1) ikuti tahap - tahap berikut :</p> <p>a Kencangkan Bolt (A) and Bolt (C) torque of <math>70 \pm 4</math> N-m (<math>52 \pm 3</math> lb ft).</p> <p>b Putar Bolt (B) and Bolt (D) <math>70 \pm 4</math> N-m (<math>52 \pm 3</math> lb ft) tambah <math>60 \pm 5</math> derajat</p> <p><b>NOTE :</b><br/><i>Jika Tooling (E) tidak digunakan, beri tanda indeks pada setiap baut dan Conrod Cap</i></p> <p>d Kencangkan Bolt (A) and Bolt (C) lagi, torque of <math>70 \pm 4</math> N-m (<math>52 \pm 3</math> lb ft).</p> <p>e Putar Bolt (A) and Bolt (C) Tambah <math>60 \pm 5</math> derajat (1/6 turn)</p> <p><b>NOTE :</b><br/><i>Jika Tooling (E) tidak digunakan, beri tanda indeks pada setiap baut dan Conrod Cap.</i></p> <p>f Putar Crankshaft untuk memastikan bisa berputar bebas.</p> <p><b>NOTE :</b><br/><i>Pasang Connecting rod bearing kering tanpa oli saat check End Play menggunakan Plastic Gauge.</i></p> | <p><math>70 \pm 4</math> N-m (<math>52 \pm 3</math> lb ft).<br/><math>60 \pm 5</math> derajat</p> <p><math>70 \pm 4</math> N-m (<math>52 \pm 3</math> lb ft).<br/><math>60 \pm 5</math> derajat</p> |   <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td><math>70 \pm 4</math> lb ft</td> <td></td> </tr> </table> <p>Tambah 60 Derajat <input type="text"/></p> | STANDARD         | ACTUAL | $70 \pm 4$ lb ft                                                 |                                                                                      |
| STANDARD                  | ACTUAL             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |
| $70 \pm 4$ lb ft          |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |
| 9                         | PISTON COOLING JET | <p>1 Putar Crankshaft untuk mendapatkan akses ke Piston Cooling Jet.</p> <p>2 Posisikan Piston Cooling Jet (2) di blok silinder. Pasang baut (1). Kencangkan baut hingga torsi <math>25 \pm 6</math> N · m <math>25 \pm 6</math> N.m.</p> <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td><math>18 \pm 4</math> lb ft</td> <td></td> </tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | STANDARD                                                                                                                                                                                            | ACTUAL                                                                                                                                                                                                                                                                                                                                                                               | $18 \pm 4$ lb ft |        | <p><math>25 \pm 6</math> N · m (<math>18 \pm 4</math> lb ft)</p> |  |
| STANDARD                  | ACTUAL             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |
| $18 \pm 4$ lb ft          |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                      |                  |        |                                                                  |                                                                                      |

| CHECK SHEET               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | WO NUMBER                                                                | DOCUMENT NO                                                                                                                                                                                                                                                                                                                              |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
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| CAT ENGINE C32 ASSEMBLING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | COMP.MODEL                                                               | PUBLIC DATE                                                                                                                                                                                                                                                                                                                              |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| ENGINE CAT C32            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | COMP.S/N                                                                 | REVISION NO                                                                                                                                                                                                                                                                                                                              |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | UNIT MODEL                                                               | PAGE                                                                                                                                                                                                                                                                                                                                     |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                          | 1 OF 3                                                                                                                                                                                                                                                                                                                                   |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| PROCESS                   | START<br>FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ASSEMBLY BY                                                              |                                                                                                                                                                                                                                                                                                                                          |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| 10 CYLINDER HEAD          | <p>1 Bersihkan permukaan Cylinder block dari kotoran, karat dan sebagainya, gunakan Tinner atau CRC, begitu juga Permukaan dari Cylinder Head.</p> <p>2 Gunakan Part Book untuk melihat Part-part apa yang harus terpasang diantara Cylinder head dan Cylinder Block seperti O-Ring, Gasket dan spacer plate.</p> <p>Setelah Cylinder Head dan Cylinder Block bersih pasang O-Ring No.3, Pasang Gasket Plate dan Spacer Plate pada Cylinder Blok.</p> <p>3</p> <p>Pasang Tool (A) dan perangkat pengangkat yang sesuai ke Cylinder Head (25). Pasang Tooling (B) ke sisi Cylinder Head (25) yang dekat dengan turbocharger. Berat kepala silinder (25) kira-kira 365 kg (800 lb).</p> <p>4</p> <p>Posisikan Cylinder Head (25) pada pin dowel di Cylinder Block.</p> <p>5</p> <p><b>Note :</b><br/><b>Posisikan Cylinder Head (25) pada pin dowel di Cylinder Block.</b></p> <p><b>Kencangkan baut dengan urutan langkah berikut:</b></p> <p>6 Passang baut (26) dan baut (27) sesuai dengan prosedur berikut :</p> <p>a Dalam urutan numerik, kencangkan Bolt (1) hingga Bolt (26) hingga torsi <math>270 \pm 15 \text{ N} \cdot \text{m}</math> (<math>199 \pm 11 \text{ lb ft}</math>).</p> | <p>270 <math>\pm</math> 15 N · m<br/>(199 <math>\pm</math> 11 lb ft)</p> |   <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td>199 <math>\pm</math> 11 lb ft</td> <td></td> </tr> </table> | STANDARD | ACTUAL | 199 $\pm$ 11 lb ft |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
|                           | STANDARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ACTUAL                                                                   |                                                                                                                                                                                                                                                                                                                                          |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| 199 $\pm$ 11 lb ft        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                          |                                                                                                                                                                                                                                                                                                                                          |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| CYLINDER HEAD             | <p>b Dalam urutan numerik, kencangkan Bolt (1) melalui Bolt (26) ke torsi <math>450 \pm 15 \text{ N} \cdot \text{m}</math> (<math>332 \pm 11 \text{ lb ft}</math>).</p> <p>c Dalam urutan numerik, kencangkan lagi Bolt (1) melalui Bolt (26) ke torsi <math>450 \pm 15 \text{ N} \cdot \text{m}</math> (<math>332 \pm 11 \text{ lb ft}</math>).</p> <p>d Dalam urutan numerik, kencangkan Bolt (27) melalui Bolt (39) ke torsi <math>45 \pm 7 \text{ N} \cdot \text{m}</math> (<math>33 \pm 5 \text{ lb ft}</math>).</p> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>138-7573</td> <td>Link Bracket</td> <td>3</td> </tr> <tr> <td>B</td> <td>1U-9200</td> <td>Lever Puller Hoist</td> <td>1</td> </tr> <tr> <td>C</td> <td>-</td> <td>Loctite Aviation Gasket Sealant</td> <td>-</td> </tr> <tr> <td>E</td> <td>-</td> <td>Loctite Graphite-50 Anti-Seize</td> <td>-</td> </tr> </tbody> </table>                                                                                                                                    | Required Tools                                                           |                                                                                                                                                                                                                                                                                                                                          |          |        | Tool               | Part Number | Part Description | Qty | A | 138-7573 | Link Bracket | 3 | B | 1U-9200 | Lever Puller Hoist | 1 | C | - | Loctite Aviation Gasket Sealant | - | E | - | Loctite Graphite-50 Anti-Seize | - | <p>450 <math>\pm</math> 15 N · m<br/>(332 <math>\pm</math> 11 lb ft)</p> <p>450 <math>\pm</math> 15 N · m<br/>(33 <math>\pm</math> 5 lb ft)</p> |  <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td>332 <math>\pm</math> 11 lb ft</td> <td></td> </tr> </table> <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td>33 <math>\pm</math> 5 lb ft</td> <td></td> </tr> </table> | STANDARD | ACTUAL | 332 $\pm$ 11 lb ft |  | STANDARD | ACTUAL | 33 $\pm$ 5 lb ft |  |
| Required Tools            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                          |                                                                                                                                                                                                                                                                                                                                          |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| Tool                      | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Part Description                                                         | Qty                                                                                                                                                                                                                                                                                                                                      |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| A                         | 138-7573                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Link Bracket                                                             | 3                                                                                                                                                                                                                                                                                                                                        |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| B                         | 1U-9200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Lever Puller Hoist                                                       | 1                                                                                                                                                                                                                                                                                                                                        |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| C                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Loctite Aviation Gasket Sealant                                          | -                                                                                                                                                                                                                                                                                                                                        |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| E                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Loctite Graphite-50 Anti-Seize                                           | -                                                                                                                                                                                                                                                                                                                                        |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| STANDARD                  | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                          |                                                                                                                                                                                                                                                                                                                                          |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| 332 $\pm$ 11 lb ft        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                          |                                                                                                                                                                                                                                                                                                                                          |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| STANDARD                  | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                          |                                                                                                                                                                                                                                                                                                                                          |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |
| 33 $\pm$ 5 lb ft          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                          |                                                                                                                                                                                                                                                                                                                                          |          |        |                    |             |                  |     |   |          |              |   |   |         |                    |   |   |   |                                 |   |   |   |                                |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |          |        |                    |  |          |        |                  |  |

| CHECK SHEET               |                    | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DOCUMENT NO |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
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| CAT ENGINE C32 ASSEMBLING |                    | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PUBLIC DATE |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
| ENGINE CAT C32            |                    | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | REVISION NO |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
|                           |                    | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PAGE        |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
|                           |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | REV 01      |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
|                           |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1 OF 3      |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
| PROCESS                   | START<br>FINISH    | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
| 11                        | REAR<br>GEAR GROUP | <p><b>Rear gear group</b></p> <p>(1) Cluster idler gear</p> <p>(2) Camshaft gear</p> <p>(3) Timing alignment marks</p> <p>(4) Camshaft bolts</p> <p>(5) Idler gear assembly</p> <p>(6) Cluster idler gear</p> <p>(7) Timing alignment marks</p> <p>(8) Crankshaft gear</p> <p>Posisikan piston nomor satu di TOP Center Posision (TC).<br/>Sejajarkan tanda "V" pada cam gear (2) dengan tanda di rumah flywheel.</p> <p><b>NOTE :</b><br/><i>Jika satu atau lebih Gear perlu dilepas untuk diperbaiki, lihat Pembongkaran dan Perakitan, "Kelompok Gear (Belakang) - Lepas" untuk melepas Gear dengan benar. Lihat Pembongkaran dan Perakitan, "Grup Gear (Belakang) - Referensi" untuk memasang Gear dengan benar.</i></p> <p>Periksa Gear apakah ada keausan atau kerusakan. Jika Gear aus atau rusak, gunakan suku cadang baru untuk penggantian.</p> |             |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
| 12                        |                    | <p><b>C32 Installation Procedure</b></p> <p>Table 2</p> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>1P-0520</td> <td>Driver Gp</td> <td>1</td> </tr> <tr> <td>B</td> <td>-</td> <td>Guide Stud<br/>5/8 - 18 by 12 inch</td> <td>1</td> </tr> </tbody> </table> <p><b>CATATAN</b><br/>Jaga kebersihan semua<br/>Componen dari Kontaminasi</p> <p><b>Note:</b><br/><i>Oleskan oli Engine yang bersih ke bearing dan shaft sebelum dipasang pada Engine.</i></p> <p>1 Pasang O-Ring Seal dan Sealing pelat pada Cylinder.Head.</p> <p>2 Pasang adapter assemblies (16), thrust plates (15), dan bolts (14).</p> <p>3 Posisikan shaft assembly (18) dan pasang bolts (17).</p>                        |             | Required Tools |  |  |  | Tool | Part Number | Part Description | Qty | A | 1P-0520 | Driver Gp | 1 | B | - | Guide Stud<br>5/8 - 18 by 12 inch | 1 |
| Required Tools            |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
| Tool                      | Part Number        | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Qty         |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
| A                         | 1P-0520            | Driver Gp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1           |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |
| B                         | -                  | Guide Stud<br>5/8 - 18 by 12 inch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1           |                |  |  |  |      |             |                  |     |   |         |           |   |   |   |                                   |   |



| CHECK SHEET               |                 | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | DOCUMENT NO |
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| CAT ENGINE C32 ASSEMBLING |                 | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PUBLIC DATE |
| ENGINE CAT C32            |                 | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | REVISION NO |
|                           |                 | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PAGE        |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | REV 01      |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 OF 3      |
| PROCESS                   | START<br>FINISH | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |
| 13                        | REAR GEAR GROUP | <p><b>Catatan:</b><br/>Sejajarkan tanda "Timing Mark" pada Gear (4) dgn "Timing Mark" pada Rear Housing dan Dowel Gear (4) dengan adaptor.</p> <p>4 Posisikan Gear (4) dan pasang baut (5).</p> <p>5 Pasang Tooling (B) pada Crankshaft, Pasang Gear (12).</p> <p>6 Jika posisi TDC sudah ditandai pada Vibratio Damper, lanjutkan ke Langkah 7.</p> <p>a Pasang flywheel Housing dan flywheel untuk sementara. Dua baut di flywheel housing dan flywheel sudah cukup.</p> <p>b Posisikan piston nomor satu di posisi Top Center TC. Engine harus berada di posisi Top Center untuk memasang Gear Group belakang. Lihat Petunjuk / Service Manual Testing &amp; Adjusting "Menemukan Posisi Top Center untuk Piston No. 1".</p> <p>c Dengan piston nomor satu di posisi Top Center (TC), buat tanda marking pada engine, dan letakkan Tanda Alignment pada Vibration Damper.</p> <p>d Lepaskan flywheel dan flywheel housing.</p> <p>7 Pasang dua nut pada dua baut untuk flywheel (19). Nut akan memberikan jarak saat baut dipasang. Pasang dua baut untuk Gear (19) di unit Gear (12).</p> <p>8 Gunakan Perkakas (A) dan pers yang sesuai untuk memasang bantalan pada rakitan roda gigi (1). Pasang bantalan hingga kedalaman <math>2,6 \pm 0,5</math> mm (<math>0,10 \pm 0,02</math> inci). Bantalan harus dikerjakan setelah dipasang ke roda gigi. Lihat Pedoman Reuse and Salvage Guidelines, , SEBF9244 untuk mengetahui dimensi Engine dan persyaratan permukaan yang benar.</p> <p>9 Pasang Stub Shaft assy (3), Gear (1), dan baut (2). Kencangkan baut (2) hingga torsi <math>55 \pm 10</math> N · m (<math>41 \pm 7</math> lb ft)</p> <p>10 Gunakan Tool (A) dan prees yang sesuai untuk memasang bearing pada Gear (6). Pasang bearing hingga kedalaman <math>2,6 \pm 0,5</math> mm (<math>0,10 \pm 0,02</math> inci). Bearing harus dikerjakan setelah dipasang pada gear. Lihat Pedoman Reuse and Salvage Guidelines, SEBF9244 untuk mengetahui machining dimensi dan persyaratan permukaan yang benar.</p> <p>11 Install stub shaft assembly (7), gear assembly (6) dan bolts (8).</p> |             |
|                           |                 | <p><math>2.5 \pm 0.5</math> mm<br/>(<math>0.10 \pm 0.02</math> inch)</p> <p><math>55 \pm 10</math> N.m<br/>(<math>41 \pm 7</math> lb.ft)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |
|                           |                 | <p>12 Gunakan Tool (A) dan press yang sesuai untuk memasang bearing pada gear (9). Pasang bearing hingga kedalaman <math>3,2 \pm 0,5</math> mm (<math>0,13 \pm 0,02</math> inci). Bearing harus dikerjakan setelah dipasang ke Gear. Lihat Pedoman Reuse and Salvage Guidelines,, SEBF9244 untuk mengetahui machining dimensi persyaratan permukaan yang benar.</p> <p><math>3.2 \pm 0.5</math> mm<br/>(<math>0.13 \pm 0.02</math> inch)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |

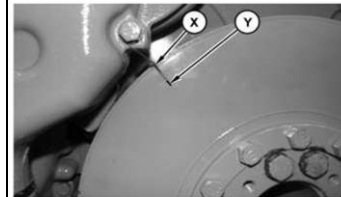
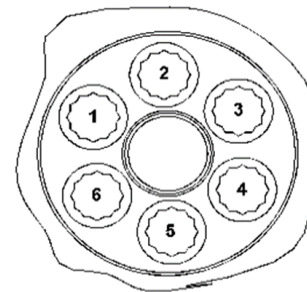
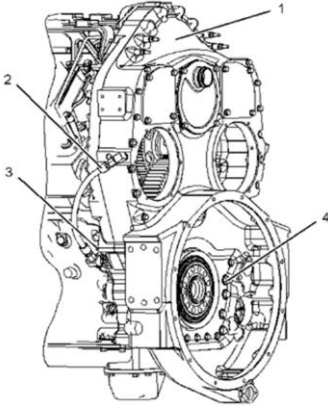


Illustration 8  
(X) Marker  
(Y) Alignment Mark

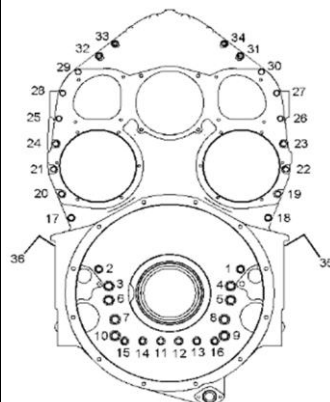
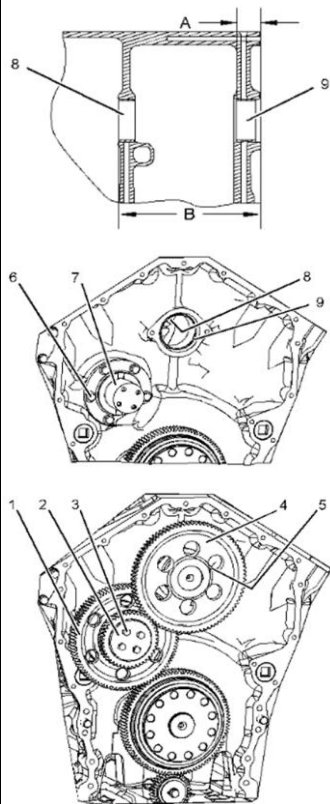
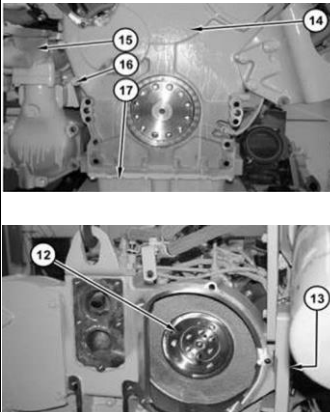


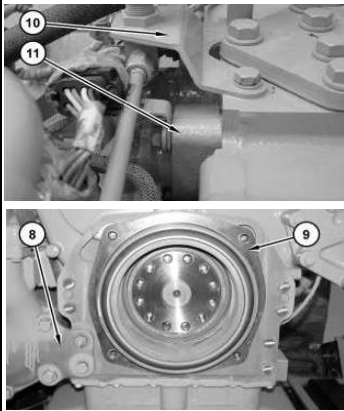
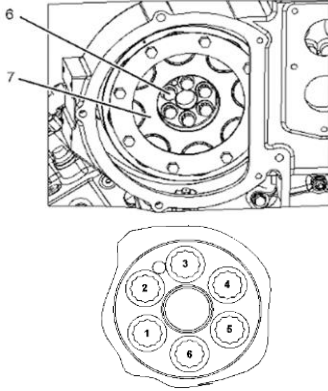
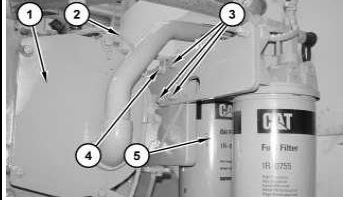
| STANDARD     | Tightening Bolt |
|--------------|-----------------|
| 41 ± 7 lb ft | ACTUAL          |

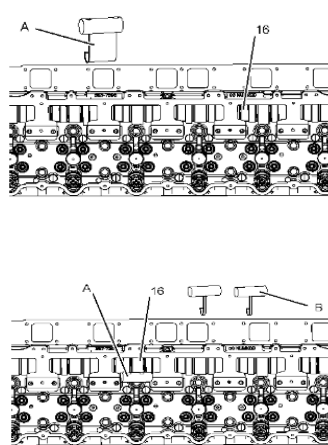
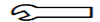
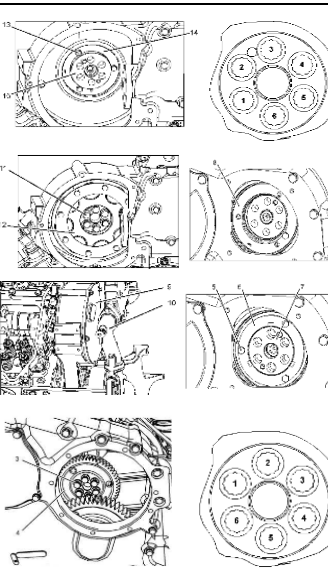


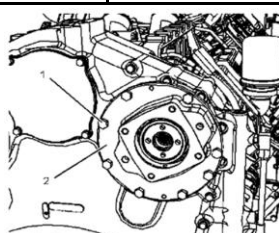
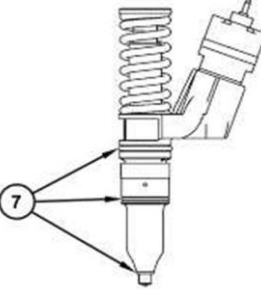
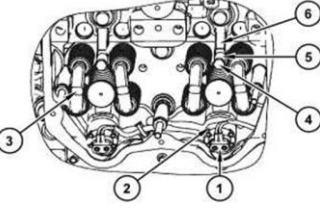
| CHECK SHEET               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | WO NUMBER                                                                | DOCUMENT NO                                                                                                                                                                                                                                      |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
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| CAT ENGINE C32 ASSEMBLING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | COMP.MODEL                                                               | PUBLIC DATE                                                                                                                                                                                                                                      |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| ENGINE CAT C32            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | COMP.S/N                                                                 | REVISION NO                                                                                                                                                                                                                                      |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
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| PROCESS                   | START<br>FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ASSEMBLY BY                                                              |                                                                                                                                                                                                                                                  |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| 14                        | <p>13 Pasang retainer assembly (11), O-ring seal, and bolts (10)</p> <p><b>Note:</b><br/>Pastikan piston nomor satu berada di posisi Top Center (TC). Pastikan tanda "Timing Mark" pada Gear (4) sejajar dengan tanda "Timing Mark" pada Housin Rear.</p> <p><b>Note:</b><br/>Jangan gunakan tanda "alignment marks" yang ada pada Gear (9) dan (12). Melakukannya akan membuat kesalahan. Gunakan langkah-langkah berikut untuk menyelaraskan gigi (9).</p> <p>14 Tahan Gear Assy (6) ke arah putaran Engine normal untuk melepaskan Backlast/reaksi balik. Tempatkan Gear Assy (9) pada shaft. Theet gear (9) akan terhubung dengan Gear (12) terlebih dahulu. Gigi Gear (9) akan berhubungan dengan gigi pada Gear (6) selanjutnya. Jika gigi pada Gear (9) tidak terpasang dengan benar, putar rakitan roda gigi ke gigi berikutnya, dan coba lagi.</p> <p>15 Putar Engine ke arah berlawanan dari putaran engine normal selama kira-kira 45 derajat. Putar mesin ke arah putaran engine normal sampai tanda kesejajaran pada Vibration Damper dan penanda pada Engine sejajar. Pastikan bahwa tanda Timing Mark Gear (4) sejajar dengan tanda pada rear housing.</p> <p>16 Kencangkan baut pada Gear Assy (4) hingga torsi <math>240 \pm 40 \text{ N} \cdot \text{m}</math> (<math>177 \pm 30 \text{ lb ft}</math>). Kencangkan baut dengan urutan berikut: 1, 4, 2, 5, 3, 6, 1, 4.</p> <p>17 Pasang Gear Assy (13) dan Thrush washer.</p> | <p>240 <math>\pm</math> 40 N · m<br/>(177 <math>\pm</math> 30 lb ft)</p> | <p> Tightening Bolt</p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td>177 <math>\pm</math> 30 lb ft</td> <td></td> </tr> </table> | STANDARD | ACTUAL | 177 $\pm$ 30 lb ft |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| STANDARD                  | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                          |                                                                                                                                                                                                                                                  |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| 177 $\pm$ 30 lb ft        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                          |                                                                                                                                                                                                                                                  |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| 15                        | <p><b>FLYWHEEL HOUSING</b></p> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>138-7574</td> <td>Link Bracket</td> <td>2</td> </tr> <tr> <td>B</td> <td>-</td> <td>Guide Stud (1/2 - 13 NC by 5 inch)</td> <td>2</td> </tr> </tbody> </table> <p>1 Pasang Tool B pada Cylinder Block</p> <p>2 Pasang Tooling (A) dan perangkat angkat yang sesuai ke flywheel Housing (1). Berat rumah flywheel (1) kira-kira 145 kg (320 lb).</p> <p>3 Posisikan Gasket dan flywheel Housing (1) pada blok silinder. Pasang baut (4) dan baut (3).</p> <p>4 Lepaskan Tool (B) dan pasang baut yang tersisa (4).</p> <p>5 Kencangkan bautnya, ikuti tahap sebagai berikut :</p> <p>a Dalam urutan numerik, kencangkan Baut 1 hingga Baut 2 dengan torsi <math>55 \pm 10 \text{ N} \cdot \text{m}</math> (<math>41 \pm 7 \text{ lb ft}</math>).</p> <p>b Dalam urutan numerik, kencangkan Bolt 3 hingga Bolt 10 dengan torsi <math>135 \pm 20 \text{ N} \cdot \text{m}</math> (<math>100 \pm 15 \text{ lb ft}</math>).</p>                                                                                                                                                                                                                                                                                   | Required Tools                                                           |                                                                                                                                                                                                                                                  |          |        | Tool               | Part Number | Part Description | Qty | A | 138-7574 | Link Bracket | 2 | B | - | Guide Stud (1/2 - 13 NC by 5 inch) | 2 | <p>55 <math>\pm</math> 10 N · m<br/>(41 <math>\pm</math> 7 lb ft)</p> <p>135 <math>\pm</math> 20 N · m<br/>(100 <math>\pm</math> 15 lb ft).</p> | <p></p> <p> Tightening Bolt</p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td>41 <math>\pm</math> 7 lb ft</td> <td></td> </tr> </table> <p> Tightening Bolt</p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td>100 <math>\pm</math> 15 lb ft</td> <td></td> </tr> </table> | STANDARD | ACTUAL | 41 $\pm$ 7 lb ft |  | STANDARD | ACTUAL | 100 $\pm$ 15 lb ft |  |
| Required Tools            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                          |                                                                                                                                                                                                                                                  |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| Tool                      | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Part Description                                                         | Qty                                                                                                                                                                                                                                              |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| A                         | 138-7574                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Link Bracket                                                             | 2                                                                                                                                                                                                                                                |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| B                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Guide Stud (1/2 - 13 NC by 5 inch)                                       | 2                                                                                                                                                                                                                                                |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| STANDARD                  | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                          |                                                                                                                                                                                                                                                  |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| 41 $\pm$ 7 lb ft          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                          |                                                                                                                                                                                                                                                  |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| STANDARD                  | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                          |                                                                                                                                                                                                                                                  |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |
| 100 $\pm$ 15 lb ft        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                          |                                                                                                                                                                                                                                                  |          |        |                    |             |                  |     |   |          |              |   |   |   |                                    |   |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |        |                  |  |          |        |                    |  |

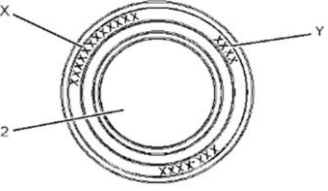
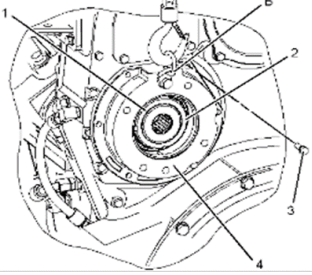
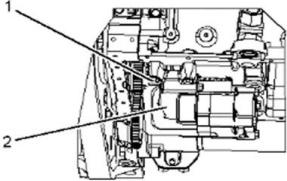


| CHECK SHEET               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | WO NUMBER                    | DOCUMENT NO                                                                        |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------|--|--|------|-------------|------------------|-----|---|---------|-----------------------------|---|---------|--------------------------------------------------------------------------------------|---|---------|------|---|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| CAT ENGINE C32 ASSEMBLING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | COMP.MODEL                   | PUBLIC DATE                                                                        |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
| ENGINE CAT C32            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | COMP.S/N                     | REVISION NO                                                                        |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | UNIT MODEL                   | PAGE                                                                               |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              | 1 OF 3                                                                             |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
| PROCESS                   | START<br>FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ASSEMBLY BY                  |                                                                                    |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
| 16                        | <div>FLYWHEEL<br/>HOUSING</div> <div><div>c Dalam urutan numerik, kencangkan Bolt 11 hingga Bolt 36 dengan torsi 47 ± 9 N · m (35 ± 7 lb ft).</div><div>6 Hubungkan Hose Assy (2) ke Flywheel Housing (1).</div><div><div><div></div><div>STANDARD</div><div>35 ± 7 lb ft</div></div><div>Tightening Bolt</div><div>ACTUAL</div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 47 ± 9 N · m (35 ± 7 lb ft). |  |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
| 17                        | <div>GEAR GROUP<br/>FRONT</div> <div><div><table><tr><th colspan="4">Required Tools</th></tr><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr><tr><td rowspan="3">A</td><td>8S-2241</td><td>Camshaft Bearing Tool Group</td><td>1</td></tr><tr><td>6V-4869</td><td>Adapter</td><td>1</td></tr><tr><td>6V-6174</td><td>Plug</td><td>1</td></tr></table><div>1 Gunakan Tooling (A) untuk menginstal bushing (8) dan (9). Pasang busing (8) ke Dimensi (B). Dimensi (B) adalah 236,7 ± 0,5 mm (9,32 ± 0,02 inci). Pasang busing (9) ke Dimensi (A). Dimensi (A) adalah 40,4 ± 0,5 mm (1,59 ± 0,02 inci). Pasang poros engkol. Lihat Pembongkaran dan Perakitan, "Crankshaft - Install".</div><div>2 Oleskan oli mesin yang bersih ke shaft (7), lubang bearing dan semua permukaan sebelum pemasangan. Pasang shaft (7) dan baut (6). Kencangkan baut (6) hingga torsi 55 ± 10 N · m (41 ± 7 lb ft).</div><div>3 Posisikan Gear (1) pada Shaft. Pasang pelat (2) dan baut (3).</div><div>4 Pasang Gear (4) dan baut (5).</div><div><div><div></div><div>STANDARD</div><div>41 ± 7 lb ft</div></div><div>Tightening Bolt</div><div>ACTUAL</div></div></div></div> | Required Tools               |                                                                                    |  |  | Tool | Part Number | Part Description | Qty | A | 8S-2241 | Camshaft Bearing Tool Group | 1 | 6V-4869 | Adapter                                                                              | 1 | 6V-6174 | Plug | 1 | 236,7 ± 0,5 mm (9,32 ± 0,02 inci)<br><br>40,4 ± 0,5 mm (1,59 ± 0,02 inci)<br><br>55 ± 10 N · m (41 ± 7 lb ft). |  |
| Required Tools            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |                                                                                    |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
| Tool                      | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Part Description             | Qty                                                                                |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
| A                         | 8S-2241                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Camshaft Bearing Tool Group  | 1                                                                                  |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
|                           | 6V-4869                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Adapter                      | 1                                                                                  |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
|                           | 6V-6174                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Plug                         | 1                                                                                  |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
| 18                        | <div>FRONT<br/>HOUSING</div> <div><div><table><tr><th colspan="4">Required Tools</th></tr><tr><th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr><tr><td>B</td><td></td><td>Loctite 598</td><td>1</td></tr></table><div>CATATAN :<br/>Jaga Kebersihan Semua Componen dari Kontaminasi Kontaminan dapat menyebabkan keausan cepat dan umur komponen pendek</div><div>1 Gunakan Tooling (B) ke bagian bawah front housing (14) pada saambungan Cylinder block dan Engine Oil Pan.</div><div>2 Pasang perangkat pengangkat yang sesuai pada Front Housing (14). Berat Front Housing (14) kira-kira 70 kg (154 lb). Posisikan Gasket Front Housing (14) pada pin dowel di Clinder Block.</div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Required Tools               |                                                                                    |  |  | Tool | Part Number | Part Description | Qty | B |         | Loctite 598                 | 1 |         |  |   |         |      |   |                                                                                                                |                                                                                     |
| Required Tools            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                              |                                                                                    |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
| Tool                      | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Part Description             | Qty                                                                                |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |
| B                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Loctite 598                  | 1                                                                                  |  |  |      |             |                  |     |   |         |                             |   |         |                                                                                      |   |         |      |   |                                                                                                                |                                                                                     |

| CHECK SHEET               |                 | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DOCUMENT NO                                                                                                                                                  |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
|---------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------|------------------|-----|---|---------|-----------|---|---|---------|---------------|---|---|--|-----------------|---|
| CAT ENGINE C32 ASSEMBLING |                 | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PUBLIC DATE                                                                                                                                                  |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| ENGINE CAT C32            |                 | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | REVISION NO                                                                                                                                                  |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
|                           |                 | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PAGE                                                                                                                                                         |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1 OF 3                                                                                                                                                       |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| PROCESS                   | START<br>FINISH | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                              |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                              |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
|                           |                 | <p>3 Pasang perangkat pengangkat yang sesuai pada Front Housing (14). Berat Front Housing (14) kira-kira 70 kg (154 lb). Posisikan Gasket Front Housing (14) pada pin dowel di Clinder Block.</p> <p>4 Pasang bonnet (15).</p>                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                              |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| 19                        | FRONT HOUSING   | <p>5 Potong gasket antara front housing , cylinder block dan bagian bawah Front housing.</p> <p>6 Pasang baut (13) dan adaptor (12).</p> <p>7 Hubungkan Elbow (11) dan braket (10).</p> <p>8 Pasang trunnion engine (9) dan braket (8)</p> <p>9 Pasang damper (7) dan baut (6).</p> <p>10 Kencangkan baut hingga torsi <math>240 \pm 40 \text{ N} \cdot \text{m}</math> (<math>177 \pm 30 \text{ lb ft}</math>). Kencangkan baut dengan urutan berikut: 1, 4, 2, 5, 3, 6, 1, 4</p> <p>11 Pasang engine oil filter (5).</p> <p>12 Reposisi unit rumah filter dan kencangkan baut (4)</p>                                              | <p>240 <math>\pm</math> 40 N · m<br/>(177 <math>\pm</math> 30 lb ft).</p>  |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| 20                        | FRONT HOUSING   | <p>13 Pasangl bolts (3).</p> <p>14 Hubungkan harness assembly (2).</p> <p>15 Pasang covers (1).</p> <div style="display: flex; align-items: center; justify-content: center;">  <div style="margin-left: 10px;"> <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td>177 <math>\pm</math> 30 lb ft</td> <td></td> </tr> </table> </div> </div>                                                                                                                                                 | STANDARD                                                                                                                                                     | ACTUAL         | 177 $\pm$ 30 lb ft |  |   |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| STANDARD                  | ACTUAL          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                              |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| 177 $\pm$ 30 lb ft        |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                              |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| 21                        | CAMSHAFT        | <p><b>C32 Installation Procedure</b></p> <p>Table 2</p> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>9U-7257</td> <td>Cradle As</td> <td>3</td> </tr> <tr> <td>B</td> <td>9U-7240</td> <td>Camshaft Hook</td> <td>2</td> </tr> <tr> <td>D</td> <td></td> <td>Loctite LB 8028</td> <td>1</td> </tr> </tbody> </table> <p><b>CATATAN:</b></p> <p><i>Jaga Kebersihan Semua Komponen dari Kontaminasi Kontaminan dapat menyebabkan keausan cepat dan umur komponen pendek</i></p> |                                                                                                                                                              | Required Tools |                    |  |                                                                                                                                                                          | Tool | Part Number | Part Description | Qty | A | 9U-7257 | Cradle As | 3 | B | 9U-7240 | Camshaft Hook | 2 | D |  | Loctite LB 8028 | 1 |
| Required Tools            |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                              |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| Tool                      | Part Number     | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Qty                                                                                                                                                          |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| A                         | 9U-7257         | Cradle As                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                                                                                                                                                            |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| B                         | 9U-7240         | Camshaft Hook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2                                                                                                                                                            |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |
| D                         |                 | Loctite LB 8028                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                            |                |                    |  |                                                                                                                                                                          |      |             |                  |     |   |         |           |   |   |         |               |   |   |  |                 |   |

| CHECK SHEET               |                 | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DOCUMENT NO                                                                                                                                                                          |
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| CAT ENGINE C32 ASSEMBLING |                 | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PUBLIC DATE                                                                                                                                                                          |
| ENGINE CAT C32            |                 | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | REVISION NO                                                                                                                                                                          |
|                           |                 | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PAGE                                                                                                                                                                                 |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | REV 01                                                                                                                                                                               |
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| PROCESS                   | START<br>FINISH | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                      |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                      |
|                           |                 | <p><b>C A T A T A N :</b><br/> <i>Jangan memutar Crankshaft atau Camshaft saat Camshaft gear dilepas. Jika Rear Gear Group tidak diatur Timing nta dengan benar selama pemasangan, dapat terjadi gangguan antara piston dan valve, yang bisa mengakibatkan kerusakan pada engine.</i></p> <p><b>C A T A T A N :</b><br/> <i>Perhatikan saat melepas atau memasang camshaft Jangan merusak permukaan akhir camshaft atau bushing camshaft.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                      |
| 22                        | CAMSHAFT        | <p>1 Pastikan Camshaft dan bushing benar-benar bersih. Lumasi lob camshaft dengan engine oil, gunakan 50/50 Tooling (C) dan Oleskan sedikit oli mesin yang bersih pada bushing camshaft.</p> <p>2 Pasang Tool (A) pada cylinder head (16) di tiga lokasi.</p> <p>3 Gunakan dua Teknisi / Mekanik untuk memasang Camshaft. Gunakan tool (B) untuk membantu menyelaraskan camshaft dengan bushing camshaft. Dengan hati-hati geser camshaft ke dalam cylinder head. Jaga ketinggian camshaft saat camshaft dipasang d cylinder head. Berat camshaft kira-kira 35 kg (77 lb).</p> <p>4 Lepas Tool (A) dari cylinder head.</p> <p>5 Pasang sealing plate (14), adapter assembly (15), and new O-ring seals. pasang bolts (13)</p> <p><b>Catatan:</b><br/> <i>Pastikan seal O-ring tetap pada posisinya pada pelat perapat (14) selama pemasangan</i></p> |                                                                                                   |
| 23                        | CAMSHAFT        | <p>6 Pasang damper (12) dan bolts (11).</p> <p>7 Kencangkan baut hingga torsi <math>240 \pm 40 \text{ N} \cdot \text{m}</math> (<math>177 \pm 30 \text{ lb ft}</math>). Kencangkan baut dengan urutan berikut: 1, 4, 2, 5, 3, 6, 1, 4</p> <p>8 Pasaang cover (9) dan elbow (10)</p> <p>9 Pasang new O-ring seals dan sealing plate (8)</p> <p><b>Catatan:</b><br/> <i>Pastikan seal O-ring tetap pada posisinya pada Sealing Plate (8) selama pemasangan</i></p> <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;"> <br/> <b>STANDARD</b><br/> <math>177 \pm 30 \text{ lb ft}</math> </div> <div style="text-align: center;"> <b>Tightening Bolt</b><br/> <b>ACTUAL</b> </div> </div>             | <p><math>240 \pm 40 \text{ N} \cdot \text{m}</math> (<math>177 \pm 30 \text{ lb ft}</math>)</p>  |

| CHECK SHEET               |                             | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | DOCUMENT NO                                                                                                                                                                     |
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| CAT ENGINE C32 ASSEMBLING |                             | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PUBLIC DATE                                                                                                                                                                     |
| ENGINE CAT C32            |                             | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | REVISION NO                                                                                                                                                                     |
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| PROCESS                   | START<br>FINISH             | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                 |
|                           |                             | 10 Pasang adapter assembly (7), thrust plate (6), and bolts (5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                               |
|                           |                             | 11 Pasang gear (4). Install bolts (3).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                 |
|                           |                             | 12 Kencangkan baut hingga torsi $240 \pm 40 \text{ N} \cdot \text{m}$ ( $177 \pm 30 \text{ lb ft}$ ). Kencangkan baut dengan urutan berikut: 1, 4, 2, 5, 3, 6, 1, 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                 |
|                           |                             | 13 Pasang Pump drive (2) dan bolts (1).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                 |
| 24                        | ELECTRONIC<br>UNIT INJECTOR | <p>1 Gunakan bahan bebas serat untuk membersihkan O-ring seal (7) dan Electronic Unit Injector (2).</p> <p>2 Lumasi dua O-ring seal teratas (7) dengan engine oil. Jangan gunakan pelumas lain.</p> <p><b>Catatan:</b><br/><i>Setelah O-ring seal (7) terlumasi, segera pasang.</i></p> <p><b>Catatan:</b><br/><i>Seal O-ring (7) harus disimpan dengan benar untuk mencegah kontaminasi.</i></p> <p>3 Pasang O-ring seals (7) pada electronic unit injector.</p> <p><b>Catatan:</b><br/><i>Hindari peregangan yang berlebihan saat memasang O-ring Seal (7).</i></p> <p>4 Saat Anda memasang injektor baru, nilai E-Trim harus diprogram ke dalam Engine Control Modul. Kode konfirmasi injektor mungkin perlu diprogram ke dalam Engine Control Modul. E-Trim Value (X) adalah kode alfanumerik 12 digit yang disertakan dengan lembar data saat membeli injector baru. Kode Konfirmasi Injector (Y) adalah kode numerik empat digit. Kedua kode tersebut ada di atas injektor unit elektronik (2). Gunakan menu berikut untuk memprogram nilai E-Trim :</p> <ul style="list-style-type: none"> <li>* ECM Summary Screen</li> <li>* ECM Summary Screen</li> <li>* Calibration</li> <li>* Injector Code Calibration</li> </ul> <p><b>CATATAN</b><br/><i>Saat Elektronik Unit Injector diganti, kode Injector baru harus diprogram ke dalam "machine personality module software" dengan menggunakan menu kalibrasi pada Electronic Technician (ET). Jika kode Electronic Injector baru tidak dimasukkan, karakteristik Injector unit lama/sebelumnya akan diasumsikan.</i></p> <p><i>Jika tidak memungkinkan untuk segera memprogram ulang kode Electronic Injector ke dalam "machine personality module software" Engine tidak akan rusak parah. Kode Electronic Injector baru harus diprogram ulang secepat mungkin untuk mengoptimalkan kinerja Engine.</i></p> | <br><br> |

| CHECK SHEET               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | WO NUMBER                                                                | DOCUMENT NO                                                                                                                                                                                                                                                                                                    |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
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| CAT ENGINE C32 ASSEMBLING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | COMP.MODEL                                                               | PUBLIC DATE                                                                                                                                                                                                                                                                                                    |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| ENGINE CAT C32            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | COMP.S/N                                                                 | REVISION NO                                                                                                                                                                                                                                                                                                    |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
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| PROCESS                   | START<br>FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ASSEMBLY BY                                                              |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
|                           | 5 Pasang Elektronik Unit Injector(2) dengan penjepit (6), spacer (5), dan baut baru (4). Posisikan injektor (2) di lubang injektor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                          |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
|                           | <p><b>Catatan:</b><br/>Perhatian saat memasang O-Ring Seal (7) agar tidak rusak saat memposisikan injektor (2) di lubang injektor.</p> <p><b>Catatan:</b><br/>Jangan mendorong atau memaksa injektor secara manual ke dalam lubang.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                          |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
|                           | 6 Kencangkan baut (4) hingga torsi $55 \pm 10 \text{ N} \cdot \text{m}$ ( $41 \pm 7 \text{ lb ft}$ ). Pastikan Electronic Unit Injektor (2) sejajar di lubang saat mengencangkan baut (4).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $55 \pm 10 \text{ N} \cdot \text{m}$<br>( $41 \pm 7 \text{ lb ft}$ ).    |  <table border="1"> <tr> <th>STANDARD</th> <th>Tightening Bolt</th> </tr> <tr> <td>41 ± 7 lb ft</td> <td>ACTUAL</td> </tr> </table>                                                                                         | STANDARD | Tightening Bolt | 41 ± 7 lb ft | ACTUAL      |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| STANDARD                  | Tightening Bolt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                          |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| 41 ± 7 lb ft              | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                          |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| 25                        | <p><b>Catatan:</b><br/>Kencangkan baut (4) dengan kecepatan yang lambat dan seirama. Jika alat pneumatik atau elektrik digunakan, perawatan harus dilakukan untuk menghindari kerusakan pada O-ring injektor yg terpasang dengan cepat.</p> <p><b>Catatan:</b><br/>Baut pengancang (4) sebagai koreksi torsi dudukan Electroni Unit Injektor dengan benar (2).</p>                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                          |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
|                           | <p>7 Periksa area di sekitar lubang injektor (2) untuk memastikan bahwa O-Ring Seal (7) tidak rusak atau terekstrusi.</p> <p>8 Pasang bridge assemblies (3).</p> <p>9 Hubungkan harnes assy (1). Kencangkan mur hingga torsi <math>2,5 \pm 0,25 \text{ N} \cdot \text{m}</math> (<math>22 \pm 2 \text{ lb in}</math>).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $2,5 \pm 0,25 \text{ N} \cdot \text{m}$<br>( $22 \pm 2 \text{ lb in}$ ). | <br> <table border="1"> <tr> <th>STANDARD</th> <th>Tightening Bolt</th> </tr> <tr> <td>22 ± 2 lb ft</td> <td>ACTUAL</td> </tr> </table> | STANDARD | Tightening Bolt | 22 ± 2 lb ft | ACTUAL      |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| STANDARD                  | Tightening Bolt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                          |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| 22 ± 2 lb ft              | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                          |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| 26                        | <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>6V-4073</td> <td>Spanner Wrench</td> <td>1</td> </tr> <tr> <td>B</td> <td>138-7574</td> <td>Link Bracket</td> <td>1</td> </tr> </tbody> </table> <p>1 Pasang Tooling (B) dan perangkat pengangkat yang sesuai ke Pump Drive (4). Berat penggerak pompa (4) kira kira 32 kg (70 lb). Pasang Pump Drive (4).</p> <p>2 Pasang baut (3) dan mur (1). Gunakan Tool (A) untuk mengencangkan mur (1). Kencangkan mur (1) hingga torsi <math>406 \pm 68 \text{ N} \cdot \text{m}</math> (<math>299 \pm 50 \text{ lb ft}</math>). Tekuk tab pengunci (2) ke dalam slot mur (1).</p> | Required Tools                                                           |                                                                                                                                                                                                                                                                                                                |          |                 | Tool         | Part Number | Part Description | Qty | A | 6V-4073 | Spanner Wrench | 1 | B | 138-7574 | Link Bracket | 1 | $406 \pm 68 \text{ N} \cdot \text{m}$<br>( $299 \pm 50 \text{ lb ft}$ ) | <br> <table border="1"> <tr> <th>STANDARD</th> <th>Tightening Bolt</th> </tr> <tr> <td>299 ± 50 lb ft</td> <td>ACTUAL</td> </tr> </table> | STANDARD | Tightening Bolt | 299 ± 50 lb ft | ACTUAL |
| Required Tools            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                          |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| Tool                      | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Part Description                                                         | Qty                                                                                                                                                                                                                                                                                                            |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| A                         | 6V-4073                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Spanner Wrench                                                           | 1                                                                                                                                                                                                                                                                                                              |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| B                         | 138-7574                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Link Bracket                                                             | 1                                                                                                                                                                                                                                                                                                              |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| STANDARD                  | Tightening Bolt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                          |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| 299 ± 50 lb ft            | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                          |                                                                                                                                                                                                                                                                                                                |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |
| 27                        | <p>1 Posisikan engine oil pump (2) dan Pasang bolts (1).</p> <p>2 Connect tube assemblies ke engine oil pump (2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                          |                                                                                                                                                                                                                            |          |                 |              |             |                  |     |   |         |                |   |   |          |              |   |                                                                         |                                                                                                                                                                                                                                                                                                                    |          |                 |                |        |

| CHECK SHEET               |                 | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DOCUMENT NO |                |  |  |  |      |             |                  |     |   |         |                |   |   |   |                             |   |
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| CAT ENGINE C32 ASSEMBLING |                 | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PUBLIC DATE |                |  |  |  |      |             |                  |     |   |         |                |   |   |   |                             |   |
| ENGINE CAT C32            |                 | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | REVISION NO |                |  |  |  |      |             |                  |     |   |         |                |   |   |   |                             |   |
|                           |                 | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PAGE        |                |  |  |  |      |             |                  |     |   |         |                |   |   |   |                             |   |
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| PROCESS                   | START<br>FINISH | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                |  |  |  |      |             |                  |     |   |         |                |   |   |   |                             |   |
| 28                        | OIL PUMP        | <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>1U-7505</td> <td>Hydraulic Jack</td> <td>1</td> </tr> <tr> <td>B</td> <td>-</td> <td>Loctite 5900 Flange Sealant</td> <td>1</td> </tr> </tbody> </table> <p><b>PERHATIAN :</b><br/><i>Jaga kebersihan semua bagian dari kontaminan.</i></p> <p><b>Kontaminan dapat menyebabkan keausan Lebih cepat dan umur komponen pendek.</b></p> <ol style="list-style-type: none"> <li>Pasang Tool(B) pada empat sambungan soket gasket oli pan dan empat sambungan T di bagian bawah Front Housing dan Rear Housing bertemu dengan Cylinder block.</li> <li>Kencangkan bolts (6) sesuai urutan sequence to a torque of <math>47 \pm 9 \text{ N} \cdot \text{m}</math> (<math>35 \pm 7 \text{ lb ft}</math>).</li> <li>Kencangkan bolts (6) torque lagi sesuai urutan <math>47 \pm 9 \text{ N} \cdot \text{m}</math> (<math>35 \pm 7 \text{ lb ft}</math>).</li> <li>Pasang drain oil plug dengan torsi <math>70 \pm 10 \text{ N} \cdot \text{m}</math> (<math>52 \pm 7 \text{ lb ft}</math>)</li> <li>Pasang Tube (1) jangan lupa O-ring seal diganti.</li> </ol> <p><math>47 \pm 9 \text{ N} \cdot \text{m}</math><br/>(<math>35 \pm 7 \text{ lb ft}</math>).</p> <p><math>47 \pm 9 \text{ N} \cdot \text{m}</math><br/>(<math>35 \pm 7 \text{ lb ft}</math>).</p> <p><math>70 \pm 10 \text{ N} \cdot \text{m}</math><br/>(<math>52 \pm 7 \text{ lb ft}</math>)</p> |             | Required Tools |  |  |  | Tool | Part Number | Part Description | Qty | A | 1U-7505 | Hydraulic Jack | 1 | B | - | Loctite 5900 Flange Sealant | 1 |
| Required Tools            |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                |  |  |  |      |             |                  |     |   |         |                |   |   |   |                             |   |
| Tool                      | Part Number     | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Qty         |                |  |  |  |      |             |                  |     |   |         |                |   |   |   |                             |   |
| A                         | 1U-7505         | Hydraulic Jack                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1           |                |  |  |  |      |             |                  |     |   |         |                |   |   |   |                             |   |
| B                         | -               | Loctite 5900 Flange Sealant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1           |                |  |  |  |      |             |                  |     |   |         |                |   |   |   |                             |   |
| 29                        | ROCKER ARM      | <ol style="list-style-type: none"> <li>Posisikan Rocker Arm Assy (4), Rocker Arm Valve (3), dan Rocker Arm Unit Injectro (1) Cylinder Head.</li> </ol> <p><b>Catatan:</b><br/><i>Oleskan oli engine bersih ke bagian bawah kepala baut dan ulir sebelum pemasangan.</i></p> <ol style="list-style-type: none"> <li>Pasang baut baru (2) dan kencangkan hingga torsi <math>55 \pm 5 \text{ N} \cdot \text{m}</math> (<math>41 \pm 4 \text{ lb ft}</math>). Kencangkan kembali baut (2) hingga torsi <math>55 \pm 5 \text{ N} \cdot \text{m}</math> (<math>41 \pm 4 \text{ lb ft}</math>). Putar baut (2) dan penambahan <math>60 \pm 5</math> derajat. Kencangkan baut dengan urutan berikut: 2, 3, 1, 4.</li> </ol> <p><b>Catatan:</b><br/><i>60 derajat tambahan dapat dicapai dengan memutar baut dua titik</i></p> <ol style="list-style-type: none"> <li>Adjust the valve lash untuk inlet valves dan exhaust valves. Refer to Testing and Adjusting, "Engine Valve Lash - Inspect/Adjust".</li> <li>Adjust the electronic unit injectors. Referensi ke Testing and Adjusting, "Electronic Unit Injector - Adjust".</li> </ol> <p><math>55 \pm 5 \text{ N} \cdot \text{m}</math><br/>(<math>41 \pm 4 \text{ lb ft}</math>).</p> <p><math>60 \pm 5</math> derajat.</p>                                                                                                                                                                                                                                                                                      |             |                |  |  |  |      |             |                  |     |   |         |                |   |   |   |                             |   |

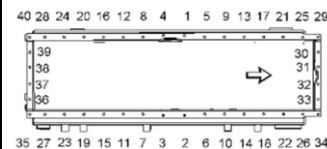
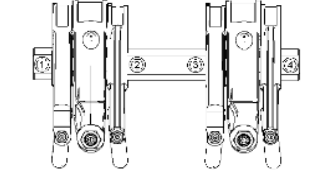
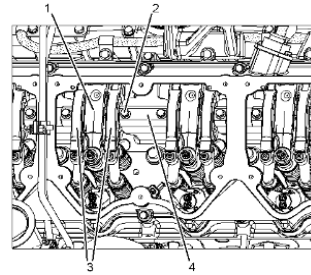
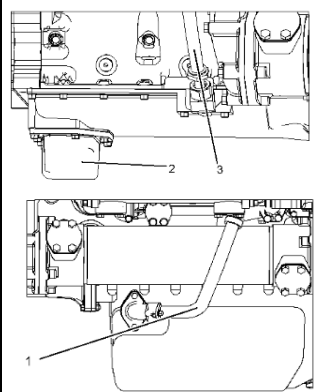


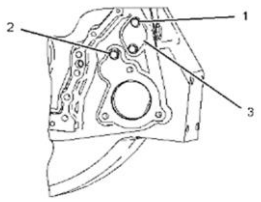
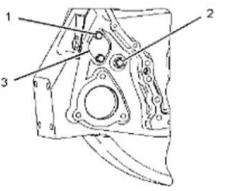
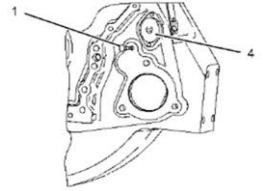
Illustration 4  
Tightening Sequence  
The directional arrow is towards the front housing of the engine.



| STANDARD    | Tightening Bolt |
|-------------|-----------------|
| 41± 4 lb ft | ACTUAL          |

Tambah 60 derajat



| CHECK SHEET                                                                                                                                                                                                                                      |                     | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | DOCUMENT NO         |         |                     |   |
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| CAT ENGINE C32 ASSEMBLING                                                                                                                                                                                                                        |                     | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PUBLIC DATE         |         |                     |   |
| ENGINE CAT C32                                                                                                                                                                                                                                   |                     | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | REVISION NO         |         |                     |   |
|                                                                                                                                                                                                                                                  |                     | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PAGE                |         |                     |   |
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| PROCESS                                                                                                                                                                                                                                          | START<br>FINISH     | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |         |                     |   |
| 30                                                                                                                                                                                                                                               | TIMING<br>TDC TOP 1 | <div style="border: 1px solid black; padding: 5px; margin-bottom: 10px;"> <p style="text-align: center; margin: 0;"><b>Needed Tools</b></p> <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="padding: 2px 5px;">9S-9082</td> <td style="padding: 2px 5px;">Engine Turning Tool</td> <td style="padding: 2px 5px; text-align: center;">1</td> </tr> </table> </div> <p>Piston No. 1 Top Center (TC) pada langkah kompresi adalah titik awal starting point of all timing procedures.</p> <p><b>Catatan:</b><br/><i>Beberapa Engine memiliki dua lubang berulir di flywheel. Lubang ini sejajar dengan Plug di kiri dan kanan depan flywheel Housing. Dua lubang di flywheel berada pada jarak yang berbeda dari pusat flywheel. Oleh karena itu, timing bolt tidak bisa dipasang di lubang yang salah.</i></p> <p>1 Timing bolt (1) adalah Cover Bolt. Baut timing dapat dipasang di sisi kiri mesin di lokasi (2) atau di sisi kanan mesin di lokasi baut timing (2). Lepaskan kedua baut (1) dan Cover (3) dari flywheel Housing. Lepaskan Plug timing di flywheel Housing.</p> <p>2 Pasang timing bolt (1) (baut panjang yang menahan Cover pada flywheel Housing) melalui lubang timing di flywheel Housing. Turning Tool P/N: 9S-9082 dan kunci pas ratchet drive 1/2 inci digunakan untuk memutar Flywheel ke arah putaran mesin normal. Putaran mesin normal berlawanan arah jarum jam. Putaran mesin yang normal dilihat dari depan Flywheel. Putar Flywheel hingga Timing Bolt terpasang dengan lubang ulir di Flywheel.</p> <p>3</p> <p><b>Catatan:</b><br/>Jika flywheel diputar melebihi titik pengaktifan, flywheel harus diputar ke arah berlawanan dari putaran mesin normal kira-kira 45 derajat. Kemudian putar flywheel ke arah rotasi normal hingga baut timing terpasang dengan lubang berulir. Prosedur ini akan menghilangkan "backlash" yang akan terjadi saat TDC piston No.1</p> <p>4 Valve Inlet dan Exhaust Cylinder No. 1 ditutup sepenuhnya jika piston No. 1 berada pada compression stroke dan Rocker Arm dapat digerakkan dengan tangan (Posisi bebas). Piston No. 1 exhaust stroke, Rocker Arm tidak dapat digerakkan dan Valve sedikit terbuka.</p> |                     | 9S-9082 | Engine Turning Tool | 1 |
|                                                                                                                                                                                                                                                  |                     | 9S-9082                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Engine Turning Tool | 1       |                     |   |
|                                                                                                                                                                                                                                                  |                     |  <p style="font-size: small;">Illustration 1<br/>Locating Top Center on the left side of engine<br/>(1) Timing bolt<br/>(2) Timing bolt location<br/>(3) Cover<br/>g01123281</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |         |                     |   |
|                                                                                                                                                                                                                                                  |                     |  <p style="font-size: small;">Illustration 2<br/>Locating Top Center on the right side of engine<br/>(1) Timing bolt<br/>(2) Timing bolt location<br/>(3) Cover<br/>g01123294</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |         |                     |   |
|  <p style="font-size: small;">Illustration 3<br/>Using 9S-9082 Engine Turning Tool<br/>(1) Timing bolt<br/>(4) 9S-9082 Engine Turning Tool<br/>g01123382</p> |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |         |                     |   |

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| CAT ENGINE C32 ASSEMBLING                                                                                                                                                                                                         |                                    | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PUBLIC DATE                            |
| ENGINE CAT C32                                                                                                                                                                                                                    |                                    | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | REVISION NO                            |
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|                                                                                                                                                                                                                                   |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 OF 3                                 |
| PROCESS                                                                                                                                                                                                                           | START                              | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                        |
|                                                                                                                                                                                                                                   | FINISH                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                        |
| <p><b>Catatan:</b><br/>Setelah posisi stroke teridentifikasi, Posisi stroke lainnya diperlukan, lepaskan Timing Bolt dari flywheel. Putar kembali Flywheel 360 derajat berlawanan arah jarum jam. Timing Bolt pasang kembali.</p> |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                        |
| 31                                                                                                                                                                                                                                | <p><b>VAVE LASH ADJUSTMENT</b></p> | <p>Adjust Valve Lash saat Kondisi Engine Berhenti. Ikuti prosedur Adjust Valve sebagai berikut :</p> <p>1 Posisikan Piston Top Center No. 1.</p> <p>Sebelum adjusment dilakukan, ketuk perlahan setiap Rocker arm di bagian atas sekrup penyetelan. Gunakan palu lembut untuk memastikan bahwa roller pengangkat bersandar pada lingkaran dasar camshaft.</p> <p>2 Lakukan adjustment valve lash pada inlet valves cylinders 1, 2, 3, 7, 8, and 10</p> <p>a Kendorkan valve adjustment locknut (3)</p> <p>b Tempatkan Feeler Gauge yang sesuai (5) antara Rocker Arm Inlet Valve Bridge. Putar sekrup penyetelan Inlet (4) adjustment locknut Valve (3) ditahan agar tidak berputar. Adjust the valve lash sampai spesifikasi yang benar tercapai. Lihat Tabel 2.</p> <p>c Setelah setiap Adjustment dilakukan, valve adjustment locknut (3) Saat valve adjustment screw (4) ditahan agar tidak berputar. Kencangkan hingga torsi <math>30 \pm 7 \text{ N} \cdot \text{m}</math> (<math>22 \pm 5 \text{ lb ft}</math>). Periksa kembali setiap adjusment.</p> <p>3 Lakukan djustment valve lash untuk exhaust valves cylinders 1, 2, 5, 6, 9, and 10.</p> <p>a Kendorkan valve adjustment locknut (1).</p> <p>b Tempatkan Feeler Gauge (5) di antara Exhaust Rocker Arm dan exhaust valve bridge. Putar exhaust adjustment screw (2) saat valve adjustment locknut (1) ditahan agar tidak berputar. Adjust Valve lash sampai spesifikasi yang benar tercapai. Lihat Tabel 2</p> | <p>30 ± 7 N · m<br/>(22 ± 5 lb ft)</p> |

| Quick Reference for Engine Valve Lash Setting       |                                         |                                     |
|-----------------------------------------------------|-----------------------------------------|-------------------------------------|
|                                                     | Inlet Valves                            | Exhaust Valves                      |
| Valve Lash Setting                                  | 0.38 ± 0.08 mm (0.015 ± 0.003 inch)     | 0.76 ± 0.08 mm (0.030 ± 0.003 inch) |
| TC Compression Stroke of No. 1 Piston               | 1-2-3-7-8-10                            | 1-2-5-6-9-10                        |
| TC Compression Stroke of No. 11 Piston <sup>2</sup> | 4-5-6-9-11-12                           | 3-4-5-8-11-12                       |
| Firing Order                                        | 1-10-9-8-5-12-11-4-3-6-7-2 <sup>3</sup> |                                     |

| Valve Lash |                      |
|------------|----------------------|
| Valves     | Thickness of Gauge   |
| Inlet      | 0.38 mm (0.015 inch) |
| Exhaust    | 0.76 mm (0.030 inch) |

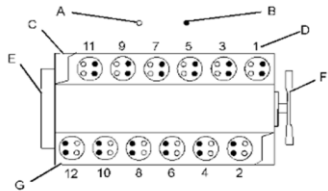
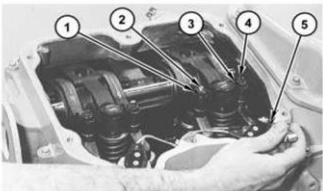
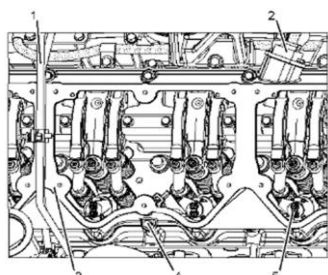
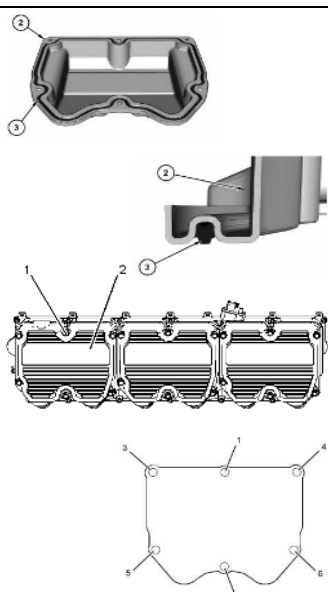
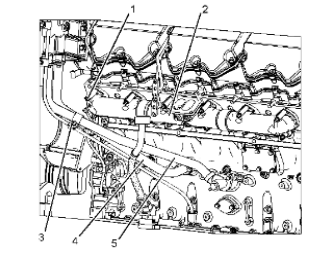


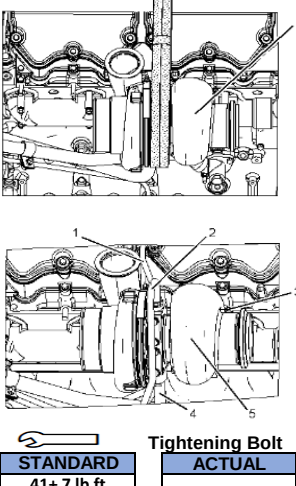
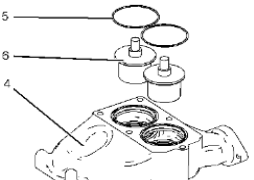
Illustration 1  
(A) Inlet valves  
(B) Exhaust valves  
(C) Left side of the engine  
(D) Cylinder number  
(E) Front end of the engine  
(F) Front of the engine  
(G) Right side of the engine

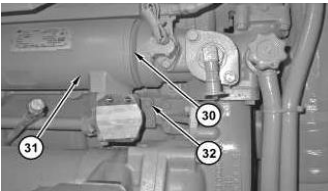
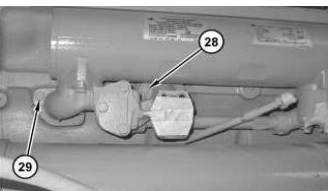
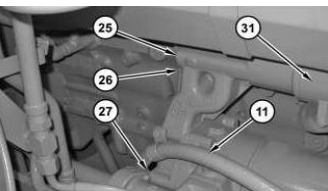
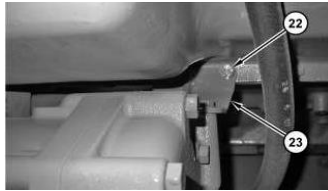
| Tightening Bolt |
|-----------------|
| STANDARD        |
| 22 ± 5 lb ft    |
| ACTUAL          |
|                 |

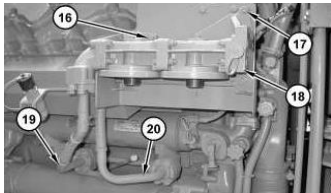
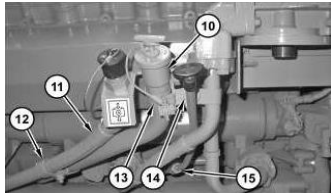
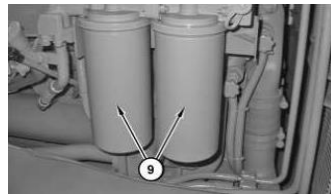
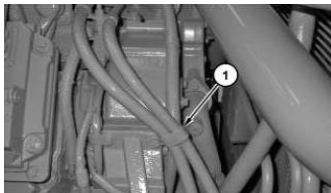
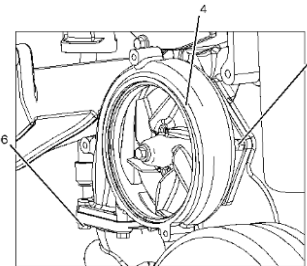
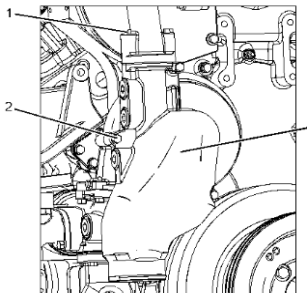


| CHECK SHEET               |                         | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                              | DOCUMENT NO |          |        |             |  |
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| CAT ENGINE C32 ASSEMBLING |                         | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                             | PUBLIC DATE |          |        |             |  |
| ENGINE CAT C32            |                         | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                               | REVISION NO |          |        |             |  |
|                           |                         | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                             | PAGE        |          |        |             |  |
|                           |                         |                                                                                                                                                                                                                                                                                                                                                                                                                        | 1 OF 3      |          |        |             |  |
| PROCESS                   | START<br>FINISH         | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                            |             |          |        |             |  |
| 32                        | VAVE LASH<br>ADJUSTMENT | <p>c Setelah setiap Adjustment dilakukan, kencangkan valve adjustment locknut (1) saat valve adjustment screw (2) ditahan agar tidak berputar. Kencangkan hingga torsi <math>30 \pm 7 \text{ N} \cdot \text{m}</math> (<math>22 \pm 5 \text{ lb ft}</math>). Periksa kembali setiap adjustment.</p>                                                                                                                    |             |          |        |             |  |
|                           |                         | <p>4 Lepaskan Timing Bolt, dan putar flywheel 360 derajat ke arah putaran Engine. Ini akan memposisikan piston No. 12 TDC compression stroke. Pasang Timing Bolt pada Flywheel.</p>                                                                                                                                                                                                                                    |             |          |        |             |  |
|                           |                         | <p>5 Lakukan adjustment valve lash pada inlet valves di cylinders 4, 5, 6, 9, 11, and 12.</p> <p>a Kendorkan valve adjustment locknut (3).</p> <p>b Tempatkan Feeler Gauge (5) antara inlet rocker arm dan inlet valve bridge. Putar inlet adjustment screw (4) saat valve adjustment locknut (3) ditahan agar tidak berputar. Adjust the valve lash sampai spesifikasi yang benar tercapai. <b>Lihat Tabel 2.</b></p> |             |          |        |             |  |
|                           |                         | <p>c Setelah setiap adjustment dilakukan, kencangkan valve adjustment locknut (3) valve adjustment screw (4) ditahan agar tidak berputar. Kencangkan hingga torsi <math>30 \pm 7 \text{ N} \cdot \text{m}</math> (<math>22 \pm 5 \text{ lb ft}</math>). Periksa kembali setiap Adjustment.</p>                                                                                                                         |             |          |        |             |  |
|                           |                         | <p>30 <math>\pm</math> 7 N · m<br/>(22 <math>\pm</math> 5 lb ft)</p>                                                                                                                                                                                                                                                                                                                                                   |             |          |        |             |  |
|                           |                         | <p>30 <math>\pm</math> 7 N · m<br/>(22 <math>\pm</math> 5 lb ft)</p>                                                                                                                                                                                                                                                                                                                                                   |             |          |        |             |  |
|                           |                         | <p>30 <math>\pm</math> 7 N · m<br/>(22 <math>\pm</math> 5 lb ft)</p>                                                                                                                                                                                                                                                                                                                                                   |             |          |        |             |  |
|                           |                         | <p>30 <math>\pm</math> 7 N · m<br/>(22 <math>\pm</math> 5 lb ft)</p>                                                                                                                                                                                                                                                                                                                                                   |             |          |        |             |  |
|                           |                         | <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td>22± 5 lb ft</td> <td></td> </tr> </table>                                                                                                                                                                                                                                                                       |             | STANDARD | ACTUAL | 22± 5 lb ft |  |
| STANDARD                  | ACTUAL                  |                                                                                                                                                                                                                                                                                                                                                                                                                        |             |          |        |             |  |
| 22± 5 lb ft               |                         |                                                                                                                                                                                                                                                                                                                                                                                                                        |             |          |        |             |  |
|                           |                         |  <p>Illustration 2<br/>Typical example<br/>(1) Valve adjustment locknut<br/>(2) Exhaust adjustment screw<br/>(3) Valve adjustment locknut<br/>(4) Inlet adjustment screw<br/>(5) Feeler gauge</p>                                                                                                                                    |             |          |        |             |  |
|                           |                         | <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td>22± 5 lb ft</td> <td></td> </tr> </table>                                                                                                                                                                                                                                                                       |             | STANDARD | ACTUAL | 22± 5 lb ft |  |
| STANDARD                  | ACTUAL                  |                                                                                                                                                                                                                                                                                                                                                                                                                        |             |          |        |             |  |
| 22± 5 lb ft               |                         |                                                                                                                                                                                                                                                                                                                                                                                                                        |             |          |        |             |  |
|                           |                         | <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <th>STANDARD</th> <th>ACTUAL</th> </tr> <tr> <td>22± 5 lb ft</td> <td></td> </tr> </table> <p>Lepas timing bolt dari flywheel setelah semua valve lash adjustments dilakukan.</p>                                                                                                                                                                                |             | STANDARD | ACTUAL | 22± 5 lb ft |  |
| STANDARD                  | ACTUAL                  |                                                                                                                                                                                                                                                                                                                                                                                                                        |             |          |        |             |  |
| 22± 5 lb ft               |                         |                                                                                                                                                                                                                                                                                                                                                                                                                        |             |          |        |             |  |

| CHECK SHEET               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | WO NUMBER               | DOCUMENT NO                                                                         |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------|-------------|----|--------------------------------------------|------------------------------------------------------------------------------------|------------------|-----|---|---------|---------------------|---|---|---------|-------------------------|---|---|----------|-------|---|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|---|---|---|---|---|----|---|---|---|----|----------|--------|-------------|--|
| CAT ENGINE C32 ASSEMBLING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | COMP.MODEL              | PUBLIC DATE                                                                         |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| ENGINE CAT C32            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | COMP.S/N                | REVISION NO                                                                         |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | UNIT MODEL              | PAGE                                                                                |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         | REV 01                                                                              |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         | 1 OF 3                                                                              |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| PROCESS                   | START<br>FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ASSEMBLY BY             |                                                                                     |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| 33                        | <p><b>VALVE MECANISME COVER BASE</b></p> <ol style="list-style-type: none"> <li>Pasang seal pada valve mechanism cover base (3).</li> <li>Posisikan valve mechanism cover base (3) pada cylinder head dan install bolts (4).</li> <li>Posisikan harness assemblies (5) pada electronic unit injectors. kencangkan cap nuts dgn torque of <math>2.5 \pm 0.25</math> N·m (<math>22 \pm 2</math> lb in).</li> <li>Connect harness assembly (2).</li> <li>Install tube assemblies (1).</li> </ol> <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <td>STANDARD</td> <td>ACTUAL</td> </tr> <tr> <td>22+ 5 lb ft</td> <td></td> </tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | STANDARD                | ACTUAL                                                                              | 22+ 5 lb ft |    | $2.5 \pm 0.25$ N·m<br>( $22 \pm 2$ lb in). |  |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| STANDARD                  | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                         |                                                                                     |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| 22+ 5 lb ft               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |                                                                                     |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| 34                        | <p><b>VALVE MECANISME COVER BASE</b></p> <p><b>PERHATIAN</b></p> <p>Jaga kebersihan semua bagian dari kontaminan.</p> <p>Kontaminan dapat menyebabkan keausan cepat dan umur komponen pendek.</p> <p><b>Catatan:</b></p> <p>Pastikan seal terpasang dengan benar ke dalam Valve Cover mekanisme. Sisi seal yang sempit harus menghadap keluar dari Valve mekanisme Cover.</p> <ol style="list-style-type: none"> <li>Pasang seals (3) pada valve mechanism covers (2)</li> <li>Posisikan Valve Mechanisme Cover (2) pada dasar Valve Cover Mechanisme. Pasang baut (1). Kencangkan baut (1) sesuai urutan yang ditunjukkan pada ilustrasi 4.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |  |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| 35                        | <p><b>EXHAUST MANIFOLD</b></p> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>5P-3931</td> <td>Anti-Seize Compound</td> <td>-</td> </tr> <tr> <td>B</td> <td>2P-2333</td> <td>High Temperature Sealer</td> <td>-</td> </tr> <tr> <td>C</td> <td>165-4638</td> <td>Guide</td> <td>3</td> </tr> </tbody> </table> <ol style="list-style-type: none"> <li>Oleskan oli engine bersih ke ujung female exhaust manifold (2) sebelum assembling. Gunakan Tooling (B) ke ujung exhaust manifold (2) sebelum perakitan. Hapus sealer berlebih dari sambungan yang dirakit</li> <li>Terapkan mesin bersih Terapkan Tooling (A) ke stud sebelum perakitan. Jika perlu, pasang stud dan kencangkan hingga torsi <math>35 \pm 5</math> N · m (<math>26 \pm 4</math> lb ft). e oli ke ujung betina manifold buang (2) sebelum perakitan. Terapkan Tooling (B) ke ujung manifold buang (2) sebelum perakitan. Hapus sealer berlebih dari sambungan yang dirakit.</li> </ol> | Required Tools          |                                                                                     |             |    | Tool                                       | Part Number                                                                        | Part Description | Qty | A | 5P-3931 | Anti-Seize Compound | - | B | 2P-2333 | High Temperature Sealer | - | C | 165-4638 | Guide | 3 | $35 \pm 5$ N · m<br>( $26 \pm 4$ lb ft) |  <table border="1"> <tr> <td>11</td> <td>3</td> <td>7</td> <td>1</td> <td>9</td> <td>5</td> </tr> <tr> <td>4</td> <td>12</td> <td>2</td> <td>8</td> <td>6</td> <td>10</td> </tr> </table> <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <td>STANDARD</td> <td>ACTUAL</td> </tr> <tr> <td>26+ 4 lb ft</td> <td></td> </tr> </table> | 11 | 3 | 7 | 1 | 9 | 5 | 4 | 12 | 2 | 8 | 6 | 10 | STANDARD | ACTUAL | 26+ 4 lb ft |  |
| Required Tools            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |                                                                                     |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| Tool                      | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Part Description        | Qty                                                                                 |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| A                         | 5P-3931                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Anti-Seize Compound     | -                                                                                   |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| B                         | 2P-2333                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | High Temperature Sealer | -                                                                                   |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| C                         | 165-4638                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Guide                   | 3                                                                                   |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| 11                        | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 7                       | 1                                                                                   | 9           | 5  |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| 4                         | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2                       | 8                                                                                   | 6           | 10 |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| STANDARD                  | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                         |                                                                                     |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |
| 26+ 4 lb ft               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |                                                                                     |             |    |                                            |                                                                                    |                  |     |   |         |                     |   |   |         |                         |   |   |          |       |   |                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |   |   |   |   |   |   |    |   |   |   |    |          |        |             |  |

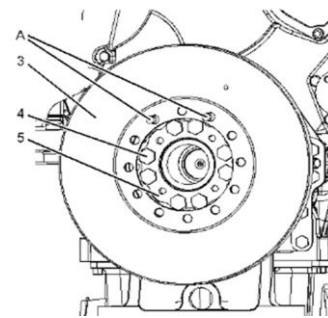
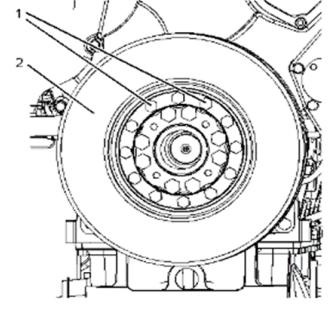
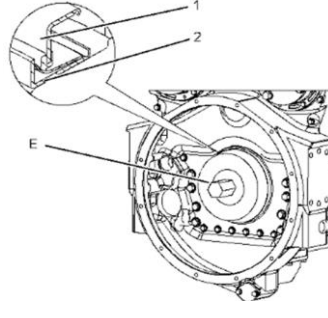
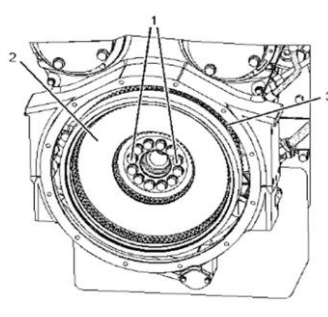
| CHECK SHEET               |                 | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | DOCUMENT NO |                |  |  |  |      |             |                  |     |   |          |                     |   |
|---------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|--|--|--|------|-------------|------------------|-----|---|----------|---------------------|---|
| CAT ENGINE C32 ASSEMBLING |                 | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PUBLIC DATE |                |  |  |  |      |             |                  |     |   |          |                     |   |
| ENGINE CAT C32            |                 | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | REVISION NO |                |  |  |  |      |             |                  |     |   |          |                     |   |
|                           |                 | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PAGE        |                |  |  |  |      |             |                  |     |   |          |                     |   |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1 OF 3      |                |  |  |  |      |             |                  |     |   |          |                     |   |
| PROCESS                   | START<br>FINISH | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                |  |  |  |      |             |                  |     |   |          |                     |   |
|                           |                 | <p>3 Posisikan exhaust manifold (2) on the engine dan jangan lupa pasang juga Exhaust Manifold Gasket nya.</p> <p><b>Catatan:</b><br/>Gunakan Tool (C) di setidaknya satu lubang besar per bagian manifold untuk menyelaraskan exhaust manifold. Jangan lepaskan Perkakas (C) sampai Locknuts kencang dan terkunci.</p> <p>4 Pasang spacer, washer, dan locknut (1). Kencangkan locknuts (1) dalam urutan numerik hingga torsi <math>20 \pm 3 \text{ N} \cdot \text{m}</math> (<math>15 \pm 2 \text{ lb ft}</math>). Kencangkan kembali locknut (1) dalam urutan numerik hingga torsi <math>38 \pm 5 \text{ N} \cdot \text{m}</math> (<math>28 \pm 4 \text{ lb ft}</math>)</p> <p>5 Connect tube assembly (4) dan tube assembly (5).</p> <p>6 Connect clamp (3).</p>                                                                                   |             |                |  |  |  |      |             |                  |     |   |          |                     |   |
| 36                        | TURBOCHARGER    | <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>5P-3931</td> <td>Anti-Seize Compound</td> <td>1</td> </tr> </tbody> </table> <p>1 Posisikan gasket pada the exhaust manifold.</p> <p>2 Gunakan Tool (A) pada threads exhaust manifold studs.</p> <p><b>Catatan:</b><br/>Pasang turbocharger (5) pada lokasi nya sesuai service manual.</p> <p>3 Pasang perangkat pengangkat yang sesuai ke turbocharger (5). Berat turbocharger (5) kira-kira 45 kg (100 lb). Posisikan turbocharger (5) pada exhaust manifold.</p> <p>4 Pasang nuts (3) dan kencangkan dengan torsi <math>55 \pm 9 \text{ N} \cdot \text{m}</math> (<math>41 \pm 7 \text{ lb ft}</math>).</p> <p>5 Pasang tube assembly (2) .</p> |             | Required Tools |  |  |  | Tool | Part Number | Part Description | Qty | A | 5P-3931  | Anti-Seize Compound | 1 |
| Required Tools            |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                |  |  |  |      |             |                  |     |   |          |                     |   |
| Tool                      | Part Number     | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Qty         |                |  |  |  |      |             |                  |     |   |          |                     |   |
| A                         | 5P-3931         | Anti-Seize Compound                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1           |                |  |  |  |      |             |                  |     |   |          |                     |   |
|                           |                 | <p>55 ± 9 N·m<br/>(41 ± 7 lb ft).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                |  |  |  |      |             |                  |     |   |          |                     |   |
|                           |                 | <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>221-8647</td> <td>Seal Installer Gp</td> <td>1</td> </tr> </tbody> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             | Required Tools |  |  |  | Tool | Part Number | Part Description | Qty | A | 221-8647 | Seal Installer Gp   | 1 |
| Required Tools            |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                |  |  |  |      |             |                  |     |   |          |                     |   |
| Tool                      | Part Number     | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Qty         |                |  |  |  |      |             |                  |     |   |          |                     |   |
| A                         | 221-8647        | Seal Installer Gp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1           |                |  |  |  |      |             |                  |     |   |          |                     |   |

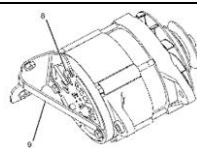
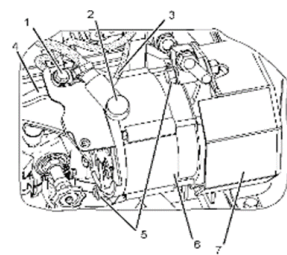
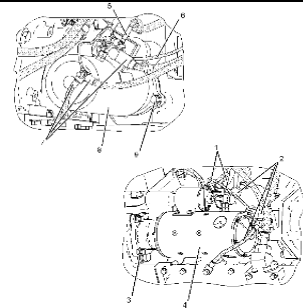
| CHECK SHEET               |                                   | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  | DOCUMENT NO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |
|---------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| CAT ENGINE C32 ASSEMBLING |                                   | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  | PUBLIC DATE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |
| ENGINE CAT C32            |                                   | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | REVISION NO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | REV 01 |
|                           |                                   | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  | PAGE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1 OF 3 |
| PROCESS                   | START<br>FINISH                   | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| 37                        | WATER<br>TEMPERATURE<br>REGULATOR | 1 Jika perlu, gunakan tool (A) untuk memasang Seal Lipetpe.<br>2 Install water temperature regulators (6) dan O-ring seals (5) pada water temperature regulator housing (4).<br>3 Posisikan water temperature regulator housing (4) dalam engine. Install bolts (3) dan bolts (2).<br>4 Connect bracket (1).                                                                                                                                                                                                                     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |
| 38                        | OIL COOLER                        | <b>Catatan:</b><br><b>Periksa seal O-ring, gasket, dan seal dari keausan atau kerusakan. Ganti komponennya, jika perlu.</b><br>1 Bersihkan engine oil cooler core assembly sesuai lubangnya satu - satu.<br>2 Install adapter (30) dan engine oil cooler (31) . Install two bolts (32).<br>3 Install two bolts (29) dan two bolts (28).<br>4 Install bonnet (25) dan six bolts (26).<br>5 Install six bolts (27) dengan hose assembly (11) dan clip.<br>6 Install six bolts (24).<br>7 Install bracket (23) dan empat bolts (22) |  | <br><br><br><br><br> |        |

| CHECK SHEET               |                 | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | DOCUMENT NO                                                                                                                                                                                                                                                                                                                                  |  |     |       |  |  |                                                                                                                                                                           |
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| CAT ENGINE C32 ASSEMBLING |                 | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | PUBLIC DATE                                                                                                                                                                                                                                                                                                                                  |  |     |       |  |  |                                                                                                                                                                           |
| ENGINE CAT C32            |                 | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | REVISION NO                                                                                                                                                                                                                                                                                                                                  |  |     |       |  |  |                                                                                                                                                                           |
|                           |                 | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | PAGE                                                                                                                                                                                                                                                                                                                                         |  |     |       |  |  |                                                                                                                                                                           |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | REV 01                                                                                                                                                                                                                                                                                                                                       |  |     |       |  |  |                                                                                                                                                                           |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1 OF 3                                                                                                                                                                                                                                                                                                                                       |  |     |       |  |  |                                                                                                                                                                           |
| PROCESS                   | START<br>FINISH | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                              |  |     |       |  |  |                                                                                                                                                                           |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                              |  |     |       |  |  |                                                                                                                                                                           |
|                           |                 | <p>8 Install filter base (18), empat bolts (17 ) and three bolts (16).</p> <p>9 Connect hose assemblies (20) dan (19)</p> <p>10 Connect dipstick tube (15) dan install bolt (14). Install tube assembly (10) dan clip (13). Connect hose assembly (11) jika dilengkapi dan two clips (12).</p> <p>11 Install oil filters (9). Referensi dari Operation and Maintenance Manual, "Engine Oil and Filter - Change".</p> <p>12 Install filter base (8), dua bolts (6) dan clips. Connect harness assemblies (7), (5), (4) dan (2). Pasang cable strap (3)</p> <p>13 Install clip (1).</p> |     |  |     |       |  |  |                                                                                                                                                                           |
| 39                        | WATER PUMP      | <p>Water pump Condition :</p> <table border="1"> <thead> <tr> <th colspan="2">WATER PUMP</th> </tr> <tr> <th>NEW</th> <th>REUSE</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> </tr> </tbody> </table> <p>1 Posisikan water pump (4) pada front housing.</p> <p>2 Pasang nuts (5) dan bolts (6).</p> <p>3 Posisikan cover (3) pada water pum</p> <p>4 Pasang bolts (2) dan bolts (1).</p>                                                                                                                                                                                       | WATER PUMP                                                                                                                                                                                                                                                                                                                                   |  | NEW | REUSE |  |  |   |
| WATER PUMP                |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                              |  |     |       |  |  |                                                                                                                                                                           |
| NEW                       | REUSE           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                              |  |     |       |  |  |                                                                                                                                                                           |
|                           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                              |  |     |       |  |  |                                                                                                                                                                           |

| CHECK SHEET               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | WO NUMBER                        | DOCUMENT NO |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
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| CAT ENGINE C32 ASSEMBLING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | COMP.MODEL                       | PUBLIC DATE |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
| ENGINE CAT C32            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | COMP.S/N                         | REVISION NO |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | UNIT MODEL                       | PAGE        |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                  | 1 OF 3      |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
| PROCESS                   | START<br>FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ASSEMBLY BY                      |             |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
| 40                        | <p><b>CRANKSHAFT FRONT SEAL</b></p> <table border="1"> <caption>Required Tools</caption> <thead> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td rowspan="4">E</td> <td>5P-1733</td> <td>Seal Locator</td> <td>1</td> </tr> <tr> <td>5P-1737</td> <td>Bolt</td> <td>3</td> </tr> <tr> <td>9S-8858</td> <td>Nut (Seal Installer)</td> <td>1</td> </tr> <tr> <td>6V-6142</td> <td>Seal Installer</td> <td>1</td> </tr> </tbody> </table> <p><b>PERHATIAN</b><br/>Jaga kebersihan semua bagian dari kontaminan.<br/>Kontaminan dapat menyebabkan keausan cepat dan umur komponen pendek.</p> <p>1 Bersihkan outside diameter pada crankshaft.</p> <p>2 Gunakan Tooling (E) untuk install crankshaft front seal (4) dan crankshaft wear sleeve (3) pada unit.</p> <p><b>Note:</b><br/><i>Install the crankshaft front seal sesuai arah panah dari yang ada pada Front Seal Crankshaft. Arah Panah menunjukkan Putaran Crankshaft atau Putaran Engine.dari depan.</i><br/><i>Dari depan Putaran Engine searah Jarun Jam.</i></p> <p>3 Install adapter (2) dan bolts (1).</p> | Tool                             | Part Number | Part Description | Qty | E | 5P-1733 | Seal Locator                     | 1 | 5P-1737 | Bolt | 3 | 9S-8858 | Nut (Seal Installer) | 1 | 6V-6142 | Seal Installer | 1 |  |  |
| Tool                      | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Part Description                 | Qty         |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
| E                         | 5P-1733                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Seal Locator                     | 1           |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
|                           | 5P-1737                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Bolt                             | 3           |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
|                           | 9S-8858                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Nut (Seal Installer)             | 1           |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
|                           | 6V-6142                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Seal Installer                   | 1           |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
| 41                        | <p><b>VIBRATION DAMPER &amp; PULLEY TYPE 1</b></p> <table border="1"> <caption>Required Tools</caption> <thead> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>-</td> <td>Guide Stud 1/2 - 13 NC by 3 inch</td> <td>2</td> </tr> </tbody> </table> <p>1 Install Tooling (A) pada adapter.</p> <p>2 Pasang perangkat lifting device pada vibration damper (5) Berat vibration damper (5) 34 kg (75 lb). Posisikan vibration damper (5) pada adapter.</p> <p>3 Install spacer (4) dan bolts (3) . Lepas Tooling (A) .</p> <p>4 Install remaining bolts (3) . Kencangkan bolts (3) dengan torque of 135 ± 20 N-m (100 ± 15 lb ft).</p> <p>5 Install pulley (2) dan bolts (1) .</p> <p>6 Install V- belt pada the engine.</p> <div style="display: flex; align-items: center;"> <div style="border: 1px solid black; padding: 2px; margin-right: 10px;"> <b>STANDARD</b><br/>100+ 15 lb ft         </div> <div style="border: 1px solid black; padding: 2px;"> <b>Tightening Bolt</b><br/><b>ACTUAL</b> </div> </div>                                           | Tool                             | Part Number | Part Description | Qty | A | -       | Guide Stud 1/2 - 13 NC by 3 inch | 2 |         |      |   |         |                      |   |         |                |   |  |  |
| Tool                      | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Part Description                 | Qty         |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |
| A                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Guide Stud 1/2 - 13 NC by 3 inch | 2           |                  |     |   |         |                                  |   |         |      |   |         |                      |   |         |                |   |  |  |



| CHECK SHEET               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | WO NUMBER                                                                                                                                                                                    | DOCUMENT NO                                                                        |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
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| CAT ENGINE C32 ASSEMBLING |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | COMP.MODEL                                                                                                                                                                                   | PUBLIC DATE                                                                        |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| ENGINE CAT C32            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | COMP.S/N                                                                                                                                                                                     | REVISION NO                                                                        |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | UNIT MODEL                                                                                                                                                                                   | PAGE                                                                               |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                              | 1 OF 3                                                                             |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| PROCESS                   | START<br>FINISH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ASSEMBLY BY                                                                                                                                                                                  |                                                                                    |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| 42                        | <p><b>VIBRATION DAMPER &amp; PULLEY TYPE 2</b></p> <p>1 Pasang perangkat lifting device pada adapter (5) . Berat adapter (5) 25 kg (55 lb). Install adapter (5) dan bolts (4) .</p> <p>2 Install Tooling (A) pada adapter (5) .</p> <p>3 Pasang lifting device pada second vibration damper (3) . Berat vibration damper (3) adalah 34 kg (75 lb). Posisikan second vibration damper (3) pada adapter (5) .</p> <p>4 Install pulley.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                              |  |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| 43                        | <p><b>VIBRATION DAMPER &amp; PULLEY TYPE 2</b></p> <p>5 Pasang lifting device pada first vibration damper (2) Berat vibration damper (2) 34 kg (75 lb). Posisikan first vibration damper (2) pada adapter.</p> <p>6 Install the spacer dan bolts (1) . Lepas Tool (A) dan install remaining bolts (1) . Kencangkan bolts (1) torque 135 ± 20 N-m (100 ± 15 lb ft).</p> <p>7 Install V-belt pada engine.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>135 ± 20 N-m<br/>(100 ± 15 lb ft).</p> <p><b>Tightening Bolt</b></p> <table border="1"> <tr> <td>STANDARD</td> <td>ACTUAL</td> </tr> <tr> <td>100+ 15 lb ft</td> <td></td> </tr> </table> | STANDARD                                                                           | ACTUAL | 100+ 15 lb ft |      |  |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| STANDARD                  | ACTUAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                              |                                                                                    |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| 100+ 15 lb ft             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                              |                                                                                    |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| 44                        | <p><b>FLYWHEEL</b></p> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td rowspan="4">E</td> <td>-</td> <td>Seal Locator</td> <td>-</td> </tr> <tr> <td>-</td> <td>Bolt</td> <td>-</td> </tr> <tr> <td>-</td> <td>Nut (Seal Installer)</td> <td>-</td> </tr> <tr> <td>-</td> <td>Rear Seal Installer</td> <td>-</td> </tr> </tbody> </table> <p>1 Bersihkan outside diameter pada crankshaft.</p> <p><b>Note:</b><br/>Install crankshaft rear seal sesuai arah panah, arah panah menunjukkan putaran crankshaft. Putaran Crankshaft dilihat dari Flywheel Housing searah Jarum Jam.</p> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>-</td> <td>Guide Stud 5/8 - 18 UNF by 11 inch</td> <td>2</td> </tr> </tbody> </table> <p><b>Note:</b><br/>Ring gear (3) Harus di installed dengan chamfered side dari teeth upward. Chamfered side dari teeth gear akan di sejajarkan dengan starting motor ketika flywheel di pasang.</p> | Required Tools                                                                                                                                                                               |                                                                                    |        |               | Tool | Part Number                                                                        | Part Description | Qty | E | - | Seal Locator | - | - | Bolt | - | - | Nut (Seal Installer) | - | - | Rear Seal Installer | - | Required Tools |  |  |  | Tool | Part Number | Part Description | Qty | A | - | Guide Stud 5/8 - 18 UNF by 11 inch | 2 |  | <br> |
| Required Tools            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                              |                                                                                    |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| Tool                      | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Part Description                                                                                                                                                                             | Qty                                                                                |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| E                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Seal Locator                                                                                                                                                                                 | -                                                                                  |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
|                           | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Bolt                                                                                                                                                                                         | -                                                                                  |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
|                           | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Nut (Seal Installer)                                                                                                                                                                         | -                                                                                  |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
|                           | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Rear Seal Installer                                                                                                                                                                          | -                                                                                  |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| Required Tools            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                              |                                                                                    |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| Tool                      | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Part Description                                                                                                                                                                             | Qty                                                                                |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |
| A                         | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Guide Stud 5/8 - 18 UNF by 11 inch                                                                                                                                                           | 2                                                                                  |        |               |      |                                                                                    |                  |     |   |   |              |   |   |      |   |   |                      |   |   |                     |   |                |  |  |  |      |             |                  |     |   |   |                                    |   |  |                                                                                                                                                                             |

| CHECK SHEET               |                 | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                     | DOCUMENT NO |                                                                                                                                                                                                                          |          |                 |                 |        |
|---------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|-----------------|--------|
| CAT ENGINE C32 ASSEMBLING |                 | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                     | PUBLIC DATE |                                                                                                                                                                                                                          |          |                 |                 |        |
| ENGINE CAT C32            |                 | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                     | REVISION NO | REV 01                                                                                                                                                                                                                   |          |                 |                 |        |
|                           |                 | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                     | PAGE        | 1 OF 3                                                                                                                                                                                                                   |          |                 |                 |        |
| PROCESS                   | START<br>FINISH | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                     |             |                                                                                                                                                                                                                          |          |                 |                 |        |
|                           |                 | 1 Naikan temperatur dari ring gear (3). Install ring gear (3) pada flywheel (2).<br>2 Install Tool (A) pada crankshaft.<br>3 Pasang lifting device pada flywheel (2). Berat flywheel (2) 86 kg (190 lb). Posisikan flywheel (2) pada the crankshaft.<br>4 Beri pelumasan dengan Engine Oil pada ujung threads bolts (1).<br>5 Pasang bolts (1). Lepas Tool (A). Pasang remaining bolts (1).<br>6 Kencangkan bolts (1) dengan torque of 270 ± 40 N·m (199 ± 29.5 lb ft).                    | 270 ± 40 N·m<br>(199 ± 29.5 lb ft). |             |  <table border="1"> <tr> <th>STANDARD</th> <th>Tightening Bolt</th> </tr> <tr> <td>199+ 29.5 lb ft</td> <td>ACTUAL</td> </tr> </table> | STANDARD | Tightening Bolt | 199+ 29.5 lb ft | ACTUAL |
| STANDARD                  | Tightening Bolt |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                     |             |                                                                                                                                                                                                                          |          |                 |                 |        |
| 199+ 29.5 lb ft           | ACTUAL          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                     |             |                                                                                                                                                                                                                          |          |                 |                 |        |
| 45                        | ALTERNATOR      | 1 Posisikan bracket (9) dan install bolts (8)                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                     |             |                                                                                                                                       |          |                 |                 |        |
| 46                        | ALTERNATOR      | 2 Posisikan alternator (6). Posisikan guard (7) and install bolts (5). Connect harness assembly (3). Connect ground strap (4). Install bolt (1). Connect cable assembly (2).<br>3 Pasang V-belt pada engine.                                                                                                                                                                                                                                                                               |                                     |             |                                                                                                                                       |          |                 |                 |        |
| 47                        | STARTING MOTOR  | 1 Pasang perangkat yang sesuai ke Starting Motor (8). Berat Electric starting motor (8) kira-kira 36 kg (80 lb). Posisikan starting motor (8) di sisi kiri Engine. Pasang baut (9). Connect harness (6) dan harness assy (5). Dan connect kabel (7).<br>2 Pasang perangkat yang sesuai ke Electric Starting Motor (4). Berat Starting Motor (4) kira-kira 36 kg (80 lb). Posisikan Electric Starting motor (4) di sisi kanan Engine. Pasang baut (3). Hubungkan kabel (2) dan harness (1). |                                     |             |                                                                                                                                      |          |                 |                 |        |





|                                                                                 |            |  |             |        |
|---------------------------------------------------------------------------------|------------|--|-------------|--------|
| <b>CHECK SHEET</b><br><b>CAT ENGINE C32 ASSEMBLING</b><br><b>ENGINE CAT C32</b> | WO NUMBER  |  | DOCUMENT NO |        |
|                                                                                 | COMP.MODEL |  | PUBLIC DATE |        |
|                                                                                 | COMP.S/N   |  | REVISION NO | REV 01 |
|                                                                                 | UNIT MODEL |  | PAGE        | 1 OF 3 |
| PROCESS                                                                         | START      |  | ASSEMBLY BY |        |
|                                                                                 | FINISH     |  |             |        |

INSPECTION RESULT :

INSPECTION & APPROVAL BY :

INSPECTION BY

APPROVE BY

MECHANIC

QUALITY ASSURANCE (QA)

GROUP LEADER (GL)



**PT SAPTAINDRA SEJATI**

**ENGINE**

**CYLINDER HEAD ASSEMBLING**

**CHECK SHEET**

**ENGINE CAT C32**

**MACHINE MODEL**

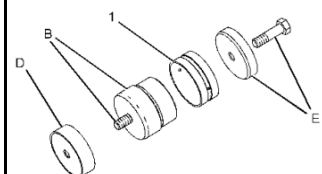
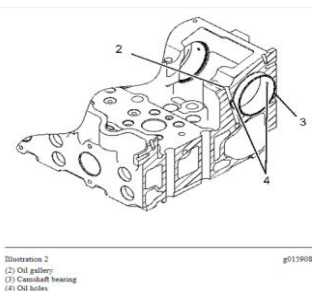
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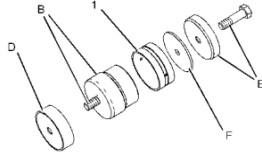
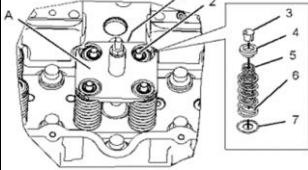
**WO NO :**

**ENGINE SECTION**

| CHECK SHEET              |                 | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | DOCUMENT NO    |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
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| CYLINDER HEAD ASSEMBLING |                 | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PUBLIC DATE    |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| ENGINE CAT C32           |                 | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | REVISION NO    |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
|                          |                 | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PAGE           |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
|                          |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 OF 3         |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| PROCESS                  | START           | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
|                          | FINISH          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| NO                       | PART TO INSTALL | CONDITION TOBE INSTALL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | SPEC           | SIGN<br>MECH GL/Q | SKETCH DRAWING OR PHOTHOES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| 1                        | VALVE GUIDE     | <ul style="list-style-type: none"> <li>Lumasi lubang Valve Guide dengan Engine Oil yang bersih.</li> <li>Pasang Valve Guide (2) pada Cylinder Head menggunakan Tool (A) dan Toll (B).</li> </ul> <p><b>Note :</b><br/><b>Tool (B) harus digunakan pada saat pemasang Valve Guide agar ketinggian dr Valve Guide rata/sama semuanya.</b></p> <p>Ketinggian Valve Guide dari Permukaan Cyl Head : 35.00 ± 050 mm.</p> <ul style="list-style-type: none"> <li>Gunakan Tool (C) untuk memasang Valve Stem Seal (1) di Valve Guide.</li> </ul> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A<sup>(1)</sup></td> <td>9U-6895</td> <td>Valve Guide Driver</td> <td>1</td> </tr> <tr> <td>B<sup>(1)</sup></td> <td>9U-6894</td> <td>Guide Collar</td> <td>1</td> </tr> <tr> <td rowspan="2">C<sup>(2)</sup></td> <td>455-0128</td> <td>Seal Installer</td> <td>1</td> </tr> <tr> <td>375-2773</td> <td>Seal Installer</td> <td>1</td> </tr> </tbody> </table> <p>(1) Part of 9U-7530 Engine Tool Group</p> | Required Tools |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                | Tool | Part Number | Part Description | Qty  | A <sup>(1)</sup> | 9U-6895          | Valve Guide Driver | 1 | B <sup>(1)</sup> | 9U-6894             | Guide Collar | 1       | C <sup>(2)</sup> | 455-0128 | Seal Installer | 1                | 375-2773 | Seal Installer | 1       |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| Required Tools           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| Tool                     | Part Number     | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Qty            |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| A <sup>(1)</sup>         | 9U-6895         | Valve Guide Driver                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| B <sup>(1)</sup>         | 9U-6894         | Guide Collar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| C <sup>(2)</sup>         | 455-0128        | Seal Installer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
|                          | 375-2773        | Seal Installer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| 3                        | INJECTOR SLEEVE | <ul style="list-style-type: none"> <li>Gunakan Tool (B) untuk membersihkan Lubang Sleeve Injector pada Cylinder Head.</li> <li>Pasang O-Ring Seal baru (2) dan O-Ring Seal (3) pada Sleeve Injector.</li> </ul> <p>Gunakan Tool (C) untuk merekatkan permukaan Sleeve Injector 1) pada Permukaan yang ada tanda "X"</p> <p><b>Note:</b><br/><b>Jangan menggunakan Tool (C) pada permukaan Cylinder Head Tool (C) hanya digunakan pada Sleeve Injector.</b></p> <ul style="list-style-type: none"> <li>Beri Pelumas O-Ring Seal (2) dengan Oli Engine bersih.</li> <li>Pasang Sleeve Injector (1) pada Tool (A) Posisikan Sleeve Injector kedalam Cylinder Head, berhati-hatilah saat memasang sleeve jangan sampai O-Ring nya Rusak/Putus. Gunakan Tool (A) dan Palu untuk Pemasangan Sleeve Injector pada Cylinder Head.</li> <li>Lepas Tool (A) dan Tool (C) jika sudah selesai Pemasangan sleeve injector.</li> </ul>                                                                                                                                                                                                                        |                |                   | <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td rowspan="3">A</td> <td>9U-6891</td> <td>Injector Tool Group</td> <td>1</td> </tr> <tr> <td>9U-6862</td> <td>Tapered Brush</td> <td>1</td> </tr> <tr> <td>9U-6863</td> <td>Small Bore Brush</td> <td>1</td> </tr> <tr> <td rowspan="3">B</td> <td>9U-7244</td> <td>End Brush</td> <td>1</td> </tr> <tr> <td>9U-7237</td> <td>Brush Extension</td> <td>1</td> </tr> <tr> <td>4C-5552</td> <td>Large Bore Brush</td> <td>1</td> </tr> <tr> <td>C</td> <td></td> <td>Loctite 620 Retaining Compound</td> <td>1</td> </tr> </tbody> </table> | Required Tools |      |             |                  | Tool | Part Number      | Part Description | Qty                | A | 9U-6891          | Injector Tool Group | 1            | 9U-6862 | Tapered Brush    | 1        | 9U-6863        | Small Bore Brush | 1        | B              | 9U-7244 | End Brush | 1 | 9U-7237 | Brush Extension | 1 | 4C-5552 | Large Bore Brush | 1 | C |  | Loctite 620 Retaining Compound | 1 |
| Required Tools           |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| Tool                     | Part Number     | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Qty            |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| A                        | 9U-6891         | Injector Tool Group                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
|                          | 9U-6862         | Tapered Brush                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
|                          | 9U-6863         | Small Bore Brush                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| B                        | 9U-7244         | End Brush                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
|                          | 9U-7237         | Brush Extension                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
|                          | 4C-5552         | Large Bore Brush                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
| C                        |                 | Loctite 620 Retaining Compound                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |
|                          |                 | <p>1 Pasang terlebih dahulu Camshaft Bearing No.7 (Belakang) dengan Prosedure urutan sbb :</p> <ol style="list-style-type: none"> <li>Masukkan Ujung Tool (B) kedalam Lubang Camshaft No.7</li> <li>Posisikan Tool (C) diatas Tool (A) melalui Cyl head Tool (B)</li> <li>Posisikan Camshaft Bearing/Bushing (1) pada Tool (B)</li> </ol> <p><b>Note :</b><br/><b>Posisikan Lubang Bushing campshaft sesuai gambar</b><br/><b>Prosedur pemasangan : (2, 3, 4).</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |      |             |                  |      |                  |                  |                    |   |                  |                     |              |         |                  |          |                |                  |          |                |         |           |   |         |                 |   |         |                  |   |   |  |                                |   |

| CHECK SHEET              |                  | WO NUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DOCUMENT NO |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
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| CYLINDER HEAD ASSEMBLING |                  | COMP.MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | PUBLIC DATE |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| ENGINE CAT C32           |                  | COMP.S/N                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | REVISION NO | REV 01 |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
|                          |                  | UNIT MODEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | PAGE        | 1 OF 3 |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| PROCESS                  | START            | ASSEMBLY BY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
|                          | FINISH           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| 4                        | CAMSHAFT BUSHING | <p>d Gunakan Tool (A) untuk menarik Camshaft Bearing (1) ke dalam lubang camshaft.</p> <p><b>Note :</b><br/><i>Ketika chamfer dari tool (E) kontak dengan permukaan dari lubang camshaft, Camshaft Bushing harus diarahkan posisi lubangnya dan pemasangannya harus rata.</i></p> <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>SS-2241</td> <td>Camshaft Bearing Tool Group</td> <td>1</td> </tr> <tr> <td rowspan="2">B</td> <td>9U-7222</td> <td>Camshaft Bearing Pilot</td> <td>1</td> </tr> <tr> <td>SM-8778</td> <td>Taperlock Stud 1/2 - 13 by 1 9/16 inch</td> <td>1</td> </tr> <tr> <td>C</td> <td>9U-7210</td> <td>Puller Plate</td> <td>1</td> </tr> <tr> <td>D</td> <td>9U-7223</td> <td>Alignment Bushing</td> <td>1</td> </tr> <tr> <td></td> <td>9U-7213</td> <td>Backup Plate</td> <td>1</td> </tr> <tr> <td>E</td> <td>OS-1621</td> <td>Bolt 1/2 - 13 by 1 inch</td> <td>1</td> </tr> <tr> <td>F</td> <td>9U-7214</td> <td>Spacer Plate</td> <td>1</td> </tr> </tbody> </table>                                                            |             |        | Required Tools |  |  |  | Tool | Part Number | Part Description | Qty | A | SS-2241 | Camshaft Bearing Tool Group | 1 | B | 9U-7222 | Camshaft Bearing Pilot | 1 | SM-8778 | Taperlock Stud 1/2 - 13 by 1 9/16 inch | 1 | C | 9U-7210 | Puller Plate | 1 | D | 9U-7223 | Alignment Bushing | 1 |  | 9U-7213 | Backup Plate | 1 | E | OS-1621 | Bolt 1/2 - 13 by 1 inch | 1 | F | 9U-7214 | Spacer Plate | 1 |
| Required Tools           |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| Tool                     | Part Number      | Part Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Qty         |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| A                        | SS-2241          | Camshaft Bearing Tool Group                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1           |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| B                        | 9U-7222          | Camshaft Bearing Pilot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1           |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
|                          | SM-8778          | Taperlock Stud 1/2 - 13 by 1 9/16 inch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1           |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| C                        | 9U-7210          | Puller Plate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1           |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| D                        | 9U-7223          | Alignment Bushing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1           |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
|                          | 9U-7213          | Backup Plate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1           |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| E                        | OS-1621          | Bolt 1/2 - 13 by 1 inch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1           |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| F                        | 9U-7214          | Spacer Plate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1           |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |
| 5                        | CAMSHAFT BUSHING | <p>2 Pasang Camshaft Bearing / Bushing No.6 sampai No.2 ikuti Langkah langkah sebagai berikut :</p> <p>a Masukkan Tool (B) kedalam Lubang Camshaft.<br/><b>Note:</b><br/><i>Gunakan Tooling (D) untuk menyelaraskan Tooling (A) dan Tooling (B) dengan lubang camshaft bushing, Pasang Tool (D) di diameter dalam bantalan camshaft yang terpasang antara Tool (A) dan Tool (B).</i></p> <p>b Pasang Tool (A) pada Tool (B) posisikan Chamshaft Bushing (1) di Tool (B) Pasang Tool (E) pada Tool (B).</p> <p>c Gunakan Tool (A) untuk menarik Camshaft Bushing (1) kedalam Lubang Camshaft</p> <p><b>Note:</b><br/><i>Ketika Chamfer Tool (E) kontak dengan permukaan Lubang Camshaft maka Camshaft Bushing harus diarahkan posisinya Lubangnya sesuai gambar prosedur diatas.</i></p> <p>3 Pasang Camshaft Bushing No.1 (Front) ikuti Langkah berikut :</p> <p>a Masukkan ujung besar Tool (B) ke lubang camshaft No.1 Pasang Tool (A) dan Tool (D) pada Tool (B).</p> <p>b Posisikan Camshaft bushing (1) pada Tool (B). Pasang Tool (E) dan Tool (F) pada Tool (B).</p> <p><b>Note:</b><br/><i>Tool (F) digunakan untuk dudukan NO.1 dari Camshaft Bushing untuk mengkoreksi kedalam lubang camshaft.</i></p> |             |        |                |  |  |  |      |             |                  |     |   |         |                             |   |   |         |                        |   |         |                                        |   |   |         |              |   |   |         |                   |   |  |         |              |   |   |         |                         |   |   |         |              |   |



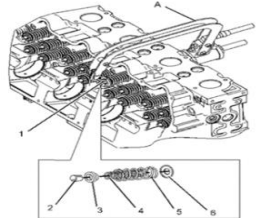
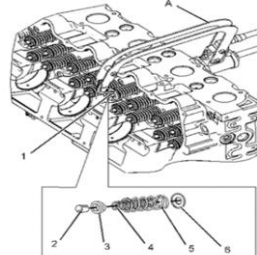
| CHECK SHEET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             | WO NUMBER                                                                                                                                                                | DOCUMENT NO |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
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| CYLINDER HEAD ASSEMBLING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             | COMP.MODEL                                                                                                                                                               | PUBLIC DATE |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
| ENGINE CAT C32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             | COMP.S/N                                                                                                                                                                 | REVISION NO |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             | UNIT MODEL                                                                                                                                                               | PAGE        |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                                                                                                                                                          | 1 OF 3      |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
| PROCESS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | START       | ASSEMBLY BY                                                                                                                                                              |             |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | FINISH      |                                                                                                                                                                          |             |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
| <p>3 Pasang Camshaft Bushing No.1 (Front) ikuti Langkah berikut :</p> <p>a Masukkan ujung besar Tool (B) ke lubang camshaft No.1 Pasang Tool (A) dan Tool (D) pada Tool (B).</p> <p>b Posisikan Camshaft bushing (1) pada Tool (B). Pasang Tool (E) dan Tool (F) pada Tool (B).</p> <p><b>Note:</b><br/><b>Tool (F) digunakan untuk dudukan NO.1 dari Camshaft Bushing untuk mengkoreksi kedalaman lubang camshaft.</b></p> <p>c Gunakan Tool (A) untuk menarik Camshaft Bushing (1) ke Lubang Camshaft No.1. Ketika Chamfer tool (E) kontak dgn Permukaan Lubang Camshaft, Camshaft Bushing harus di atur agar posisi Lubang Camshaft Busing letaknya sesuai Prosedur.</p> <p>Jika menggunakan Tool Valve Compression (A) bisa memasang Valve Intake dan Exhaust bersamaan 4 Valve sekaligus.</p> <p><b>Ikuti Prosedur sebagai berikut :</b><br/><b>Beri Pelumasan Valve Steam dengan Oli Engine bersih.</b></p> <ul style="list-style-type: none"> <li>Pasang Tool (A) diantara Lubang Sleeve Injector dan 4 Valve steam sesuai Gambar.</li> <li>Pastikan Spring Washer (7) Terpasang pada Vallve Guide.</li> <li>Pasang Install Valve Spring (5) dan Outer Valve Spring (6) di Valve (2).</li> <li>Pasang Valve Rotator (4) pada Valve (2).</li> <li>Gunakan Tool (A) untuk mengkompres Valve Spring (5) dan Outer Valve Spring (6). Pasang Spring Keeper (3) pada Valve Outer Valve Spring (6). Pasang Spring Keeper (3) pada Valve Stem Intake dan Exhaust.</li> <li>Kendorkan Nut Tool (A) perlahan sampai Tool (A) naik keatas dan Keeper mengunci Valve dan Rotator.</li> </ul> <p><b>Note :</b><br/><b>Setelah Valve semua terpasang pada Cylinder Head beri tanda pada Rotator menggunakan Marker kemudian Lakukan Pemukulan pada Valve menggunakan Palu Karet,Pastikan tanda Marker</b></p> |             |   |             |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
| <table border="1"> <thead> <tr> <th colspan="4">Required Tools</th> </tr> <tr> <th>Tool</th> <th>Part Number</th> <th>Part Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>9U-7241</td> <td>Valve Spring Compressor</td> <td>1</td> </tr> <tr> <td>B</td> <td>441-0714</td> <td>Tool As</td> <td>1</td> </tr> </tbody> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             | Required Tools                                                                                                                                                           |             |  |  | Tool | Part Number | Part Description | Qty | A | 9U-7241 | Valve Spring Compressor | 1 | B | 441-0714 | Tool As | 1 |  |  |
| Required Tools                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |                                                                                                                                                                          |             |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
| Tool                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Part Number | Part Description                                                                                                                                                         | Qty         |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
| A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 9U-7241     | Valve Spring Compressor                                                                                                                                                  | 1           |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 441-0714    | Tool As                                                                                                                                                                  | 1           |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |

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| <b>CHECK SHEET</b>              |  | <b>WO NUMBER</b>  |  | <b>DOCUMENT NO</b> |        |
| <b>CYLINDER HEAD ASSEMBLING</b> |  | <b>COMP.MODEL</b> |  | <b>PUBLIC DATE</b> |        |
| <b>ENGINE CAT C32</b>           |  | <b>COMP.S/N</b>   |  | <b>REVISION NO</b> | REV 01 |
|                                 |  | <b>UNIT MODEL</b> |  | <b>PAGE</b>        | 1 OF 3 |

|                |               |  |                    |  |  |
|----------------|---------------|--|--------------------|--|--|
| <b>PROCESS</b> | <b>START</b>  |  | <b>ASSEMBLY BY</b> |  |  |
|                | <b>FINISH</b> |  |                    |  |  |

|                | <p>Jika menggunakan Tool Valve Spring Compressor (5S-1330) Hanya bisa 1 Valve tidak bisa memasang 4 Valve sekaligus.</p> <p><b>Ikuti Prosedur sebagai berikut :</b><br/><b>Beri Pelumasan Valve Steam dengan Oli Engine</b></p> <ul style="list-style-type: none"> <li>• Pasang Washer (6) pada Valve Guide.</li> <li>• Pasang Inner Valve Spring (4) dan Outer Valve Spring (5) pada Valve (1).</li> <li>• Pasang Valve Rotator (3) pada Valve (1).</li> <li>• Gunakan Tool (A) untuk mengkompres Valve Spring (4) dan Outer Valve Spring (5). Pasang Spring Keeper</li> <li>• Berhati-hati dalam melepas Tool (A).</li> </ul> <p>Note :<br/>Setelah Valve semua terpasang pada Cylinder Head beri tanda pada Rotator menggunakan Marker kemudian Lakukan Pemukulan pada Valve menggunakan Palu Karet,Pastikan tanda Marker berjalan/berputar.</p> <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <tr> <th colspan="4">Required Tools</th></tr> <tr> <th>Tool</th><th>Part Number</th><th>Part Description</th><th>Qty</th></tr> <tr> <td>A</td><td>5S-1330</td><td>Valve Spring Compressor</td><td>1</td></tr> <tr> <td>B</td><td>441-0714</td><td>Tool As</td><td>1</td></tr> </table> | Required Tools          |     |  |  | Tool | Part Number | Part Description | Qty | A | 5S-1330 | Valve Spring Compressor | 1 | B | 441-0714 | Tool As | 1 |  |  | <br> |
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| Required Tools |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                         |     |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |                                                                                                                                                                            |
| Tool           | Part Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Part Description        | Qty |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |                                                                                                                                                                            |
| A              | 5S-1330                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Valve Spring Compressor | 1   |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |                                                                                                                                                                            |
| B              | 441-0714                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Tool As                 | 1   |  |  |      |             |                  |     |   |         |                         |   |   |          |         |   |  |  |                                                                                                                                                                            |

**INSPECTION RESULT :**

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INSPECTION & APPROVAL BY :

INSPECTION BY

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MECHANIC

APPROVE BY

\_\_\_\_\_

QUALITY ASSURANCE (QA)

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GROUP LEADER (GL)